Artificial Intelligence in Management

Self-learning and Autonomous Systems as Key Drivers of Value Creation

Andrzej Wodecki, 04.2020

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Full-stack Development Platforms ..... ***Błąd! Nie zdefiniowano zakładki.***

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Introduction

One of the sources of human 'competitive advantage' over other animals stems from our capacity for communication and abstract thought. This allows us not only to collectively perform certain tasks but also to transfer years-worth of cumulative experience and knowledge to our children and other members of the community. This knowledge constitutes a key factor in our development: the more we acquire and learn to utilise, the faster we are able to grow.

Thousands of years of history allowed humans to develop many effective methods of transferring knowledge, particularly by teaching it to others. Still, those methods are not without certain inherent limitations: they are time-consuming and highly dependent on the aptitude and motivation of the respective people involved. But even despite this fact, the dynamics of societal development, be it in the technological, economic (welfare) or social dimension (level of education, health, value systems) remains staggeringly high.

But let us imagine structures that would 1. have the ability to gather and process considerably larger amounts of information from a far broader spectrum of stimuli than that available to humans, 2. be able to extract knowledge therefrom and learn from the experiences gained, and 3. have the capacity to instantaneously and reliably transfer the same to all members of their community. As soon as any one of the thousands of elements comprising such a structure were to learn a certain skill – all other members thereof would also gain access to the same. And if a certain mistake were made and then successfully remedied in a given situation – all members would learn the ability to avoid it in the future. How quickly would such a structure be able to develop? And what would be the direction of its evolution?

For years, artificial intelligence has excited the imaginations of Science Fiction writers and business managers alike. On the one hand, there are many hopes for potential future benefits, and on the other, exaggerated fears of less than welcome repercussions. Optimists see it as a chance for accelerated growth, lower production costs, or improved safety. Pessimists fear the expected disappearance of job opportunities, excessive dependence on technology, deterioration of human cognitive abilities, a world controlled

by elites with exclusive access to intelligent technologies and ultimately ripped from our grasp by AI itself. Meanwhile, researchers are busy looking for new methods and value generation structures applicable to organisational and networked context alike.

As follows from the author's experience, many managers and owners of medium and large enterprises are open (often eagerly so) to the possibility of implementing such solutions and have the necessary resources at their disposal, but find themselves incapable of developing a portfolio of projects that would rationally utilise the potential of artificial intelligence (AI) in generating tangible business value in short, medium, and long-term perspective. At the same time, local and global markets are rife with providers offering such systems (or components necessary to create them) to both mature companies and newly established start-ups. The situation is further complicated by the rapid growth of industries, methods and technologies fundamental to the development of autonomous solutions. The situation is not unlike that observed in the context of integrated IT systems several dozen years ago: both the technologies themselves and the related demand existed, but there were still no reliable methods of building viable business cases or experienced teams capable of effectively implementing such solutions.

In order to discover and effectively utilise the potential of a system, one needs to have at least some intuitive grasp, if not full understanding, of the way in which it operates. This provides an insight into both the abilities (explicitly stated in e.g. the provider's offer or derived from application scenarios, as well as still uncovered) and limitations of a given solution. With his knowledge, one is able to better project the particulars of implementation, necessary resources (human, financial, infrastructural), and feasible time frames. A more in-depth intuition, coupled with industry-specific experience and knowledge, will also be helpful when determining the potential impact that a given technology can have on markets and future trends.

The aim of Chapter 1 will be to familiarise the reader with the basic methodologies that constitute the foundation of autonomous systems. Firstly, the general landscape of artificial intelligence and machine learning will be presented. We will discuss the key methods and techniques involved, describe their underlying principles, and suggest possible applications. The next sections will explore in greater detail the ways in which

intelligent systems are able to learn through interactions with their environment. A set of such methodologies, jointly described as reinforcement learning, is fundamental to many contemporary intelligent systems – an intuitive understanding of the basic ideas related to this area will prove very useful when trying to discuss the potential and limitations inherent in autonomous solutions. Chapter 1 will close with a presentation of various definitions of autonomy applicable to currently employed systems, thus naturally paving the way for the deliberations included in the subsequent parts of the book.

As already mentioned, this publication is an attempt to describe a fairly universal model of value generation in organisations utilising the potential of autonomous technologies. Decision-makers (leaders, managers, owners) in companies that collaborated with the author, typically identified two primary needs: to identify the advanced technologies that offer the greatest potential for facilitating them in the achievement of already defined strategic goals, and, in the case of those more versed in the topic, to modify their existing strategies with the view to better utilising the potential of AI and preparing for imminent changes (e.g. in their markets or competitive milieu). In either case, even with the knowledge of various models of value generation gained from e.g. scientific publications, anyone attempting to satisfy both those needs would need to be familiar with the viable uses of intelligent technologies, if not all of them than at least in the context of organisations similar to one's own.

This is one of the reasons why such a large part of this book (entire Chapter 2 in fact) will be devoted to real-life examples of practical AI applications. Firstly, the key technologies employed in intelligent systems will be presented. Subsequently, dozens of AI applications in various functional areas of organisations will be discussed: e.g. in logistics, manufacturing, sales, or customer service. Finally the possible impact on various industries will also be considered.

This compiled factographic material will provide the basis for the subsequent identification of best practices applicable to intelligent system implementations, a categorisation of the possible values attainable, and proposal of an implementation strategy developed through analogy with processes inherent in human resources management. Those conclusions, presented in Chapter 3 of the book, will hopefully

contribute to achieving the primary purpose of this publication: to aid managers in reliably and effectively generating value from intelligent systems and solidifying the resulting changes in their organisations.

Chapter 4 will be dedicated to future trends: the possible impact of autonomous technologies on market sectors, the concepts of distributed AI (DAI) and cognitive networks, as well as some particularly fascinating studies conducted at research centres and corporate laboratories. Some of those ideas may set the trends for future technological development and help to create a new niche for systems that are not only autonomous but also capable of continuous learning and self-improvement.

Bibliographical note

Anyone interested in gaining a deeper insight into AI methods and technologies would benefit greatly from studying, above all, some of the great textbooks pertaining to the topic. One of the best and most comprehensive analyses was offered by Norvig and Russell (Russell et al. 2010). This admittedly rather bulky publication (nearly 1,000 pages) provides a solid mathematical foundation before proceeding to discuss the methods of both so-called *Good Old Fashioned AI (GOFAI)* and the most recent advances in the areas of machine learning or deep learning. Valuable insights into the concept of machine learning can also be derived from publications by Geron (Géron 2019), Witten et al. (Witten 2017), and Goodfellow et al. (Goodfellow et al. 2016). The foundations of reinforcement learning methodologies were analysed in considerable detail by Sutton and Barto (Sutton and Barto 2017), while practical implementations (including algorithms and software fragments, as well as some really intuitive insights) were discussed in the books by Morales (Morales 2020) and Lapan (Lapan 2020).

Most of the content included in Chapter 2 was based on case studies. Initially, over 500 companies were identified that either utilised or marketed intelligent, autonomous systems. The search was facilitated by services such as index.co or Crunchbase as well as various market reports (e.g. Gartner.com as well as major advisory companies). Subsequently, a preliminary analysis of the identified enterprises allowed the selection 186 most pertinent examples (cf. Appendix 1). The same were then analysed in terms of

the employed technologies, spectra of applications, and values generated, with the view to classifying them under respective categories (functional areas in organisations and industry sectors). At the stage of summarising and classifying the values generated by intelligent systems (Chapter 3), information published by data science providers such as DataRobot, Rapid Miner, H2O, or Dataiku proved very helpful.

Chapter 4 makes numerous references to scientific research conducted in the area of artificial intelligence. This was primarily facilitated by results presented in the course of major conferences in this field, e.g. *Neural Information Processing Systems: NeurIPS.* References were also made to research results published by companies such as DeepMind (Deep Mind 2020) in *Nature, Science* and other periodicals.

Artificial Intelligence and Machine Learning Landscape

Artificial Intelligence: Basic Concepts

Machine intelligence

Artificial intelligence is the central focus of this book –it is therefore only fitting that we begin our deliberations by reflecting on the term itself.

The first word denotes something fairly easy to define: *artificial* simply means *man-made*. Naturally, any attempt to delve on specifics such as e.g. *form, physical location,* or *architecture* of such a creation would significantly complicate things. As will be shown in the subsequent chapters of this book, at present, intelligent systems are basically computer programmes running on devices utilising silicon chips. They can operate 'autonomously', take advantage of remote intelligent services (e.g. in the Cloud Computing model), or engage with other objects to establish networks. Those seemingly trivial observation are in fact anything but that. These days, highly advanced operations can already be performed by quantum computers (inherently different from contemporary integrated circuitry), data can be stored in DNA codes (rather than on digital memory chips), and calculations can be performed with the use of protein structures (and one does not mean animal brains). As we can see, there is currently more than one way to potentially approach the issue of artificiality – anyone interested in the topic could do far worse than to read through Bostrom's compelling analysis of the same [@bostrom2014].

A concept far more difficult to put a finger on is *intelligence*. There have been many definitions over the years, mostly in the field of psychology. It has been a long standing dream to create structures capable of assisting or replacing people in solving various day-

to-day problems. One of human 'competitive advantages' over animals is the ability to use tools – objects enhancing people's innate potential, e.g. strength or speed. In time, such tools evolved into machines – still, their contributions continued to be limited to physical activities. The development of higher cognitive functions, in particular the ability to formulate complex goals, adopt and implement strategies, and plan actions necessary to that end, inspired people to search for methods of enhancing their abilities also in those specific areas (i.e. no longer exclusively physical but also mental). A sort of a 'prototype' in this context came in the form of mathematical theorems and algorithms which (due to their abstract nature) helped in categorising and correlating certain seemingly diverse problems and subsequently fairly easily solving them (e.g. the laws of geometry applied in the construction of the Tunnel of Eupalinos in ancient Greece). The next natural step was to 'implement' those originally abstract algorithms in physical devices: initially simple (e.g. the abacus) and then increasingly complex (e.g. Babbage's difference engine) (cf. e.g. [@davis1949]).

As algorithms continued to be developed, so did the range of problems to which they could be applied, which on the one hand, created a demand for increasingly advanced computing machines, and on the other, generated entirely new problems. There were certain rather narrow areas (e.g. arithmetic) where machines fairly quickly proved themselves to be more capable than humans. However, as computers began to become more popular, in particular more available to a wider group of programmers, the spectrum of problems that could be possibly solved by man-made machines grew exponentially – which soon inspired people to reflect on their (the machines') intelligence.

Before committing to a definition of *intelligence*, let us first take a somewhat closer look at the problems one may reasonably expect intelligent machines to solve. To that end, we will use the slightly adapted classification first proposed by Norvig and Russel in their fundamental textbook on artificial intelligence [@russell2010a].

Let us begin by **characterising the environment** wherein our machine might operate. As the given task is being performed, the same can remain unchanged (**static** as e.g. in the

game of chess) or evolve – i.e. be **dynamic**[1]**.** Moreover, the next consecutive state of the environment may depend solely on the actions of the machine (in which case the environment is **deterministic**) or be influenced by factors independent of the agent (a **stochastic** environment). The environment can also be **discrete** (described by variables with a finite set of possible values) or **continuous**. The latter distinction is particularly important from the computing perspective: in a world of infinite choices, the system must have the capacity for generalisation. Finally, our machine (often described as the agent) may act within the given environment either **independently** or be accompanied by other 'beings' (be it machines, animals or humans) – in which case we can talk of a **multiagent** environment.

The next distinction is related to **knowledge about the environment**. The agent may **know** the rules and laws applicable in the given context (e.g. in board games) or **not know** them (as is often the case in real life). Another aspect of knowledge about the environment is access to information about its state: the agent may either have **full knowledge** of the environment's state, only **partial knowledge** thereof ,or **no knowledge whatsoever** (be blind).

The final classification relates to the character of **tasks** that our agent might aim to perform. They may be either **episodic** (have a clearly defined beginning and end, like in the case of most games) or **continuous** (have no natural end point).

As follows from the above, the complexity spectrum of potential problems is very wide: from the performance of episodic tasks in single-agent, deterministic, fully observable worlds with readily available rulesets, to continuous tasks performed in stochastic environments without any knowledge of the applicable rules or ability to perceive anything.

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[1] It is also possible that while the environment remains unchanged, the task itself evolves – such an environment is described as semi-dynamic.

The above classification is a good starting point when attempting to define intelligence. With full appreciation for the complexity of this issue, for the purposes of the deliberations to follow, intelligence will be defined as the **ability to effectively act under new circumstances**. A closer analysis of the proposed definition will also facilitate a first look at the concept of autonomous, self-learning systems whose business potential will be discussed in the subsequent parts of this book.

Let us begin with the wold *effectively*. It assumes the existence of a certain goal function: in most cases, intelligent systems are expected to perform certain missions whose success or failure is possible to define. Measures of success are usually an integral part of the task and help the machine to monitor, on an ongoing basis, the progress of the learning process as well as the status of mission completion.

Formulating a viable goal function is a considerable challenge in itself. One of the most popular methods (discussed below in greater detail) is the trial and error approach: the agent decides to perform a given action and receives feedback from its environment containing data on the new situation and applicable reward or punishment. The reward, often described as *enhancement* – a term derived from psychology, is one of the primary drivers of the learning process. On the one hand, the imperfection of many teaching methods manifests itself in particular in the necessity of engaging in a large number of interactions between the system and its environment – i.e. many *trials* and *errors*. This can prove not only costly but also risky.

For this reason, algorithms are often trained in virtual simulators. The creators of such solutions must not only make sure to carefully recreate the given environment but also construct a precise reward system to promote various behaviours, usually by relating the state of the environment resulting from the agent's activity to mission-specific conditions. In practical terms, this can pose a considerable challenge: the system will operate on the premise of being motivated while learning, consequently any errors in the definition of the reward function will have far-reaching consequences.

A separate, more and more commonly discussed question is *who* should determine the goal function. If we can learn anything from human history, the story of king Midas and the like, fulfilling one's dreams can actually have rather tragic consequences. Therefore,

maybe it would be in people's best interest if an AI were to decide what is and what is not good for them? I.e., at the operational level, independently map out its own way towards the target? This question has been the focus of research on so-called *inverse reinforcement learning*: the purpose being to teach a system to identify goals, values and rewards applicable to a given object by observing its behaviour (cf. e.g. [@zotero-1710]). Such an approach may not only impact the ways in which AIs are designed but also redefine the very concept of intelligence. It may soon be the case that we will expect such systems to effectively solve problems posed to them as well as accurately identify our even unexpressed needs. Should this happen, *the future won't be what it used to be* (to paraphrase Paul Valery).

For now, however, let us go back to our definition of *intelligence*. We described it as the *ability to effectively act under new circumstances*. As we have now briefly discussed *effectiveness*, it is time to look into *action*.

Action entails not only the ability to make decisions but also access so-called actuators: physical devices or communication interfaces facilitating interactions with the environment. In the next section focusing on the basics of reinforcement learning, the concept of the so-called agent will be introduced with his role limited solely to control. Our definition of intelligence expands the concept of action by including the ability to affect the environment.

Nonetheless, interaction is not the only type of *action* necessary for the successful performance of a mission. It is equally, possibly even more important to be able to effectively extract data, information, and knowledge from the environment, and then use it for the improvement of one's efficiency. The ability to pass on experience and knowledge, not just from generation to generation (e.g. through the process of evolution) but also within the same age group, in a scope far broader than mere genetic changes, has long constituted another element of human 'competitive advantage' over other living creatures. This allows our societies to learn at an accelerated pace, which, combined with the capacity for memory (both short-term and historical) has also been the basis of out technological advancement. Given the above, another desirable skill of an intelligent

system would be to have the ability to gather information and knowledge and use experiences, not only originating from oneself but also from others.

The final aspect of activity pivotal to our understanding of intelligence is the *capacity for self-learning and self-improvement*. Intelligent machines should not only effectively use their externally provided 'brains' but also relatively quickly gather new experiences and be able to independently learn from them. This is the central focus of machine learning (whose key premise will be discussed below) as well as studies conducted in the areas of so-called *meta-learning* (teaching machines to recognise the optimum strategies of self-learning) and *self-supervised learning* (teaching machines to supervise their own learning processes). These are but a few fascinating areas of research that may strongly influence the future of the world we live in.

The final element of our definition refers to *new circumstances*. In this context, '*new*' should be understood as synonymous to the notions of complexity and uncertainty, whereas '*circumstances*' refer to the given context (environment, world) wherein the intelligent system operates. The phrase is used here to facilitate reference to the aforementioned classification of problems with which machines may have to contend.

As already stated, the spectrum of such problems is vast: from things well-known, observable and deterministic, to everything that is incomprehensible, non-observable and changeable. In our definition, *new circumstances* refer to the latter extreme, bearing in mind that it may be difficult to naturally distinguish between intelligence and automation stemming from the implementation of simple algorithms. An intelligent agent ought to effectively perform its tasks under circumstances that are inherently uncertain (stochastic environments where one cannot be certain about the viability of one's actions), prone to change (with the rules of the game changing dynamically), and only partially observable ('with limited visibility'). In this context, the ability to operate in *continuous* environments seems relatively trivial (after all, this is the natural human environment: the images we perceive are not composed of large 'pixels', our movement is fluid, etc.), but in practical applications this requirement (decision-making when faced with information overload) turns out to pose a very significant challenge. Our definition of *machine intelligence* is, in a way, not particularly far removed from our intuition about

human intelligence. The people we consider to be intelligent are often those capable of performing effectively when faced with completely new situations – the condition of newness is therefore crucial to the distinction between *intelligence* and *teachability*.

Many of the technologies described in the next chapter fail to meet the requirements of such a strong definition of intelligence – nonetheless, their developemnt is a necessary step if the same is ever to be achieved. An intelligent system should be able to receive information from its surroundings – it will therefore have to heavily rely on various sensor technologies. It must also know how to interpret the same: recognise objects and situations, identify patterns – which will require the use of algorithms facilitating image recognition or natural language processing. A machine will also have to develop strategies and plans, learn, memorise, and eventually also interact with its environment – not unlike us, humans. It is therefore hardly surprising that the trend of developing systems equipped with such higher abilities is referred to as *cognitive computing*.

Machine learning and programming

How then does one build an intelligent machine?

Let us imagine a manager responsible for the performance of a small business unit. How would he ensure the effective operation of his department, in particular timely performance of the designated goals?

Our hypothetical manager has a few available options. The first, somewhat crude but sometimes the only viable solution, is to do all the important work himself. This minimises the risk of failure (a highly experienced expert can often complete a given task much faster than a relative novice) but is unsustainable in a long-term perspective. The subordinates will quickly get accustomed to having his support, which will prevent them from developing their own competences. Soon, exertion will have the better of our hero and he will himself become the proverbial bottleneck. Clearly, this is not the best solution.

Another option is to issue precise commands and enforce their subsequent execution. This is already a step forward: instead of doing everything himself, the manager coordinates the work of others. The risk here? Employees will quickly get accustomed to

working only under the influence of external stimuli, they will not have an internal locus of control and the entire structure will collapse at the first sign of leadership deficiency.

A somewhat better idea is to manage responsibilities. Employees are no longer assigned with specific tasks but rather areas of responsibility. We formulate goals and leave it to the employees to make them happen.

Are there even higher conceivable levels of self-reliance? In a way, there are: one could teach employees to formulate performance strategies and learn independently - if their competences are lacking, even to formulate their own goals consistent with the organisation's vision. This way, they could set their own targets and motivate others in the process. If all goes well, they will be able to independently solve new problems, create new rules of procedure, and constantly improve their competences wherever necessary.

And how does this relate to *building intelligent machines*? Let us now imagine an engineer who aims to design an autonomous helicopter. Navigation would be a considerable challenge here: it would require adjusting the speed of the main (horizontal) rotor as well as the vertical rotor at the back. And all that not only while maintaining a certain fixed position but also while travelling in a certain direction in fast-changing (e.g. gusts of wind) conditions. So how should he approach building such a machine?

Our engineer has to chose between two options. The simplest he can think of would entail developing computer software that would instruct the machine telling it how to behave under given circumstances, naturally based on experiences of professional pilots. At the end of the day, this could be boiled down to rudimentary *what-if* rules, and in more refined environments to dedicated functions capable of approximating such algorithms. What are the risks? It could soon turn out that the helicopter finds itself in a situation not supported by the algorithm – i.e. one in which it will not receive any instructions. In real world applications this would be a sure recipe for disaster. This approach to machine building is referred to as **programming**.

So, what else can one do? Rather than program the machine to perform tasks, he could ensure that it can actually learn how to perform them. In other words, the machine would need to be provided not so much with information on how to behave under given

circumstances but rather the capacity to learn the same. This is the approach commonly referred to as **machine learning**.

To look back to our analogy of a manager responsible for a team: issuing commands can be compared to programming (the employees perform clearly defined tasks), while encouragement of independent work and self-improvement – to machine learning. An intuitive understanding of this distinction will prove very helpful when navigating the complex world of data science, machine learning and artificial intelligence.

We could also use a somewhat different example to illustrate yet another aspect of the need to *teach machines how to self-learn*. Let us imagine a machine performing simple arithmetic calculations, e.g. a common calculator. In order to build it, a mathematician had to create an algorithm: procedures to follow for three input objects (for the sake of simplicity, let us assume two digits and an operator), and one output object (a number). Each digit can take 10 different values, there are four possible operators (+, -, x, :), and 82 possible results (0 to 81). A programmer then implemented this algorithm using a particular programming language, and an engineer uploaded it to the final device which is further composed of such elements as a keypad, processor, or screen. Nonetheless, the key component remains the algorithm: procedure of processing three 'small' (in terms of the number of dimensions and possible values) objects in order to produce the desired output. It is so simple in fact that mathematicians have long been able to utilise it even without machines.

Now, let us consider the process of recognising objects in photographs – for the sake of simplicity only two: a dog and a cat. In this case, there are considerably more input 'objects' to consider: so-called pixels, i.e. the most elementary components of any digital image. Each pixel is described by three numbers (separately for the red, yellow, and blue channel) with the possible values ranging from 0 (perfect black) to 225 (perfect white). Should we assume an average resolution of 800 x 600 pixels, the image would therefore be characterised by 1,440,000 numbers, each with the value range from 0 to 255 – that is considerably more than 3 objects with a dozen or so possible values… Consequently, we end up with an algorithm that has to process over 1.4 million 'objects' to provide an output of merely two digits, each with the range of possible values from 0 to 1: the first number

to indicate the probability that the image is a cat, the other the relative probability of the image depicting a dog. As we can see, an algorithm tasked with distinguishing between a cat and a dog will have to handle drastically more input information than a simple calculator. One could immediately ask: **would a human be able to come up with a way of processing such amounts of information and distinguishing those two important animals?** Obviously, our own brain is more than capable of doing that, but what we really want to know is if a programmer could implement the skill in software and an engineer get it to work on a dedicated device. It does seem that creating an algorithm of this type would be beyond human ability – but if that is the case, why not have a machine do it? **This is a task exactly suited for machine learning**: typically, the process yields its effects in the form of 'brains', nowadays increasingly taking the form of so-called neural networks capable of not only recognising objects in images, but also understanding spoken language and interpreting human emotions. A machine begins its 'education' with a completely fresh 'brain' and then proceeds to improve it in a variety of 'ways'. The entire process is typically described as 'machine learning' and this particular form of teaching as the 'learning algorithm'. Both will be described in greater detail in the subsequent section.

Methods of machine learning

The most popular criterion used for classifying machine learning methods is their **level of autonomy**: from supervised learning, through semi-supervised, unsupervised, and reinforcement learning, all the way to self-supervised and meta-learning. Each of the above will be briefly characterised below, including their most important algorithms and common applications.

Let us start with **supervised learning**. In this case *supervision* means the presence of a **teacher** who supervises the learning process, usually by evaluating the correctness of the answers given. The basis for learning is provided by a so-called **teaching set** containing data characterising the problem as well as the correct answers. The task of the **teaching algorithm** is to guide the entire process in such a way that the structure trained (e.g. randomly initiated neural network or another set or rules, e.g. a decision tree) could quickly evolve out of the random answer stage (beginning of the learning process) and up

to a satisfactory level of accuracy (end of the learning process). This is not unlike teaching skills to humans: a beginner will have problems with e.g. recognising constellations on the night sky, but once trained, they will do it practically flawlessly.

As mentioned above, the starting point in this case is a **man-made dataset**. It usually comprises **objects**, their **properties** (explanatory variables, predictors), and **labels** (dependent variables). The *properties* correspond to questions and the *labels* to answers. For instance, if we want to teach our system to recognise dog breeds from photographs, the teaching set should contain a large number (e.g. several thousand) photographs (predictors) with descriptions of the respective breeds (labels). In the course of the teaching process, we 'show' the system one photograph at a time, asking it to identify the breed. After each answer feedback is provided to indicate how correct the given answer was, which the teaching algorithm then uses to 'streamline' the trained structure. The process is then repeated until the incidence of errors is satisfyingly low.

The key element of the process is dividing the set of the correctly described objects (in our case photographs) into those that serve as the teaching base (training set) and those that will be used to test the system (test set). There is no point in testing the system's ability to provide answers that were previously taught to it. Again, a parallel to human learning could be drawn: using problems solved during lectures as test question yields only an illusion of competence.

The most common uses of supervised learning include **discrete variable prediction** (so-called **classification**) and **continuous variable prediction** (so-called **regression**). Classification may be binary (when dealing with only two properties, e.g. SPAM or NOT-SPAM in classifying e-mail messages) or multi-label (e.g. the dog breed classification). In regression, the system is taught how to diagnose a variable whose potential values may even be infinite. Good examples here could include apartment prices (e.g. with location and floor area serving as dependent variables) or stock price forecasts.

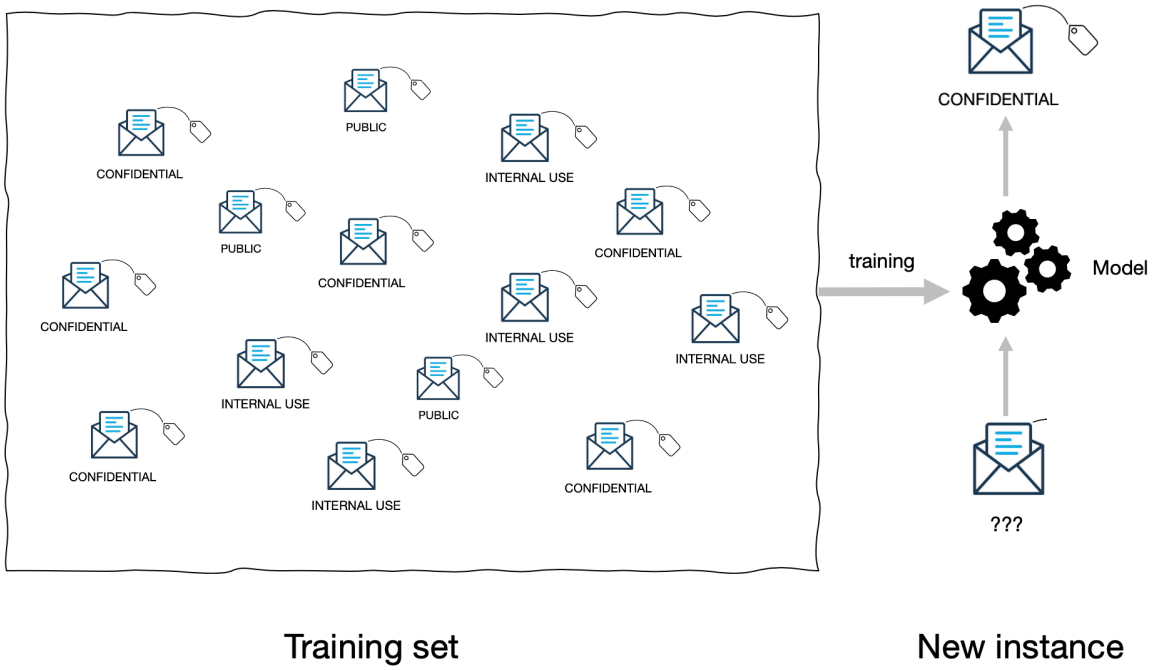

Figure. 1. Illustration of the multi-label principle. The task of the model is to predict a feature whose potential values are infinite (labels). In the presented example, the feature relates to types of email messages: they can be classified as confidential, internal or public. The model trained with the use of training data is later used to prognose the properties of other instances (in our case email messages). Source: own elaboration.

In a sense, the objective of variable learning is to achieve better **generalisation**. Generalisation is the foundation of abstract though, it allows us to simplify our surrounding reality by reducing it to a limited subset of concepts. Every time I look at my reflection in the mirror, my brain receives an entirely different image: facial expressions will change, as will angles of view, lighting, elements in the background, etc. And yet, I invariably identify this set of visual stimuli as my face. Not someone else's but mine, not a leg but a face. Hence, my brain generalises: various images are equated to a single object.

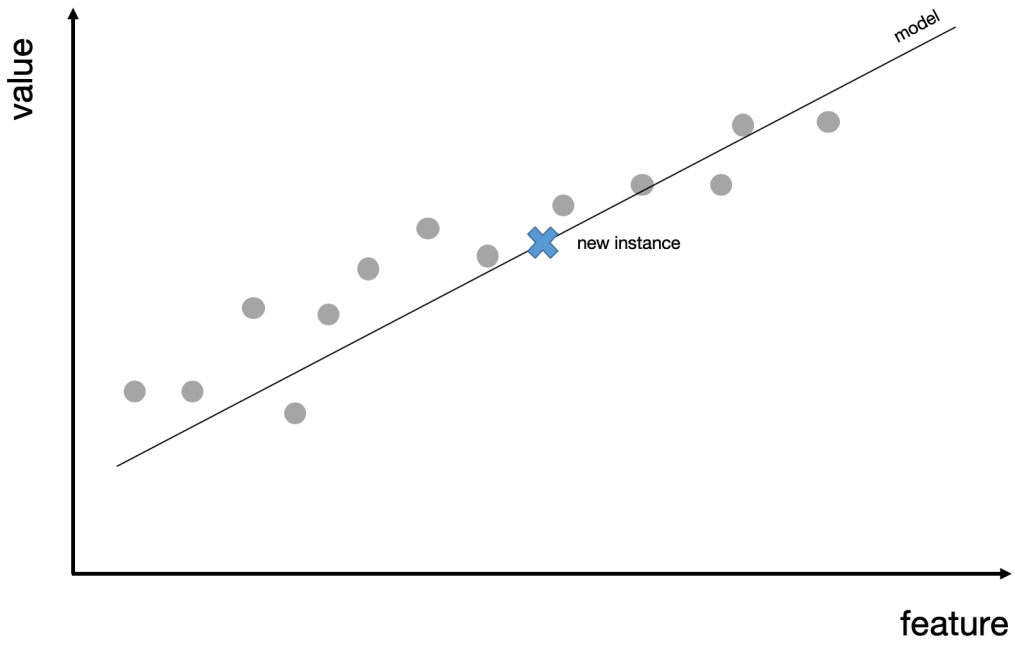

Figure 2. Illustration of a regression model. Similarly to classification, the purpose of the model is to prognose a certain feature, in this case, however, it is the explanatory variable that can have infinite values. For instance, when projecting the price of an apartment depending on its floor area. Source: own elaboration.

A similar ordinal function is served by definitions and, more generally, many linguistic parts of speech. It is a matter of certain convention that an object slightly smaller than a person, with a small horizontal area and several (but at least three) elements attached below will be called a table. There are literally millions of the concept's possible 'iterations', and yet we will still recognise each of them as a table. Such generalisations can certainly make one's life easier, so their utility is considerable. As discussed below, a similar ability should be available to intelligent systems – without it, each of such states would be treated differently, placed in a different reality. **One of the purposes of supervised learning is to develop the capacity for generalisation**.

The most important algorithms (or 'teaching methods') used in supervised learning include linear regression, logical regression, support vector machines, decision trees, random forests, and generally neural networks.

In summary: **the key element in supervised learning is the man-made dataset containing pairs of questions and answers.** It is used in the teaching process as a source

of feedback on the system's progress. The dataset is divided into training and testing subsets, the latter being used to evaluate the system's quality.

It is often the case that no set of questions and answers is readily available, or that rather than prediction, our goals are focused on e.g. pattern recognition, limiting the number of variables, or simply visualisation. In such cases, it is common to turn to the methods of **unsupervised learning, i.e. learning unlabelled data.**

One of the main roles of unsupervised leaning relates to **data compression,** a skill that is, in a way, complementary to generalisation. As we contemplate reality, our brain is always searching for certain patterns, underlying principles of the world we live in. Such patterns emerging e.g. as 'formulas' allow us to 'compress' reality. For instance, rather than build a tabular listing of the possible terminal speeds of a body falling from different heights, we use the formula for speed in uniformly accelerated motion and save ourselves the need to do literally millions of calculations (individually for each of the possible elevations) and just do one - using the discovered formula. Once identified, a pattern compresses information. And in order to find it, we can turn to the methods of unsupervised learning.

A typical application of unsupervised learning entails **clustering**. For instance, a telecom with access to information concerning details of its customers' phone calls may be interested in identifying and clustering the same. The grouping could be done based on behaviour with the members of a single group sharing a certain common feature and clearly standing out from other groups in some respect. What is important, the company does not know the character of the respective groups: the customers are not labelled, there is no training dataset to use – it is the machine learning model that is expected to identify the same.

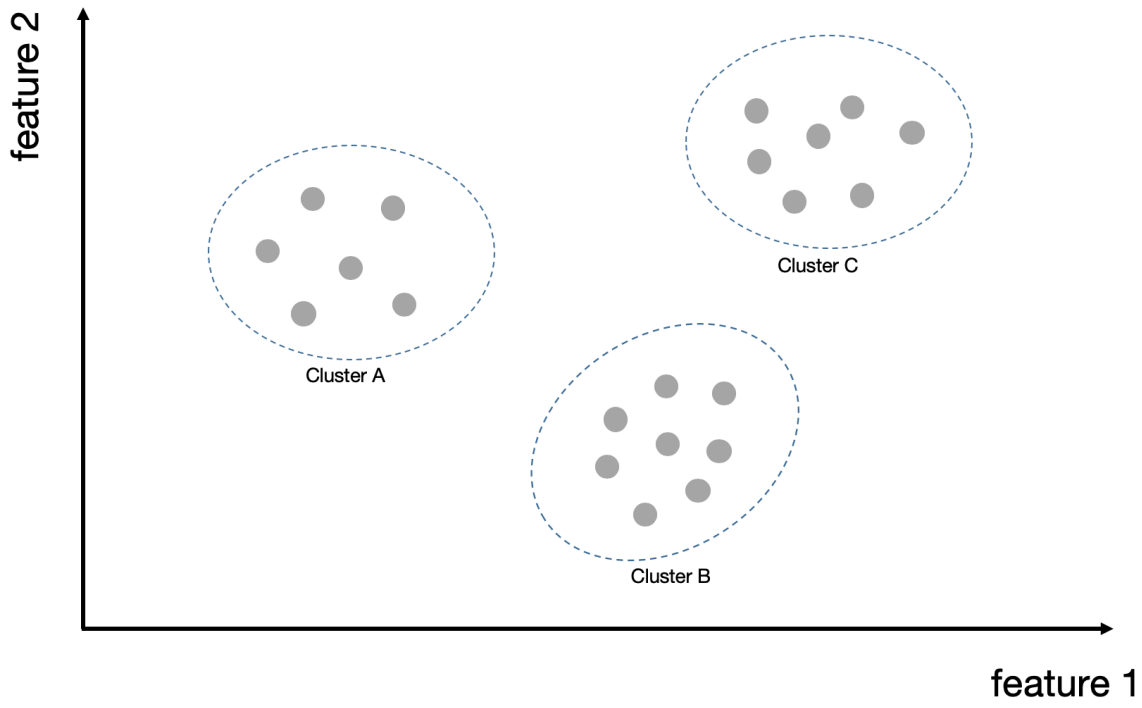

Figure 3. Illustration of a clustering model. Here, the model's task is to identify a pattern: divide objects into groups of objects that are similar to each other and different from other groups. Unlike in the case of regression, the Y-axis (vertical) does not correspond to a dependant variable but another feature of the object (naturally, there may easily be more than two features concerned). The model identifies groups (clusters)and labels them with its own, abstract designations. It is the job of the human to interpret the significance of the groups and rename them where appropriate. Source: own elaboration.

What is the point of clustering customers? The first obvious benefit stems from **data compression**: the human brain is equipped to better understand a situation with fewer qualitatively different elements (e.g. 5 to 10) rather than, say, millions of individual customers. Another benefit stems from **personalisation of offerings**: it is easier to develop several distinct packages of diverse product offers than millions of almost identical ones[2].

[2] It is noteworthy that the benefits for the customer are also considerable: a vast majority of people actually tend to opt for fewer rather than more options.

‘Compression’ can be applied not only to objects (e.g. customers) but also the number of features that describe them. Operators of popular social media sites can easily produce hundreds of different statistics about their users – the difficulty lies in extracting valuable information form such raw data. The human brain has its inherent limitations: We find it relatively easy to analyse two variables (by plotting two objects in e.g. a two-dimensional graph), three make the process somewhat more difficult (a three-dimensional graph or two-dimensional with the addition of e.g. another colour), and with every additional dimension, out capacity to understand reality suffers significantly. And what if the dimensions come in their hundreds? It turns out that supervised learning, in particular the methods of so-called dimensionality reduction, can significantly reduce the number of variables. When combined with constantly improving methods of visualisation, such solutions assist people in more efficient interpretation of facts.

The primary difference between supervised and unsupervised learning stems from the aforementioned lack of the teaching set. However, this does not mean that the algorithms have to learn without data: they are still provided but without preassigned labels (dependent variables) – it is up to the model to interpret them. The most popular algorithms used in unsupervised learning (i.e. methods of developing models) include the K-means method, Hierarchical Clustering, DBSCAN, and Principal Component Analysis (PCA).

An ‘in-between’ method combining elements of both of the above is called **semi-supervised learning**. The method is useful in cases where a teaching dataset (i.e. question and answer pairs) is available but relatively limited – and our aim is to generalise that knowledge to many more cases. For instance, social media websites may be interested in implementing a system capable of identifying people in photographs uploaded to the system. How could this be achieved? In the first stage, we use supervised learning methods to group people in millions of untagged photographs. The system learns to identify a given person in many photographs (in clustering, a person is the equivalent of a *group*), still without knowing their name (the group code is always abstract, generated by the model and otherwise meaningless). Once this stage is complete, the system can conclude that e.g. person A115 appears in photographs no. 10, 14, and 167. In

the second stage, the method of supervised learning is employed: based on the existing training set (in this case photographs of people associated with specific personal information), the model obtained in the previous step is taught how to associate abstract codes with specific people. Consequently, the combination of the two methods (supervised and unsupervised learning) yields a recognition model trained with a relatively small training set.

Having discussed the methods of supervised, semi-supervised, and unsupervised learning, it is now time to move on to somewhat more 'autonomous' methods. But what exactly does autonomy entail here? Was the system not required to independently discover patterns and rules already in the unsupervised approach?

In machine learning, intelligent systems learn from data. In the three cases described above, the data had to be prepared and provided by humans. Even though in unsupervised learning only the features of objects were provided (no labels i.e. no explanatory variables), but they were nonetheless fed to the system by a human operator – like food to a small child. Meanwhile, autonomous systems are expected to act, in particular learn, without external human support. The question is how to design a 'didactical process' where the system is capable of independently providing feedback.

The first such method is referred to as **self-supervised learning** and is an interesting hybrid of supervised and unsupervised approaches. Similarly to supervised learning, the system learns to recognise objects based on labelled datasets (i.e. *question* <> *answer* pairs). The key difference is that in this case, the labels are assigned by the system itself, without outside support. The system supervises itself. There are many interesting applications for this particular approach. Image recognition systems learn to independently classify objects based on contextual and spatial data (cf. e.g. [@doersch2016]), and robots train their visual systems to recognise objects based on data from other perception systems [@nava2019]. The method also allows intelligent systems to explore and learn about the world on the strength of actual curiosity rather than enhancement stimuli (punishment and reward] [@pathak2017].

Another interesting approach, referred to as **reinforcement learning**, was inspired by advances made in psychology. As humans, we are taught by parents and teachers from an

early age, but also explore our surroundings by ourselves and learn by trial and error. In order to achieve a certain goal, we perform various possible actions and observe their results. In time, we are either able to develop an efficient sequence of steps or, which ultimately may prove more effective, learn about the operating principles of a given device or rules applicable to a given environment. Autonomous systems using the methods or reinforcement learning follow similar learning patterns. They will be described in more detail in a further section, but for now it is worth pointing out that the trained system does not have access to a man-made dataset. Instead, it independently extracts the necessary information from its environment, usually in the course of the following cycle: *observation of the sate of the environment > action > new information (new sate of the environment and punishment/reward*). The method is used in training and improving a large percentage of the solutions discussed in the subsequent chapters of this book, with the use of dozens (if not hundreds) of different algorithms.

Last but not least, **meta-learning** is a very interesting class of machine learning solutions. In this approach, the intelligent systems learns to find the best learning strategies, which could be described as *learning to learn*. The idea is to accelerate learning processes, limit the amount of crucial data and number of necessary experiments (*trial and error*). It has been observed that the methods described above tend to require large sets of good quality data, which can be both effort and cost intensive. Hence, there is a demand for more effective learning methods – in the case of meta-learning, this is achieved by *learning to learn*. In particular, meta-learning can be used in conjunction with reinforcement learning (so-called Unsupervised Meta-Learning for RL, see e.g. [@gupta2019]) which basically entails teaching the agent how to apply algorithms developed in the performance of previous tasks to solving new problems. In other words, the system is taught how to identify and use previously learnt skills in the performance of new tasks, which is not unlike *transfer learning* in the context of psychology and traditional education.

As follows from the above, machines can be trained in a variety of ways. One of the ways in which such methods can be classified relates to their level of autonomy understood as the extent of human supervision and support. The *supervised, semi-supervised and unsupervised learning* methods require data to be provided by humans. In turn, *self-*

supervised, reinforcement, and meta-learning methods entail autonomous extraction of data from the environment. In each case, the result of the 'didactic process' is an imparted model capable of effectively solving new problems. But this begs the question: how does one go about improving it further?

Much like in the case of machine learning, several possible strategies could be adopted here. Presented below are two of the most popular ones: *batch-learning* and *on-line learning* (cf. e.g. [@geron2019]).

In the case of batch learning, the system is taught based on all currently available data. Once the data is collected and processed, the algorithm is trained, tested, and implemented in production upon approval. If new data is acquired (externally or on the basis of prior experience, the cycle needs to be repeated: training a new model and replacing the one currently in operation. Of course, the process can be automatized (and repeated e.g. on a daily basis) but such an approach would not be practical in the conditions of continuous inflow of new data significant to the quality of the model, e.g. in systems forecasting stock exchange price levels. The primary drawbacks of batch-learning include the need for constant supplementary learning based on large sets of gathered data, high computing power requirement (the model is trained from the ground up every time), and lengthy learning process. For those reasons, such processes are usually performed 'in the background' – on external servers – i.e. **off-line** from the perspective of the business process.

The other option entails training with the use of **on-line** systems. In this approach, the model is improved incrementally, based on new incoming data. This can be done continuously (data stream) or in short intervals (data divided into mini-batches). The strategy tends to be employed in situations where data significant to the model's quality are received in a continuous manner (e.g. a production line or stock quotations), or if hardware limitations are a factor: not enough computing power to retrain the entire model time after time, or not enough memory (operating or storage) to effectively manage large datasets.

The efficiency of on-line learning is affected by two key factors. Firstly, a decision has to be made as to the extent to which model updates will rely on new experiences as opposed

to previously gathered data. This factor is determined by the so-called *learning-rate*. If the value setting is high, new data will have a considerable impact on the model, effectively forcing it to ignore historical experience. If the setting is low, the *learning rate* will minimise the impact of new data on the current mode of operation. This can be rather naturally referred to the way in which we, humans update our knowledge: some people are quick to forget about the past and able to flexibly adapt to changes in their environments. Others are more conservative: they survey the world, take new experiences and phenomena into account, but take considerably more time to alter their behaviour. Each of the options has advantages and disadvantages. As nearly with everything else, the trick is to strike a reasonable balance, which naturally strongly depends on the current situation and environment.

The other factor very important in the context of on-line learning (although also significant in the off-line approach) related to **data quality**. As data influence the model, they also impact the decisions and behaviours of the intelligent system. Bad quality data (incorrect, incomplete, etc.) will significantly hinder system performance – not unlike bad quality food weakening the organism. For this reason, machine learning has to ensure not only high efficiency of the learning algorithms but also the quality and proper preparation of input data.

To recapitulate the key points presented above, let us once again refer to the metaphor of human learning.

The **process of machine learning** usually includes the stages of gathering and preparing data (or developing mechanisms of acquiring the same), selecting the learning algorithm, training the model, and testing its performance. The same is mostly parallel to the classical **didactic process** of e.g. school teaching.

The learning subject, or student, is the **model**. The **teacher** is usually a human although, as mentioned above, intelligent machines should also have the capacity to teach themselves. The model (i.e. our 'student') learns based on **data**. The data may be provided externally (e.g. by the teacher) or obtained independently: through active interactions with the environment or passive exploration.

Just like in a real classroom, *the primary goal of the educator should be to make the student independent of the teacher.* Similarly, the **teaching goal** in our context is to 'train' the model to be able to effectively solve new problems – ones never encountered before (e.g. during the training process). Moreover, a good teacher will also instil the capacity for independent self-improvement (see off-line and on-line learning mechanisms). Similarly to human education, machine learning takes advantage of a varied range of teaching (model training) methods. **In our context, machine learning algorithms** (e.g. k-means or Hierarchical Clustering) **are a direct counterpart of teaching methodologies.**

At the beginning of the process, the model ('student') usually has no competences to speak of (with 'supplementary learning' being a notable exception). '**The didactic goal**' is to develop a very specific set of competences. '**The educational results**' can be precisely measured at practically every step of the way by evaluating e.g. prediction errors, classification quality, or levels of punishment/reward received. The exact measures depend on the type of skill one wishes to train. For instance, in the case of classification, the same will be the surface area under the receiver operating curve (ROC) (so-called AUC ROC), while for regression, it will be the root mean square error (RMSE).

The purpose of evaluating learning progress is to determine the model's ability to act under new circumstances. Hence, in supervised learning models, evaluation should be based on data not previously used in the learning process – similarly to a school where examined students should not be asked to solve questions previously done in class.

As follows from the above, one could draw many, more or less justified, analogies between the methods of human and machine learning. The reason they were included in these deliberations was to help readers less interested in exploring the mathematical and technological particulars of the problem to nonetheless develop an intuitive grasp of how intelligent systems develop and operate. They also provide a good starting point for the subsequent section which will discuss in greater detail various concepts applicable to reinforcement learning – a crucial contributing factor in the development of autonomous solutions.

Basic principles of reinforcement learning as a foundation for autonomous, self-learning systems

In the preceding deliberations related to machine learning, the 'machine' (hereinafter the agent) was trained with the use of previously prepared data. In supervised learning, such data were labelled and in unsupervised learning they were not. The agent learnt to be able to cope with a new situation and correctly identify the 'features' of new (previously unencountered) data (in supervised learning) or identify certain patterns (in unsupervised learning). At the end of the day, the problems solved by the methods described above could be reduced to three primary categories: classification (prognosis of a discrete variable), regression (recognition of a continuous variable), and segmentation. Whatever the case, however, the same always boiled down to 'recognition', i.e. interpretation of externally provided data: the agent took no action, did not interact with its environment, and had no specific 'mission' to perform.

But intelligent structures are expected to do more than simply recognise and interpret their surroundings. An autonomous robot performing a complex mission has to function within a specific environment, integrate with its surroundings, receive feedback on the results of its actions, and use all of that to achieve a certain ultimate goal. Such a process is considerably more complex and involves qualitatively new elements such as 'actions', 'response from the environment', or 'goal'. Presented below is the concept of so-called reinforcement learning – a method that underlies a vast majority of contemporary autonomous systems, be it robots, drones, or conversation bots. The following paragraphs will discuss the primary concepts involved, interactive learning methods, and key algorithms, while the most important challenges faced by such systems will be mentioned in the summary. An intuitive grasp of these methods will facilitate a better understanding of solutions discussed in the subsequent chapters as well as the rather fascinating relationships between mechanisms constituting the basis of decision-making as observed in humans, organisations, and machines.

Agent, environment, actions, observations, and rewards

Our analysis should start with the explanation of some key concepts.

The computer program that we wish to teach how to effectively act under new circumstances is referred to as the **agent**. In this context, *effective performance* is understood as the ability to solve sequential decision-making problems under conditions of uncertainty in order to maximise a pre-defined goal function.

It is noteworthy at this point that the agent is only the decision centre, a piece of software: its sole task is to **make decisions, not implement them**. Consequently, a robot working autonomically on a production line is not an agent; the agent is the software controlling its operation (which, notably, may just as easily be running remotely, e.g. in the cloud). Arms, motors and other actuators are considered **elements of the environment** as far as the decision centre is concerned. This distinction may become more compelling once we consider that robots often operate within stochastic environments, where a decision to move e.g. right does not guarantee (it may only provide a certain level of probability) that a move to the right will indeed be executed: in practical applications it may indeed happen that the resulting motion will be to the left. We have all experienced a similar situation at one time or another, e.g. when trying to walk on ice: the decision is one thing, but actual execution is a different story . . .

The Agent has a certain inherent structure: interfaces that allow it to communicate with its environment (receive information about the state of the environment and transmit decisions concerning respective tasks) as well as the decision centre itself (the 'brain') which makes decisions with the aim of maximising the so-called goal function, evaluates their effects, and learns so that each subsequent iteration of the action yields even better results and contributes to mission execution. Therefore, the key components of the agent include **interfaces** and the '**brain**', while the primary processes involved are **interaction, evaluation**, and **improvement**.

The Agent operates within a certain **environment**. It encompasses everything external to the agent, of which it has no complete control. In particular, the arm of a robot is an

element of its environment: as already mentioned, the intention of moving it to the right cannot guarantee such motion (e.g. due to failure or an obstacle).

From the perspective of the agent, **the environment is represented by a set of variables related to the problem.** For a robot, these could include e.g. position of the object that is to be moved, position and speed of the arm it uses. For a drone, it could be its speed (in three dimensions), position and rotational speed of the respective rotors. **The state of the environment** is defined by a set of values assigned to said variables at a given time. It is a subset of the so-called **state space**: the set of all variables describing the environment and all of their possible values. In turn, **observation** performed by the agent at any given moment is a part of the state it perceives at the time. The distinction between the state of the environment and observation is very important: it is often the case that the agent has access only to a small subset of information regarding everything that is happening to it and around it. A good parallel would be the spectrum of electromagnetic waves we are able to perceive: only a fraction is visible but that does not mean we are not surrounded by e.g. radio waves.

The agent usually has a specific **task** to perform within the given environment. The task may have a definable conclusion (e.g. in the game of chess) or not (e.g. navigating a given space). The former are referred to as **episodic** and the latter as **continuous tasks**. In the case of episodic tasks, the sequence of steps from the beginning to the end of the task performance is called an **episode**, while the total of rewards collected in the course of a single episode (e.g. gold picked up while navigating a labyrinth) is referred to as **return**. The agent is usually programmed to make decisions aimed at maximising the return, it may, however, repeat the episode several times to learn the optimum pattern. In the process of learning, continuous tasks are often limited by imposing certain finite stages (e.g. number of steps), which brings them somewhat closer to their episodic counterparts.

The **goal** of the task is indicated by a special signal transmitted to the agent by the environment and described as the **reward**. Its value may be positive, negative, or zero – whatever the case, the reinforcement learning terminology will refer to it as a reward. The manner in which the environment rewards the agent is critical to the learning progress – not unlike in human learning in fact. The issue will be discussed in greater detail in the

subsequent sections (cf. also [@morales]) but it should be noted at this point that rewards may be delayed in time to a varying degree, awarded with varying frequency, or have a varying nature (evaluative or instructive).

In some cases, this delay can lead to a situation where the actual consequences of the action taken will only become apparent after some time: immediately after eating a cookie, we feel great (high reward), but after a month of having 10 cookies for supper every day, we will start receiving troubling signals. The rewards evaluating our actions may be given very frequently (e.g. after each activity) or more rarely (e.g. once in every ten actions). The former case could be compared to a mother praising or criticising her child every other minute, this can accelerate learning but puts the agent at risk of copying the behavioural patterns of the teacher (mother) therefore hindering its future self-reliance – in the context of machine learning such a situation is described as *overtraining*. Less frequent feedback will have the opposite effect: slower learning but more space for experimentation, which helps to develop independence and discover new, more effective methods. Finally, **evaluative** feedback refers to the values of states achieved after the performance of actions, while **instructive** feedback reflects the values of the actions performed. The function determining the reward that the agent is to receive in the given state of the environment is called the **reward function**.

The goal of the agent is to perform a task posed to it by the environment. In order to do so, it must have the capacity to make decisions maximising the **total** reward (in the case of episodic tasks: the sum total of the returns accumulated during the task), which in turn requires **long-term planning**. The agent should also be able to **gather information** and, due to the fact that it cannot explore all the states of the environment, **generalise** its experiences to account for other, similar states. At the end of the day, it should have the ability to create the optimum **policy: method of choosing actions in response to the given situation with the view to perform the task.** From the mathematical point of view, it means the ability to learn the function mapping observation into action.

Tasks are performed through the agent's **interactions** with the environment. This usually takes place in cycles comprising **action, feedback, reflection, and update**. In the first stage, the initial state of the environment is surveyed. Next, based on the available

knowledge, a decision is made to choose the optimum action. The action is performed, and feedback is received to determine the state of the environment resulting from the action (i.e. the influence exerted) and the reward received. Next, the effects are evaluated, policy is updated, and the cycle begins anew.

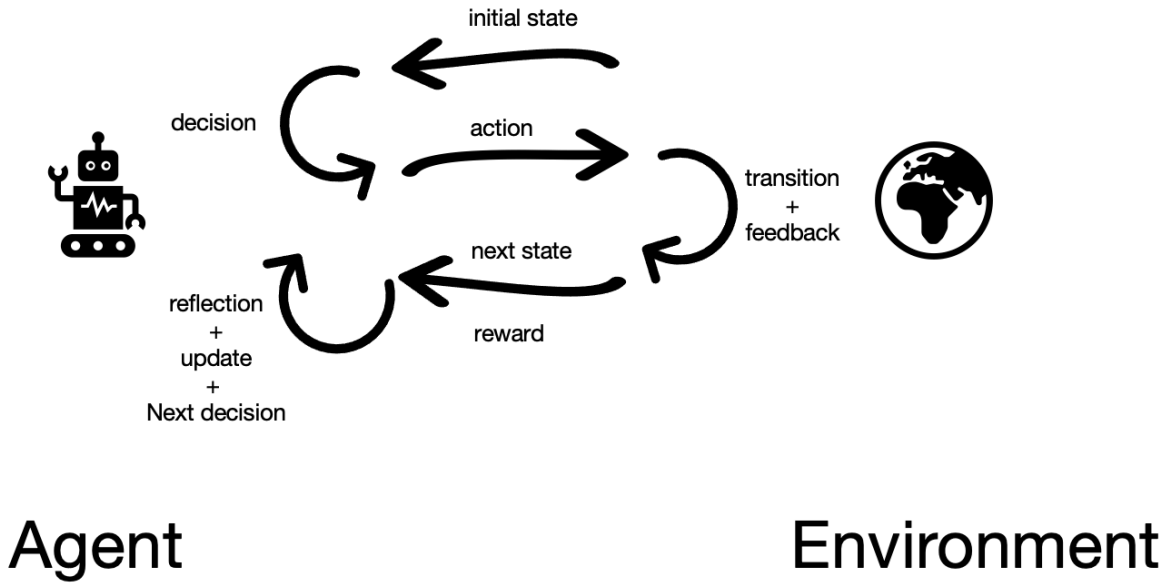

Figure 4. Simplified model of interaction between an agent trained through reinforcement learning and its environment. Source: own elaboration.

The cycles can be scaled to very diverse time intervals: from milliseconds, through hours, to days – depending on the given problem and environment. After each cycle, the agent gains a certain **experience** represented by a specific dataset (often referred to as *tuple* in IT jargon) pertaining to the initial state, undertaken action, resulting state, and reward. The experience is usually gathered as a separate dataset used in the process of self-improvement, both by individual agents and entire teams.

In each state, the environment **allows** the agent to undertake certain actions – the scope of which can vary in time. As already mentioned, the action may (but does not have to) influence the environment. The function describing the way in which a given action affects a given state (i.e. mapping action > state) is called the **transition function**.

Now that we have discussed the basic terms and pattern of agent behaviour, we can turn our attention to two other, very important concepts: **Markov Decision Process** and **world model**.

When updating its performance, the agent has to make certain assumptions. One of the most important of the same is the so-called Markov property which can be explained in a single sentence: "*The future is independent of the past given the present".* In other words, the future of a system depends solely on the decisions made in the present – the past is no longer a factor. This rather extreme premise proves to be very effective in practice, it greatly simplifies calculations and facilitates solving complex problems. Decision processes consistent with the Markov property are described as **Markov Decision Processes** (MDP).

A **world model** can be compared to a map of a new territory: having the same will greatly increase the chances of finding the optimum solution. With a map of a city, one can easily find the shortest route from point A to point B without having to explore, which reduces the scope of the problem to simple planning. If one has no map, one is forced to try different variants and gradually create one, which takes time and, if obstacles emerge, can also prove risky.

In summary: actions performed by an agent alter the state of the environment (which is described by the **transition function**) and in the given state, reward is provided as dictated by the **reward function**. Within the framework of reinforcement learning, knowledge of the world model comes down to knowing those two functions: transition and reward. In other words, we could say that we know the world model if 1. We know what state s' will result from our action a performed in the state s, and 2. What reward r will be received in the state s'.

Learning through interaction

With the basic nomenclature explained, it is time to take a closer look at some key concepts related to the methods of teaching agents how to efficiently perform tasks in an environment.

As already discussed, interactions between the agent an its environment take place in the cycles of observing the state of the environment, performing an action, and gathering feedback (new state and reward). Each cycle is described as a **time-step**, while the compete set of information gathered during each time-step (initial state, action, final state, reward) is called **experience**. It was also mentioned that the gathered experience is evaluated by the agent (**reflection**) in order to **update** the decision-making policy. **In other words, the agent improves its performance by learning through trial and error based on the experience gathered, which constitutes the essence of** *reinforcement learning*.

Feedback is crucial to the learning process. The forms and types of such information depend strongly on the given task and environment, which poses considerable challenges and greatly affects the learning strategies. The issue will be discussed in detail below – gaining an intuitive grasp thereof is not only key to understanding the problems of RL but may also prove useful in personal life.

Firstly, feedback from the environment can be **staggered**. The consequences of choosing a given action in the present state may only become apparent in a more distant future (not necessarily in the next time-step). The goal of the agent is to undertake *here and now* actions aimed at maximising the total of rewards gained during the episode, and not at maximising the rewards applicable to subsequent states. The agent must therefore be able to learn from such staggered feedback as well.

Strategies can vary and are highly dependent on the problem. For instance, in the case of episodic tasks with the possibility of performing multiple episodes, the agent may learn the world model on an episode-by-episode basis (i.e. discover the transition functions and rewards), and based on the continuously perfected model, make decisions with the aim of maximising all the rewards (from all the time-steps). This becomes somewhat problematic if one has only a single 'life' at one's disposal – this calls for a different strategy, e.g. relying on the experiences and value systems of other agents (in our case: humans), which as we know, is hardly easy and in no way guarantees success.

Another factor diversifying information received from the environment is the **character of feedback**. In their acclaimed textbook, Sutton and Barto [@sutton] compared

reinforcement learning to other machine learning methodologies and concluded that a reward constitutes a signal evaluating the quality of action (so-called evaluative feedback) rather than an instruction indicating the correct action (so-called instructive feedback). This interesting classification of machine learning methods based on distinguishing between two types of feedback received in the learning process: *evaluative* (assessment of the consequences of action) and *instructive* (recommendation of the best action in the given state regardless of the actual action of the agent) may be easily applied also to the methods of human education.

Instructive learning basically boils down to imparting a certain set of *what-if* procedures: *in this state you should do this*. The role of the teacher is to determine whether the action was consistent with the procedure and evaluate the discrepancy (i.e. error). The main limiting factor in this context is the experience of the teacher (or the available training dataset): future problems may emerge in situations that the teacher was not aware of (or simply did not think to account for in the process of generalising his knowledge). In the case of **evaluative learning**, the role of the teacher is assumed by the environment itself (i.e. the situation in which the agent finds itself). It informs the agent of the consequences of a given action (good, bad, neutral) without suggesting the optimum course of action. It is the learner that has to evaluate the action taken in the given state (situation), which is not easy as it requires disciplined experimentation and honest self-assessment. At the same time, however, it creates the potential for exploration and finding new, more effective policies. In short: one can either be taught through instruction, by pointing out the adequate behaviour in a given situation, or through encouragement to undertake varied action, with the only support coming in the form of feedback on the consequences thereof. The former approach is characteristic of supervised learning, the latter of reinforcement learning. Naturally, in practical applications there are various methods seeking to find a middle ground between these two extremes, nonetheless, the above classification will be very useful to us.

To recap, in the case of reinforcement learning, the agent must be able to learn from evaluative (rather than instructive) feedback. The knowledge as to which action is optimum in the given state should be generated by the agent itself, and then applied to

making effective decisions. Interestingly and quite paradoxically, knowing the values of respective actions is, in itself, insufficient: were we to always make decisions that our experience indicates as best in the given situation, we would never have a chance of verifying the possible consequences of suboptimum actions.

In reinforcement learning this phenomenon is referred to as the **dilemma of exploitation and exploration**. *Exploitation* is the policy of always choosing the action that our experience tells us will yield the highest reward possible. Unfortunately, this approach has two inherent risks. The first is the already mentioned inability to test the effects of actions other than the alleged 'optimum': the value of an action can be determined through experimentation, and exploitation prevents it by its very definition. The second risk stems from the possible changes to the environment: a decision that was optimum 10 years ago may be less than so today. Similarly: a given incentive system may be optimum in one group of employees but may fail completely and even disincentivise others. As evident from the above, striking a good balance between *exploration* and *exploitation* is quite the challenge, not only in the context of machine learning.

Another key competence of a digital agent is the ability to **generalise**. To reiterate the relevant point, the agent learns by interacting with its environment, which generates feedback. It surveys the state of the environment, takes action, checks the state resulting from the action and evaluates the potential reward. Let us now consider the same process from the perspective of action and state space.

Again: the state space is the set of all possible states which the environment can have, and the action space is the set of all possible actions available to the agent at a given time (or more precisely: that the environment allows in its given state). Spaces can be both discrete and continuous, and this characteristics applies to all possible values assumed by variables describing the given environment or action. This sounds somewhat abstract, so it would be prudent to illustrate the point with a few examples.

Firstly, let us consider the so-called grid-world. Imagine an agent travelling across a 5x5 square board, starting its journey from the top left square. The goal is to reach the bottom right corner while maximising the reward collected on the way: there are various 'positive' (let's say gold) or 'negative' (e.g. traps) rewards waiting on the respective

squares of the bard. In this case, both the state space and action space are discrete. Each state is described by two variables (x and y) with a finite range (1 to 5). The available actions are also discrete: up, down, left, right (in some extreme positions only e.g. right or down).

Another 'classic' example is the so-called **inverted pendulum** problem. Imagine an agent whose task is to control a horizontal platform on which a vertical stick is balanced. The idea is to prevent the stick from tipping over for as long as possible (the scenario is performed with software simulating the laws of physics).

In this case the state space is continuous: even though the state of the stick can be described with only four parameters (horizontal position, horizontal speed, angle of inclination, and angular velocity), each can assume an infinite number of values. Meanwhile, depending on the environment, the space can be discrete (e.g. move left or move right) or continuous (e.g. move left with speed v_l). The question whether the state space is discrete or continuous is immensely important when choosing the RL method of training the agent. In the case of discrete spaces with a limited number of dimensions, one could easily imagine a situation where the agent, given sufficiently many episodes (e.g. when training on a simulator) will ultimately 'visit' all the states and perform all the actions possible therein. This may take some time but (assuming that the environment is deterministic) the agent will be able to develop a complete world model based on experience, which in turn will practically reduce the problem of exploration to simple planning. After the subsequent update, the policy will be reducible to a function which, given sufficient input data describing the given state, will return parameters describing action. Notably, due to the limited number of possible values attributable to the parameters of states and actions alike, that function will be multivariate but with a finite number of dimensions, matrices. In other words: the policy will be reduced to a finite number of *what-if* rules, where both *what* (action) and *if* (state) will have a finite number of possible values.

However, things tend to get significantly more complicated if one of the spaces, be it states or actions, becomes continuous. In this case, the variables describing e.g. the state may take an infinite number of possible values – which means that compiling a complete world

model (i.e. two functions: (*state; action*) > *new state* and *state > reward*) would require an infinite number of experiments (due to the need to 'visit' all the possible states and try all the possible actions). And clearly, that is an impossibility.

This is exactly why we expect our agent to have the capacity for **generalisation**. It allows the agent to use its limited experience to formulate policies capable of recommending optimum action in a given state[3] even if it is potentially infinite. Generalisation most commonly entails attributing a recommended action to a certain value range of variables describing a given state. For instance, in the case of the *inverted pendulum*, the policy could include rules such as: *if the inclination is in the range of 85-90 degrees left and the speed does not exceed 10 cm/s, move left.* Notably, the agent should be able to generalise not just the policy (function of *action > state*) but also the transition function (map of *(state, action) > next state*)) as well as the so-called value function (map of *(state, action) > value*).

In practical implementations, strong, non-linear approximate functions are used for this purpose: multiple mathematical transformations of input variables (cf. e.g. [@nielsen2015a]). The methods and techniques of machine learning, in particular neural networks, prove rather useful in this context. Reinforcement learning models utilising deep-learning neural networks go by the name of deep reinforcement learning (DeepRL).

To recapitulate, in order to effectively operate in environments where continuous spaces of states and/or actions are present (i.e. nearly all practical applications), the agent should have the capacity to learn from fragmentary feedback: i.e. information reflecting only a small subset of all possible state <> action configurations. This can be achieved if the agent is able to generalise, which in turn is facilitated by approximators such as e.g. artificial neural networks.

Before we proceed to present the strategy of developing the abovementioned skills in RL agents, let us reiterate (after [@russell2010a] and [@morales]) the classification of

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[3] I.e. a function of state > action that ensures the maximum possible reward yield for the episode.

environment features, levels of agent knowledge, and types of feedback, which will significantly aid us in navigating the dozens (if not hundreds) different possible problems and RL algorithms.

Let us first have a look at **environment features**. As already mentioned, **the spaces of states** and **actions** can be either **discrete** (finite) or **continuous** (infinite). In turn, the functions of **transition** (i.e. map of *(state, action) > new action*) and **reward** (map of *state > reward*) may be either **deterministic** (when it is certain that e.g. action *a* in state *s* will produce state *s'*) or **stochastic** (when there is no such certainty and one can only estimate, based on previous experience, the probability *P* of action *a* performed in state *s* producing state *s'*).

Another element facilitating the classification of RL problems is the **level of the agent's knowledge about the environment**. As far as the **space of available action** is concerned, in practice it is always assumed that the agent had complete knowledge of the same. It is another matter entirely, however, when it comes to knowledge of the **state of the environment** – here, there are three possible scenarios: **the agent has exhaustive knowledge about the sate of the environment** (i.e. *observation = state of the environment*), **the agent has only partial knowledge** (*observation* is a subset of the *state*: the agent knows something but does not have full insight), or **the agent has no knowledge of the state** (it is 'blind'). When it comes to knowledge of the **world model** (i.e. of the transition and reward functions), the agent can either know them (each taken individually) or not (in which case it will be forced to learn). Agents can also have varying access to information about the **reward**: again, this knowledge can either be exhaustive or sampled.

Finally, the determination of **reward** characteristics is also helpful in classifying RL problems. Firstly, the **moment** of receiving it is a factor: the agent may be rewarded immediately after completing an action (so-called one-shot reward) or over a certain period of time (reward is delayed) – in which case it is said that the reward is sequential. Another factor is the actual character of the reward: as already mentioned, it can reflect the consequences of actions (so-called **evaluative** feedback) or simply constitute an **instruction** (supervised feedback).

The above classification can significantly facilitate identification of optimum strategies and algorithms applicable to the agent's training. For instance, the agent can operate in an **environment** characterized by a discrete state space, continuous action space, stochastic transition function and deterministic reward function, with only sampled **knowledge** of the current state, no knowledge of the transition function, but exhaustive knowledge of the reward function, with an **evaluative** reward being provided **immediately after** the action.

	feature of the environment	**agent's knowledge**
states	discrete / continuous space	exhaustive / limited / none
actions	discrete / continuous space	known
feedback	immediate / sequential	exhaustive / sampled
feedback	evaluative / instructive	exhaustive / sampled
transition function	deterministic / stochastic	exhaustive / none
reward function	deterministic / stochastic	exhaustive / none

Table 1. Classification of reinforcement learning problems.

Before we proceed to present the main RL algorithms, let us first summarise our deliberations up to this point by indicating the most interesting challenges faced by an agent aiming to learn how to effectively perform in a given environment. Above all, the agent **must resist the temptation to immediately aim for the maximum reward**. It often needs to assume that although the choice of a given action promises to yield a high reward in the short-term, all hell might break loose later on . . . If the reward is to be maximised in the perspective of the entire episode ('life'), adequate balance must be struck between fast but minor immediate gratification and delayed but overall much larger reward in the future. Naturally the key word in this sentence is *balance*: excessive ascesis is also not without many inherent dangers – indeed, this is the very source of the challenge.

Another problem relates to the **infinite multitude of states and choices**. Too much choice is not a good thing: a fact that any sales person or aspiring violinist will readily

corroborate. Hence, due to often limited possibility of experimentation, the agent must learn the art of wise **generalisation**: to classify states and actions so that ultimately, a finite, manageable number of choices is returned. In this case *wisely* is the key word: overly simplified classification, e.g. binary - reduced to dichotomies such as *good-bad* or *red-blue*, tends to be a dead end (leading e.g. to overly stereotypical thinking). On the other hand, an excessive number of categories yielding thousands of potential options will prolong and complicate decision-making processes, thus negatively affecting performance. In short, the agent should be able to generalise, but wisely.

The last important dilemma derives from the type of feedback and comes down to the choice between **exploitation** and **exploration**. *Should I always act based on knowledge, backed with experience of what is the best for me? In other words, should I always choose actions which, based on my knowledge, are optimum? Or should I go wild from time to time: take a risk in hope of unexpectedly arriving at new, even more attractive states? And If I were to combine the two, how do I decide on the ratio?* Clearly, this is yet another case where the agent should strive for a certain optimum balance – between following the safe route and taking risks in search of better ones.

Such inherently human dilemmas are approached, in a variety of ways, at the level of algorithms which, when implemented in software and end devices (e.g. a robot controller), allow the machine to become increasingly autonomous and, in a sense, more rational. The specific business applications thereof will be discussed in the subsequent chapter, for now, let us focus on the most important algorithms used in reinforcement learning.

Key RL methods and algorithms

As already mentioned above, reinforcement learning algorithms come in a wide variety of shapes and forms, while the dynamic growth of this field of study continuously stimulates the development of new ones. To gain some measure of understanding of their possible applications, strengths and weaknesses, one needs to first adopt a simple, manageable taxonomy thereof.

One of the most popular classifications of RL algorithms (cf. e.g. [@zotero-1702]) distinguishes them first in terms of the level of knowledge about the environment, and then based on the object of learning as such. Knowledge about the world can be as based on a certain world model (*model-based RL*) or not (*model-free RL*). Furthermore, *model-based* algorithms can be divided into those where the model is externally provided (*given the model*) and those that require it to be leant (*learn the model*). In turn, *model-free* algorithms come in *value learning* and *policy optimisation* variants. Each of the mentioned categories will be briefly characterised below.

Let us first consider algorithms teaching the agent how to evaluate its decisions in a situation where it has **exhaustive knowledge on the state of the environment, the reward is delayed in time**, and the **agent uses a world model** when making decisions. It is a rather rare and untypically convenient situation for the agent, nonetheless, it will provide a good starting point in the introduction of the ideas underlying RL algorithms. A classic example of such a problem is the already discussed task of navigating a labyrinth: travelling from point A to point B in such a way, so as to maximise the reward in a situation where the rewards waiting on respective squares can be either 'positive' (e.g. gold) or 'negative (traps) [4]. At any given moment, the agent has exhaustive knowledge about the state of the environment (agent's position), its task is to maximise the reward sum total, and it can make an unlimited number of attempts in the learning process (*it has endless lives*).

The goal of the agent (i.e. also the 'didactic goal' of the reinforcement learning process) is to identify the sequence of events that yields the maximum cumulative reward in the given episode (return). The goal can be reformulated as finding the optimum policy, where *optimality* refers to maximum return in the course of the episode, and *policy* means the manner of choosing the action to be performed in each of the possible states.

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[4] Again, in RL nomenclature feedback is referred to as reward regardless of whether its value is positive, zero, or negative.

Policy is a key concept in this context. From the perspective of mathematical formalism, it is the function (map) of *state > action*[5]. Naturally, the agent will have a choice of several possible policies within a single environment: e.g. in our labyrinth it can determine that it will always start by turning left, then proceed two squares forward, or turn right and move three squares forward, etc. Notably, such policies can be **deterministic** (specifying only a single action in a given state) or **stochastic** (recommending the probability distribution of actions in a given state).

Another vital element of the training process is the **level of knowledge about the world model** (i.e. the functions of transition and reward). As already stated, the function of transition is the map of *(state, action) > state'*, whereas the reward function is the map of *state > reward*. If the agent knows both these functions, it has exhaustive knowledge of the possible consequences of actions in the respective states as well as the rewards it can expect to obtain. If complete and accurate, such information reduces the problem to mere **planning**[6]: **identifying** the optimum policy, rather than **learning** the optimum course of action.

So how does one define the policy in a situation where one has full knowledge of the consequences of any action in any state? The problem basically boils down to the following question: *of all the possible policies, which one is the most valuable*?, which in turn leads to the problem of *determining the value of a given policy*.

The policy for each possible state indicates the course of action that the agent should take: in the case of the example *grid-world*, knowing the world means being able to identify the route for traversing it in accordance with any given policy. At the same time, the agent is

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[5] In a general case: the map of state > actions probability distribution

[6] As a result, one cannot talk of reinforcement learning relying on the trial and error approach. However, this example is often evoked in RL handbooks as it facilitates a fairly simple introduction of certain key concepts, which will later provide the basis for developing learning methods applicable to environments that do require exploration.

perfectly aware of the reward that any given state can yield. Consequently, assuming that we always start on square A and finish on square B, and that the world does not change in the course of the model (admittedly, some very strong assumptions . . .), the value of each strategy can be calculated as the sum total of all rewards collected by navigating the route indicated thereby. And that value provides the basis for identifying the optimum policy.

Naturally, in practical applications the calculation of policy value tends to be somewhat more complex than that. Two new concepts come into play in this process: the **state-value function (V)** and the **action-value function (Q)**.

The state-value function V_π (s) allows a certain **value** to be attributed to a state. This value is understood as the expected sum total of all the rewards that will be collected by an agent beginning its 'journey' from the given state, provided that it proceeds in accordance with the given policy. In our case (*grid-world*) calculating the function will assign each square on the board with a certain value, which informs the agent: *if you start your action from this square and follow the policy π, this is the rewards you will get.*

An interesting question worth considering when calculating the state-value is the delay of reward. The dilemma is this: when we find ourselves in a given state, **to what extent should we account for rewards obtainable in a very long-term perspective when assessing its value?** RL methodology solves the dilemma as follows. The value of a given state is calculated as 1. the reward obtainable immediately after achieving the same, plus 2. the value of the subsequent state (i.e. the next state to be reached when following the given policy) times the discounting factor which can assume values from 0 to 1. The decision on the actual level is, in a way, a measure of the agent's (or more specifically, the teacher's) hedonism. If the value is 0, it means we proceed solely based on the benefit generated by the given state (extreme hedonism): subsequent rewards have no importance whatsoever. If the value is 1, all possible future rewards are equally important as the immediate reward obtainable upon reaching the given state: this extreme means maximising the reward in the perspective of the entire episode ('life').

Another fundamental function in many algorithms is that of **action-value** Q_π (s, a). It determines the total cumulative reward to be obtained by an agent beginning its journey

in the state s and taking action a in this state. At first glance, it would seem that the function's role is only auxiliary: knowledge of the value function and policy should be sufficient to evaluate a given policy. But this approach has been found to work only in the situation of exhaustive knowledge of the world model, which is rather uncommon in practical applications.

With the defined and calculated functions of states and actions (without complicating things with many important methods that allow the same to be actually plausible in practice, e.g. Bellman equation or dynamic programming), we can now turn to the problem of identifying the optimum policy.

To once again use our example, let us imagine an agent in square s and faced with the choice of one of four available actions: go up, down, left, or right. Which one should it choose? The answer can be fairly easily obtained on the basis of 1. one's knowledge of the world (in this case knowledge of the state one will end up in after choosing each of the options, and 2. the pre-calculated function of the state-value. The first policy that immediately springs to mind is simple: let us choose the action that will lead us to the state with the highest value. In fact, this is so inherently tempting that the RL nomenclature tends to refer to it as the *greedy policy* – but, as will be demonstrated next, it does not necessarily have to be the best choice . . .

In our previous example, the agent knew the world model: it was able to precisely predict the consequences of its actions in all the states, in particular: it knew what states would follow from all of the possible actions and what rewards would be gained as a result. Hence, the problem of identifying the optimum policy came down to proper planning: using the functions of state-value and action-value, the agent could identify the optimum policy without the need for experimentation (exploration). However, the process becomes somewhat more complex in a situation where the agent cannot use such a model. We then have only two possible options: either find a way to learn the same or find another method of identifying the optimum actions. Methods following the former approach are classified as model-based (the agent initially does not know the model but decides to learn it), while algorithms that do not rely on a world model at all are called model-free (as it turns out, one can do pretty well even without a 'map' . . .).

Let us first discuss the methods of cognising the world. As mentioned before, knowledge of the world is equivalent to knowledge of the transition function (map of *(state, action) > state'*) and reward function (map of *state > reward*). But how does one acquire this knowledge?

Firstly, we could consider a rather convenient situation where the task is episodic in nature (it has a beginning and an end), and the agent can learn by performing an unlimited number of episodes (there are no limitations to exploration). In this case, learning can be based on complete episodes by repeating the experiment numerous times and registering all the states reached by the agent, consequences of the actions taken, and rewards gained. In time (with sufficiently many repetitions), we would end up with the complete set of information needed to build the world model (transition and reward functions), which in turn would be sufficient to identify the optimum policy (calculate the function of state and action value, etc.). The methods allowing calculation of the action-value function based on complete episode iterations are referred to as Monte-Carlo methods.

However, in the real-world, the agent's situation is hardly ever convenient enough to allow actually 'visiting' all the states and testing all the possible options (imagine trying to taste all the dishes offered in all the restaurants in your city . . .). This may be due to time or resource limitations (e.g. the power of computers used for simulations), and naturally, the greater the 'density' of states (number of possible values of the respective parameters describing a state) and actions (number of actions possible in a given state), the greater the number of options that need to be examined. Consequently, it becomes crucial to develop a strategy for **approximation** (rather than exact calculation) of the action-value function. RL terminology refers to this as the **problem of prediction**: the quality of prognoses relative to the value of a given action in a given state is key to identifying the optimum policy, which comes down to the simple observation that *a policy is only as good as the prognosis used in its development*. A well estimated action-value function (i.e. knowledge of the value of the given action in the given situation) constitutes the basis for identifying the optimum policy (solving the so-called **control problem**).

In situations where estimating the functions of state-value and action-value based on complete episodes is found to be unfeasible, methods other than Monte Carlo (MC) have

to be employed. The first of such methods entails prediction on the basis of temporal difference (aptly called the temporal-difference method (TD)). The initially randomly initiated function of value is later updated step-by-step (and not at the end of the episode as in the case of MC), whereby a step is understood as any single interaction between the agent and the environment (i.e. a sequence of observation > action > feedback). Updating the value of a given action in a given state comes down to adjusting the current value by accounting for the so-called prediction error: the difference between the current estimate and the actual experience. In considerably simplified terms, this means that if the value of the new state calculated based on the current state-value function is at the level V (estimate) and the actually experienced value is at the level T (target), we should adjust the value function for the given state by T-V (i.e. so-called prediction error). The difference between MC and TD methods stems from the fact that in the former, the value function is updated only at the end of each episode, whereas in the latter: after each consecutive step, which significantly reduces the time needed to learn what is preferable (the value function).

Methods such as TD and its variants (e.g. n-step TD or TD-Lambda) aim to address one of the major limitations of reinforcement learning - that is limited availability of samples or possibility of conducting experiments (so-called **sample efficiency**). Learning by trial and error is safe in environments simulated by computers (although even then we can encounter limitations such as available computing power) but can be less than that when conducted in the actual physical world: just imagine using this method when training an autonomous vehicle, with the length of one episode defined relative to the life-expectancy of the driver or other road users . . .

The methods of identifying the optimum policy described so far were based on **learning the value** (so-called Q-learning) of both **states** and **actions**. It turns out, however, that an effective policy can also be developed by directly optimising the policy itself – some of the major methods following this approach will be presented below.

First, however, let us reiterate some of the main points already discussed. Knowledge of the world model (i.e. functions of transition and reward) allows one to identify the optimum strategy (one that maximises the total attainable reward) without the need to

interact with the environment. However, the practical applicability of such an approach is negligible, hence the agent is forced to learn by trial and error. As it explores the environment, in each new situation it encounters it must answer the following questions:

1. **What do I ask the environment?** What information about the given state are the most important to obtain?
2. How good are my current estimated action values?
3. What task should I engage in now?
   - Should I act based on my previous experience (choose the action with the highest value score? < **exploitation**
   - Or should I take a chance and attempt a sub-optimum action in this situation? < **exploration**

As clearly follows from the above, the *exploit or explore* dilemma plays a very significant role in the process, becoming one of the key elements of the ultimate policy[7]. The key to finding the right balance can be expressed as follows: **our goal is to develop the best possible policy, so that we can later exploit it; but in order to do so, we must first engage in intensive exploration.** In this sense **exploration builds the foundation for future, effective exploitation**. Morales [@morales] illustrates this with a rather apt analogy: *It is like searching for a gold vein in Eldorado: before you can effectively exploit it, you must first spend a lot of time looking for it (perform a lot of explorations).*

Exploration strategies in reinforcement learning can be divided into **random, optimistic, and based on the space of information states**. An agent following the policy of **random exploration**, **in most cases** chooses the action with the highest value (maximum result of the state-value function Q), but **at times** will also opt for a lower value action. The term *at times* is defined by the parameter ε which can take values from 0 to 1 and denotes the probability of taking a chance and selecting an action other than the one known to yield

[7] As already mentioned, the strategy entailing the choice of the highest value action in each state (described as greedy) is one of many rather than the only available choice.

the highest value (based on our current experience). The value of ε is a policy parameter. If it is set to 0, the agent will follow the greedy policy (no experimentation), whereas if ε=1, the agent will completely ignore previous experience and always opt for exploration. Notably, after each such experiment, the function of Q is updated – the learning process continues. A good example of a practical application of the 0.25 value could be a policy where in 75% of the cases, we select our favourite food at the restaurant, while in the other 25% of the cases, we follow the recommendation of the waiter. Clearly, similarly to the dilemma of delayed pleasure determined by the parameter of (hedonism <> salvation after death), here too (conservatism <> recklessness) many analogies can be found with everyday choices. The primary difference, however, lies in the number of available episodes: reinforced learning usually provides many more than we can hope to enjoy in real life.

Naturally, random policies come in many diverse variants. In practical applications, training a model begins with a high level of exploration, which tends to be gradually reduced as the process progresses through subsequent episodes (by lowering the ratio). A somewhat more refined model (so-called SoftMax exploration strategy) correlates the probability of choosing a given sub-optimum action with the value of such action – one could describe it as *random with a preference for . . .*

Optimistic strategies such as e.g. o*ptimistic initialization strategy, upper confidence bound* or *Thompson sampling strategy* are based on the assumption that high uncertainty is welcome in the decision-making process (somewhat along the lines of *no risk, no fun*). By randomly assigning the initial values of the value function, we initiate it together with the function associating each action with a certain level of confidence. Next, in each subsequent step, we quantify the uncertainty related to respective decisions made and promote the decisions with higher uncertainty in the process of exploration. In **information state-space strategies**, the information states (including uncertainty) of the agent are treated as features of the environment. It perceives uncertainty as an additional dimension of information about the state, and consequently less explored states are perceived differently.

How can one approach the task of streamlining polices? The most popular method employed in a variety of algorithms is the so-called **General Policy Optimisation**, which entails iteration of three, deceivingly simple steps:

{repeat until you reach satisfactory results}

1. Take the current policy
2. Evaluate it
3. Improve it.

Although seemingly trivial, these steps have been practically employed in the development of many inspiring methods and techniques.

Let us then have a closer look at what exactly is meant by *evaluation* and *improvement.*

Starting from the beginning, the **'entry level' policy determines the agent's actions in a given state**. It could be e.g. the greedy policy: where in each state the agent always opts for the highest-value action[8]. Or one of the more or less explorative ones: the highest-value option is chosen in most cases but, now and again with the probability of ε, one lets luck decide the outcome. It can also be any of the other possible variants. What matters is that the policy is 'accompanied' by estimation of the action-value function Q_π (s, a) which assigns each of the possible actions in each of the possible states with a certain value understood as the total reward expected when performing the given task in the given state in accordance with the initial policy.

The second stage entails evaluation of the policy's effectiveness. The key word in the previous paragraph was **estimation**. Our goal is to improve the current policy – i.e. we humbly acknowledge that we have less than complete confidence in the estimated value of function Q and agree to conduct an experiment to verify it. This is particularly important in the context of early iterations when estimations of action-values are pretty

[8] Technically, it comes down to searching among the available actions for one with the highest value of function Q in the given state.

much random (or assumed based on experience from other projects, available heuristics, etc.).

And how does one go about verifying the accuracy of one's estimates as to what is and what is not correct? By experimenting of course! The agent **gains new experiences**, usually through interactions (i.e. sequences of *observing the state > taking an action > gathering feedback*) but also, in the case of the *model-based method*, by consulting the world model. After conducting the experiment, i.e. gathering new experiences, **the agent updates its assessment of the value function** using methods such as Monte Carlo, TD Learning, or others. This is a key stage in the learning process: we begin with the 'old' assessment of the action value and state value (function Q) and arrive at a new assessment backed by experimental results (function Q').

The final, third stage entails **updating the policy**. It usually comes down to replacing the original function of Q with the updated value function Q' and/or modifying other parameters e.g. ε. The thus modified policy becomes the 'entry level' policy in the subsequent iteration.

At this point it is worth underscoring the critical value of stage two: estimating the quality of the policy. This could be compared to broadly understood critique: customer opinions, feedback from the trainer, or self-reflection. The idea is simple: we have certain rules, we implement them, ask for feedback from the environment, and change accordingly. In this context, it is very important to remain aware of certain factors such as that our *critic* may be biased or have bad habits himself, as well as remember that pleasing everyone is a widely recognised impossibility. For these and other reasons, the process of improving value function estimations should always be meticulously planned and executed.

There are many other aspects of *General Policy Iteration* that ought to be considered, for instance, which policy to employ when generating new experiences: our current policy (in which case methods are categorised as *on-policy*, a good example being the SARSA algorithm) or an entirely different one (e.g. someone else's experiences – such methods are categorised as *off-policy* – e.g. the *Q-learning* family of algorithms). There is also the question of when to update the policy value estimation: after each episode (as in MC

methods – the approach is described as *off-line* learning) or after each step (*on-line* learning).

The limitations of this analysis prevent a more in-depth look into those interesting but very technical issues. One should note, however, that **continuous state and action spaces** cerate an entirely separate category of problems forcing agents to develop a capacity for generalisation and stochastic policies (i.e. ones that recommend only a certain probability distribution of action in a given state). The method of generalisation takes advantage of deep neural networks and such methods as deterministic policy gradients, including an entire range of *actor-critic* algorithms.

Challenges

There are still many challenges related to reinforcement learning. At the same time, the field continues to dynamically develop and RL algorithms are used in a growing range of applications, which only adds to the complexity of the matter. Dulac-Arnold et al. [@dulac-arnold2019] provided an interesting synopsis of problems currently (2019) faced by autonomous system developers:

1. **Effective learning off-line**, i.e. based on already gathered experience rather than direct interactions between agents and their environments. Trial and error methods are not always safe (e.g. vehicles) or effective (e.g. experiments with various strategies in on-line marketing where the perceivable effects of actions can be significantly delayed)
2. **Fast learning based on limited data** (the problem of sample efficiency)
3. Learning in multivariate continuous state and action spaces
4. **Safety** issues in learning processes
5. **Learning tasks performed in conditions of high uncertainty** (e.g. only partially-observable or stochastic environments)
6. Learning in situations where the goal functions (reward systems) are underdefined, susceptible to risk or oriented towards multiple goals
7. The ability to explain the learnt action rulesets to operators supervising autonomous systems (so-called self-eXplainable AI: XAI)

8. Inference that has to take place in near real time
9. **Extensive and unpredictable operating delays** related to operation of actuators, sensors, or reward signals.

Clearly, there is no shortage of challenges. But one hopes that weighed against the rapid development technologies auxiliary to AI (e.g. computing power) and undeniable business potential, those challenges will ultimately contribute to the development of this very interesting field of research.

Studies analysing methods of training agents to effectively act under new circumstances and often in fast-changing environments are very inspiring: they bring together many seemingly distant fields such as algorithmics, optimisation, psychology, cognitive neuroscience, management and decision-making. At the same time, many mechanisms developed to improve RL processes could also be successfully applied to everyday life or management. For this reason, this field of study is certainly worth learning about, even if one were not to pursue a career in the development of autonomous, self-learning systems – the entry barrier may be high, but the effort is still likely to prove worthwhile.

Self-learning, autonomous systems in the modern world

Before we proceed to discussing the business potential of self-learning, autonomous systems, let us briefly summarise the main points related to the development of intelligent solutions.

The goal of studies on artificial intelligence is to create systems capable of performing at least as well as their human counterparts. Such solutions are able to operate in a variety of environments, at different levels of predictability and complexity. In order to be effective, they should be provided with a clearly defined function of purpose, capacity for effective perception (of both the environment and internal parameters), build-in or developable behavioural rules, and the ability to undertake action influencing the environment.

The advancement of a solution and the related complexity of algorithms are most strongly dependent on the predictability of the environment's behaviour and the agent's own perception capacity. The methods currently employed in AI aim to minimise the impact of such uncertainties. They aid digital agents in reducing the uncertainty related to registration of parameters (e.g. image, text, and speech recognition algorithms), environmental dynamics (predictive algorithms), evaluation of actions taken, and the consequences thereof (machine learning methods). Additionally, precise motor solutions contribute to the capacity for adequate response, while new methods of gathering experience and forms of presenting knowledge facilitate the process of determining the impact of specific actions in specific environments.

One of the main goals in this context in to provide systems with autonomy. But what does that mean in practice?

The first aspect of autonomy was already discussed in the section focusing on machine learning. The methods described therein were categorised relative to the level of human supervision: from supervised learning to meta-learning. The level of human involvement in the learning process is a natural measure of its autonomy – the best autonomous solutions not only do not require human support but can also select the methods of learning that best suit their respective purposes (*learning to learn*).

Varying levels of autonomy can be observed in the context of Business Intelligence (BI) or more generally decision-making systems. One of the methods employed in the classification of different types of business analyses assumes the distinction between **descriptive** (*What happened?*), **diagnostic** (*Why did it happen?)*, **predictive** (*What might happen?*), **prescriptive** (*What should be done about that?*), and **adaptive analyses** (*How do we build a system that would effectively adapt to the environment?*). Let us consider this classification from the perspective of a decision-maker. Contemporary managers often make decisions based on descriptive and diagnostic analyses whose forecasts are typically treated as a point of interest (with the possible exception of stock exchange brokers for whom effective prediction in the very basis of decision-making). Overall, they rarely base their choices on the results of prescriptive analyses, and hardly ever on adaptive analyses. Meanwhile, even the full inclusion of only predictive analyses in the

decision-making processes can significantly change the playing field: let us imagine a board meeting during which the decision-makers, rather than consider a course of action dictated by the past, look instead to the future and decide what would be the optimum choices should one assume that the algorithm's predictions are indeed accurate . . . Naturally, this would go against some deeply ingrained habits but could also bring an entirely new quality to the decision-making table.

Prescriptive analyses go even a step further: they recommend the optimum course of action. A system developed by Aera Technologies, which will be discussed further in the book, utilises advanced analytical and predictive systems processing large amounts of data gathered from various other systems operating in an organisation to provide managers with recommendations of future action in many functional areas: from logistics, through production and sales, to finance. It is not difficult to imagine that as such solutions become more effective and flexible, the role of managers may ultimately be reduced to formulating relevant questions, evaluating the quality of recommendations, and their ultimate approval, modification or rejection. In turn, this may eventually allow materialisation of the so-called *self-driven enterprise* concept. At present, this vision obviously still remains somewhat utopian but as such systems continue do dynamically develop, one would be wrong to entirely dismiss it as a viable future possibility.

To a varying extent, autonomous systems have long had a place in industry. In fact, robotics has already been heavily studied in terms of interesting taxonomies of autonomy levels. Beer et al. define robot autonomy as "the extent to which a robot can sense its environment, plan based on that environment, and act upon that environment with the intent of reaching some task-specific goal (either given to or created by the robot) without external control"[@beer2014b]. The authors proposed helpful questions (guidelines) applicable when evaluating the autonomy level of a given robot, e.g.: *What tasks are to be performed by the robot? What aspects of said tasks are to be included? To what extent are those tasks to be performed independently?* Answering the same will help one to determine the *autonomy category of the robot*. In this context, Beer et al. propose 10 distinct levels (from minimum to maximum autonomy): *manual operation, remote operation, assisted remote operation, batch processing, decision-making support, joint human-supervised*

control, joint robot-supervised control, operating supervision, supervision, and *full autonomy*. The resulting evaluation of the autonomy level is later used to determine its potential impact on human-robot interaction (HRI) with due consideration for the perspective of the robot, the human, and the social milieu.

Surgical robots constitute a particularly interesting subset of autonomous machines. Ficuciello et at. [@ficuciello2019] identify four levels of autonomy applicable thereto, based on the level of so-called meaningful human control (MHC). At level 0, a surgent is in full control of the robot – its role is reduced to increasing the efficiency and precision of surgical operations with the view to compensating for certain sensory and motor limitations of the human operator. At level 1, the robot assists in the surgery – serving the role of a so-called *robotic surgery assistant*. Its primary function, particularly valuable in the context of so-called micro-surgery, is to actively limit and modify the movement trajectory of surgical instruments. At this level, the surgeon has the option of disabling this functionality, thus forcing the choice of the human-defined trajectory. This could be compared to mechanisms evening the movement of pencils or brushes in drawing applications on e.g. a tablet. At level 2, the human operator defines a task and the robot proceeds to autonomously perform it. The surgeon's role is reduced to hands-free monitoring of the task's performance and stepping in to adjust it where necessary – in this case, the robot is under discrete rather than continuous control (as opposed to level 1). Finally, at level 3, the robot independently prepares strategies for the performance of surgical tasks and the role of the human comes down to selecting and approving the optimum strategies and then supervising their execution. Clearly, we are now only a step away from medical devices previously only seen in Science Fiction movies.

Let us now proceed to autonomous cars. This industry is currently probably the most advanced in terms of practical application of artificial intelligence solutions. Hence, quite naturally, it has long developed its own dedicated taxonomy. The Society of Automotive Engineering (SAE) identified six levels of vehicle autonomy based on a detailed identification of respective human and vehicle-controlled tasks (cf. e.g. [@zotero-1739]).

And so, at levels 0, 1, and 2, the human driver is responsible for controlling the vehicle's movement (acceleration, breaking, steering, etc.), even if certain assistance functionalities

are engaged. At level 0, the vehicle itself provides little support, mainly in the form of alerts or rapid response, e.g. automatic breaking in hazardous situations, blind spot or lane departure warnings. Level 1 provides assistance in terms of steering OR acceleration/breaking, e.g. line centring or adaptive cruise control. Level 2 is similar, although in this case steering AND acceleration/breaking assistance is provided simultaneously.

Higher levels of autonomy are available at levels 3 through 5. Here, the driver is no longer actively involved in the functional processes: only at level 3 is his or her input required when the vehicle 'demands' it. At levels 3 and 4, the vehicle operates autonomously as long as certain conditions are met. Level 5 entails full autonomy of the vehicle, irrespective of conditions: driver input is unnecessary and sometimes even impossible (e.g. if steering implements are not even provided).

It is equally interesting to consider autonomy from the perspective of autonomous aerial vehicles. Derenick [@2020b] defines drone autonomy as the ability to answer three question: *where am I?, where am I going?,* and *How do I get there?* Apparently, in the sometimes very difficult conditions that such machines have to cope with, these problems turn out to be anything but trivial. Let us imagine a drone used for mapping out a mine. Firstly, it will have no access to geolocation (e.g. GPS), it will therefore have to rely on advanced analyses of data received from various sensors and maps, often generated in real time, to independently answer the first of the questions: *Where am I?* The answer to the question of *Where am I going?* requires planning, which basically entails decomposition of the mission goal into partial tasks, and the latter into elementary behaviours. At the level of behaviours, optimum trajectories are established to help the drone to reach its designated destination. Finally, the question of *How do I get there?* requires autonomous performance of the task, without any external supervision, often without viable communication systems. When this is coupled with obvious weight limitations, which also translates to limited computing power (a drone cannot carry a powerful computer), limited energy (battery), and susceptibility to external hazards such as gusts of wind, we can begin to appreciate the considerably more convenient 'situation' of autonomous vehicles . . .

As follows from the above, the specific definition of autonomy is strongly dependent on the context and application of an intelligent system – nonetheless, it always comes down to a gradual reduction of dependence on the human operator. Still, it is up to the human to decide what competences are to be delegated to the machine. So far, these have been limited to the performance of tasks – soon, however, we might allow machines to also identify goals, and not only particular (e.g. when planning the course of a surgery) but also strategic ones. There is much to suggest that researchers are now gradually laying the groundwork for just that (cf. e.g. inverse reinforcement learning methods described in Chapter 4). One should remain aware of these trends and be able to recognise the various methods of extracting value from intelligent systems, which will be the topic of the subsequent parts of this book.

Summary

At the beginning of this chapter, intelligent systems and technologies were defined as *capable of effectively operating under new circumstances*. A detailed explanation of the respective components of this definition was then provided, with a particular focus on the ability to respond to fast-changing circumstances characterised by high uncertainty and only partial availability of information. Next, a whole range of machine learning methods were discussed, with a more in-depth consideration of reinforcement learning, which demonstrated the complexity of issues related to training effective solutions intended for real world applications.

One could get the impression that advanced machine learning algorithms remain the domain of researchers testing their effectiveness in technical simulations or at laboratories, research centres, and industrial facilities. This could not be further from the truth: autonomous robots have long been successfully implemented on production lines, and autonomous cars have already travelled millions of miles on 'real' roads. Indeed, one could hardly think of a better test for an intelligent system (within the meaning of our definition) than a crowded highway travelled at night in heavy sleet . . . The systems are already so mature that regulators have initiated legislative processes aimed at officially

allowing such vehicles to participate in public traffic (cf. e.g. reports from the work of The National Highway Traffic Safety Administration (NHTSA) in the US [@2020a]).

Clearly, the vision of intelligent systems *capable of effective operation under new circumstances* is no longer a utopian idea. Their gradual proliferation is commonly expected – which begs the question: *how does one effectively utilise their potential to generate value in one's organisation?* This problem will be the focus of the subsequent Chapters of this book.

The impact of autonomous systems on technologies, processes and industry sectors

Technologies laying the groundwork for autonomous systems

Introduction

The impact of autonomous systems on Information and Communications Technologies (ICT) has been rather complex: on the one hand, dynamically developing autonomous systems (e.g. in the context of transport or robotics) stimulate the development of new technologies (sensors, algorithms, security measures, etc.), and on the other, the methods of reinforced learning help to facilitate the operation of already existing IT systems. This section will be limited to only the first of said aspects: we will discuss the technologies necessary for the development of effective autonomous solutions, while their impact on ICT technologies will be presented in the subsequent part of this chapter.

Sensing, Perception and Situational Awareness

The ability to perceive the environment is of key importance to the effectiveness of autonomous systems. Not unlike in people, such systems rely primarily on senses (mostly visual, albeit in a spectrum far exceeding that available to humans, and audio) and perception (recognition of objects, sounds, as well as more complex constructs such as behaviour), which jointly facilitate so-called situational awareness. Strict requirements applicable to autonomous systems, particularly with respect to the safety thereof, impose very high standards of precision, efficiency, and reliability which necessitate constant development and improvement of methods and techniques employed in this context.

Presented below are the key technologies facilitating sensing of the surroundings (including LIDAR, radar, infrared and thermal detectors), perception (object, speech and

behaviour recognition), as well as situational awareness solutions. In each case, the most important concepts will be presented along with examples of application and key features of the respective solutions.

Sensing

LiDAR (Light Detection and Ranging) is certainly one of the most popular sensory techniques available. The heart of the system is a laser which emits strong pulses of light at a specific wavelength through a special optics. It then registers and analyses ambient light scattering with the use of dedicated optical and computer systems.

LiDARs are utilised by most (if not all) manufacturers of autonomous vehicles. They are commonly employed in advanced driver assistance systems (ADAS) for the purposes of blind spots identification, lane keeping, autonomous speed control and breaking, as well as collision prevention. LiDARs manufactured by companies such as Blickfield are also present in fourth and fifth autonomy level vehicles, where they serve the purposes of classifying objects emerging on the road: pedestrians, vehicles, cyclists, etc., with their key benefits including high precision, considerable range, wide-angle vision, and real-time object location.

The technology is also extensively used in industry. Solutions offered by Velodynelidar are commonly employed in e.g. delivery or inspection robots, agriculture (e.g. spraying), interior mapping, or maritime economy. Combined with drones, they are prevalent in the energy industry (e.g. in wind turbine monitoring), construction, rescue systems, agriculture, geography, or archaeology. The technology is also increasingly incorporated in smart cities: LiDARs facilitate the monitoring of traffic intensity, identification of available parking spaces, as well as surveillance and security measures in public spaces. They are now also commonly employed for the purposes of generating high resolution maps: when mounted on vehicles or drones, they are able to precisely (with centimetre accuracy) measure the topography of large 3D environments, both indoors and outdoors, including industrial or agricultural areas. The SLAM (Simultaneous Localisation and Mapping) technology utilised in this context allows LiDAR data to be combined into consistent maps which can later be used for the purposes of autonomous vehicle guidance, investment planning, or monitoring and surveillance of construction works on

roads, bridges, factories, etc. In order to properly perform their tasks, LiDAR technologies must provide the adequate level of precision, reliability, and availability. Precision does not depend solely on the properties of the sensor itself, but also the efficiency of the algorithms processing the data it provides. Luminartech is a company specializing in the supply of so-called perception engines for LiDARs which, combined with good quality sensors, are capable of facilitating the range of up to 500 m, 120 degree field of visions, resolutions similar to those of modern cameras (up to 800 px/Deg2), and precision at the level of up to 1 cm. Consequently, vehicles utilizing such technologies are able to effectively recognise road lines from up to 150 m and objects on the road from up to 250 m. In turn, reliability refers to the capacity for effective perception in difficult conditions. Luminartech and Innoviz.tech emphasise the high effectiveness of their solutions even in high intensity light (e.g. sunlight), bad weather (snow, rain) or difficult environmental conditions (dust, vibration, impact). One of the reasons for the relatively low reliability of LiDAR solutions, which is often identified as one of the key disadvantages of this technique alongside high cost and excessive amounts of data generated), stems from its rotational mechanism (the sensor mirror is rotated along its axis at high speed). One solution to that problem may come in the form of LiDARs developed to work without any movable parts, such as those offered e.g. by Xenomatics, which significantly improves stability and the system's overall fault resistance. The company's products additionally utilise the so-called multi-bean technology, which facilitates time savings and reduces error incidence in the process of signal analysis. Currently, it seems that availability is probably the most significant problem with LiDAR technologies. In 2018, the price of a single sensor could reach even several dozen thousand dollars, it was not until 2019 that Waymo corporation managed to reduce that cost to a somewhat more manageable level of USD 5-8 thousand.

Another technique allowing analysis of environmental data is RADAR: detection systems using radio waves to determine the range, angle, or velocity of objects. Arberobotics radars used in autonomous vehicles are capable of concurrent processing of data from up to 2,000 virtual channels, which allows them to simultaneously track hundreds of different objects within a wide field of view, at high time resolutions (up to 30 frames per second), and in four dimensions: the three spatial dimensions plus speed. The company

markets its offering as a significant complementation to the perception systems installed in modern autonomous vehicles.

Another class of sensors are those capable of detecting infrared light (SWIR: ShortWave InfraRed). They provide good environmental perception under low visibility conditions (e.g. bad weather, night-time, excessive dust). Moreover, dedicated image sensors facilitate very low energy consumption, reliability, and small size (cf. e.g. Trieye solutions). They can be complemented by thermal sensors which are also characterised by relatively good reliability, efficiency, as well as low price and small size. Adasky, one of the providers of this type of sensors, further advertise their products by evoking their comprehensiveness (integration with dedicated deep learning algorithms), adaptability to the high requirements of the automotive industry, and scalability (thanks to relatively low costs).

As shown above, autonomous situational awareness systems utilise numerous, mutually complementary 'senses': LiDARs, RADARs, infrared or thermal sensors. Each of such subsystems is capable of generating a variety of data types. For an autonomous system to be able to operate effectively, it is therefore crucial to be able to properly integrate such data – a task handled by so-called sensor-fusion solutions. These are either offered directly by sensor manufacturers (e.g. Exyn or Neptec Technologies) or providers of dedicated platforms (e.g. Accelerated Dynamics).

Summary (frame) 1. Key features of solutions: precision, reliability, availability. 2. Dimensions of integration: sensor, algorithm, sensor fusion.

Object, image and behaviour recognition

Good quality sensor data are the basis for functionalities such as image, object, and behaviour recognition. Solutions in this class most commonly utilise artificial intelligence algorithms based on deep learning networks. It is noteworthy at this point that even the most capable object recognition algorithm will not be sufficient by itself: if it is to work effectively (e.g. fast, using minimum computer resources and with limited energy consumption), a dedicated IT architecture needs to be provided as well.

Consequently, this fact creates a niche for companies supplying often very narrowly specialised solutions capable of detecting specific types of objects[9].

Robosense is a company specializing in the development of dedicated algorithms processing LiDAR data. Its solutions facilitate precise (up to 20 cm), real-time location of objects, road curbs/driving area detection, road sign recognition (both vertical and horizontal), identification and classification of obstacles (distance, location, size, shape, and category), as well as tracking of moving objects (including in terms of speed and acceleration). All those functionalities are critical to the effective and safe operation of autonomous vehicles.

Stradvision offers similar solutions, with the subdivision of systems into those recognizing images inside and outside of the vehicle. The algorithms responsible for analysing the external environment provide capabilities similar to those offered by Robosense, with the additional ability to identify empty space, measure distance from other vehicles, and recognise registration plates. On the other hand, internal image analysis systems can determine certain characteristics of the driver such as height, width or weight (as well as even concentration levels), and adjust various elements of the vehicle's operation accordingly, e.g. in terms of airbag air pressure, seat position, etc.

Nauto offers solutions that, on the one hand, detect the overall condition and behaviour of the driver, and on the other monitor the situation on the road. They constitute the basis for an interesting range of dedicated services addressed not only to drivers but also owners or operators of vehicle fleets (not just autonomous, in fact primarily traditional vehicles). For instance, detecting the condition of the driver (e.g. level of distraction, tiredness) and their current driving style (e.g. driving too close to the preceding vehicle, speed, acceleration, other manoeuvres) can serve as the basis for not only a variety of driver warning notifications but also driver profiling systems. Such profiles combined

[9] In fact, the problem is so complex that one could even talk of the emergence of an entire new industry sector – as discussed in Chapter 4.

with driver statistics (e.g. accidents) can in turn contribute to analytical systems offered to fleet operators who may employ them as the basis for e.g. dedicated training or employee evaluation systems. Moreover, camera images combined with accident detection and damage class identification systems can significantly accelerate insurance related processes, e.g. claims management.

Algorithms capable of recognizing images, behaviours and situations allowed Radius.ai to develop solutions identifying various aspects of so-called consumer journey without infringing on their privacy. Radius.ai systems identify customers and use information about their behaviour as well as other contextual data to recommend products, events, important notifications, etc. This, in turn, allows companies to improve the sales figures of their brick-and-mortar outlets, increase revenues, generate better customer loyalty, optimise product distribution, and plan more effective campaigns.

As mentioned above, a key factor in the context of image recognition algorithms and their effectiveness is their optimum implementation vis-à-vis IT infrastructures. Pilot.ai is a company whose primary competitive advantage stems from the development of proprietary methods and techniques of implementation to facilitate effective image analyses with the use of relatively simple devices such as digital cameras, door systems, or household vacuum cleaners. This necessitates the development of dedicated image recognition algorithms, methods of training them based on selected datasets specific for a given industry or practical application, and techniques of implementing the same without interfering with the primary infrastructure. Consequently, Pilot.ai systems are currently employed in various commercial contexts (customer behaviour analytics), smart homes ('intelligent' household devices), and public spaces (e.g. for monitoring the behaviour of people at airports or in shopping malls).

Another interesting use of image recognition algorithms entails precise identification of objects in storage or production spaces. Mujin, a company specializing in the production of logistic robots, had to equip its devices with systems facilitating recognition of packaging edges and their distribution on pallets in such a way so as to allow the robot to selectively move particular boxes, e.g. in the process of depalletization. The employed

algorithms are also crucial to the effective handling of products inside a warehouse or their packing for delivery.

The use of artificial intelligence is also common in medical imaging analyses. This particular area of AI applications is, on the one hand, a source of major challenges (any potential error can have tragic consequences), but on the other, large sets of well diagnosed images allow the generation of very high quality algorithms. For years, companies such as Zebra Med or Enlitic have been offering solutions capable of supporting radiological analyses, facilitating early identification of high risk patients, aiding healthcare authorities in developing effective disease prevention programs, and helping hospitals in the implementation of quality control systems.

Summary (frame) 1. Key features of solutions: image, situation, behaviour recognition. 2. Competitive dimensions: algorithm, effective hardware implementation, accompanying services.

Situational Awareness

Situational awareness is another level of environmental perception of key significance for the effective operation of autonomous systems. It is understood as the ability to perceive elements and situations in the surroundings (space and time), recognise their significance, and anticipate their condition in the future. Some of the technologies that make the respective elements of situational awareness possible have already been described in the previous section – discussed below, are some of its more advanced applications.

Terrain maps created by systems utilizing e.g. LiDAR technologies may be enhanced through the use of so-called semantic mapping, where the representations of recognised objects include e.g. certain conditional characteristics thereof. This allows autonomous systems (e.g. robots or vehicles) to answer questions such as: *Are there any open doors in this room? Where are there people in this building?*, and monitoring systems in public spaces to identify individuals, determine if a person is e.g. unconscious (the man has been lying in the street for over a minute – he may have fainted), or identify potentially

hazardous objects (that package has been left unattended for over 20 minutes and nobody seems to be interested in it).

Clearly, semantic mapping can provide a basis for systems capable of behaviours far exceeding simple programming (autonomous rather than automated). For instance, solutions offered by Exyn used for drone mapping in the mining industry, contribute to a significant reduction of manhours required to monitor underground safety, allow access to otherwise inaccessible areas, and consequently not only improve the safety of employees but also allow them to spend their time on more vital tasks and create the space for hiring people with entirely new competence profiles.

In turn, Naptec Technologies are marketing a solution described as 3D Real-Time Intelligence: this dedicated software and algorithm package (segmentation, identification, tracking, change detection, etc.) allows real-time LiDAR data extraction and conversion into information necessary for effective (self)control of autonomous vehicles. The Neptec platform not only allows acquisition of such information but also development of proprietary applications that utilise the same.

A somewhat different approach to intelligent mapping has been adopted by Carmera. Their system gathers current geospatial information with the use of cameras mounted on the fleet vehicles of partner organisations, thus ensuring virtually continuous map updates (e.g. reconstructions, failures, etc.). Thanks to this "crowdsourcing' approach, the costs of map development and maintenance (updating) is relatively low, while at the same time the product maintains a very high quality. DeepMap.ai offers a system oriented towards very high precision (within several centimetres). Apart from high quality maps, DeepMap also offers simulation-grade data: based not on artificially generated datasets but rather on genuine maps of the surroundings. This is very important in the process of teaching algorithms to control autonomous vehicles.

One of the key purposes of situational awareness systems is to provide as detailed information as possible about the current position of an object in order to be able to effectively control it afterwards. One of the main limitations of autonomous vehicles, and particularly flying devices (e.g. drones), is the extent of their ability to process large amounts of data. While in the case of a car, one could imagine having a powerful computer

with the needed power supply installed e.g. in the boot which would be dedicated to processing large amounts of information gathered from tens of sensors in real time, the same would be considerably less practical in the case of flying devices. For this reason, one should consider other, centralised systems of situational awareness, wherein data are processed and analysed on cloud servers. In a solution offered by Scape.io, the localised object sends camera images to a central server where they are compared to previously rendered 3D models of the surroundings and converted into precise geolocation information. A similar method is employed in the dedicated systems offered by Naverlabs in the context of indoor spaces (rather than urban spaces as is the case with Scape.io). In their product, Neverlabs engineers took the concept one step further: they created a prototype of a so-called 'brainless' robot controlled by an external server established with the use of a 5G telecommunications network. Its operating principle is similar to that developed by Scape.io: firstly, a dedicated robot creates a detailed map of a building's interior, which is later used to operate other devices.

Yet another approach to positioning objects inside buildings (an important aspect of situational awareness) was adopted by Accerion. Using advanced surface scanning solutions allows very precise (down to millimetres) localisation of objects in e.g. storage areas or production floors, and navigation with the use of virtual lines. What sets this system apart is the fact that no changes to the existing infrastructure are necessary - the positioning requires no additional devices or markings (e.g. QR or physical lines) on the relevant surfaces, which allows for quick and cost-effective implementation as well as navigation under dynamically changing conditions. Additional benefits include savings in terms of space taken up my routes designated for goods transport (which directly contributes to cost cuts) and the possibility of implementing so-called micro-fulfilment systems that support, rather than replace, the warehouse staff.

An interesting solution for precise positioning of vehicles in open spaces is offered by SwiftNav. The company's algorithms improve on the positioning data provided by GNNS (Global Navigation Satellite Systems) to allow positioning of vehicles within the United States with up to 20 cm accuracy, which proves practically sufficient for autonomous

vehicles. The solutions are implemented not only in automotive products but also in railroad transport as well as in agriculture.

Finally, one should also mention one other context of solutions offered by companies such as Scape.io or Neverlabs: they indicate a potentially very interesting direction for the future development, even if not a new 'philosophy', of autonomy. In the 'classic' approach, autonomy can be a quality of a separate physical entity (a vehicle, robot, another device). However, in the 'brainless' concept, autonomy applies to an object in the "logical' sense: sensors and actuators are installed on one device, while the control centre operates elsewhere (on a server) – the connection between the two is not physical, it takes advantage of the ultrafast 5G wireless communication. It is certainly possible hat the two philosophies may, in time, compete against each other in the context of e.g. autonomous transport: one extreme solution might entail the development of fully autonomous end products, while the proponents of the other might strive to develop ecosystems (e.g. intelligent roads) which will control vehicles travelling along them.

Communication and Connectivity

Apart from perception, another 'skill' of autonomous systems necessary to allow their effective operation in an environment is the ability to **communicate**. Due to the volume limitations of this book, the below deliberations will focus only on communications systems utilised by autonomous vehicles as they provide an example that is not only interesting but also fairly universal – similar solutions are commonly employed in other sectors as well.

An autonomous vehicle navigating a public space must be able to receive signals from its surroundings. The same emerge not only in the form of sounds and images, but also signals actively transmitted by other road users (e.g. vehicles) as well as elements of road infrastructure (e.g. traffic lights or transmitters installed on highway barriers). Therefore, when analysing autonomous solutions, one should approach them as a whole, as a certain fairly consistent architecture. In the context of communication as discussed in this section, the issue boils down to at least 4 separate channels commonly referred to as V2X, where X can stand for another vehicle (V2V – Vehicle-to-Vehicle), road infrastructure (V2I -

Vehicle-to-Infrastructure), a pedestrian (V2P - Vehicle-to-Pedestrian), or a communications network (V2N - Vehicle-to-Network). The currently dominant technology, based on the IEEE 802.11p standard, provides a range of up to 1 km, low latency (approx. 2 ms), and high reliability – it facilitates communication between fast moving vehicles and in extreme weather conditions (wind, rain, snow, fog), while remaining outside the wavelength range of telecommunications networks.

The most important purpose served by V2X communication is to ensure safety. Example scenarios of application include collision warning systems (alerts such as *Do not overtake!* generated based on signals received from a vehicle coming from the opposite direction), assistance in navigating crossroads (particularly when turning left) and turns in conditions of limited visibility, speeding alerts, or pedestrian/cyclist incursion warnings. Oher potential uses (not requiring ultrafast communication and therefore usable in cloud computing-based solutions) include e.g. information about roadworks, parking assistance (including identification of available parking spaces), processing highway fee payments, or general traffic control.

The technological capacities listed above suggest two possible, extreme philosophies of implementing autonomous vehicle systems. The first entails the development of independent vehicles capable of effectively, safely and autonomously navigating in various conditions (the full burden of autonomy is placed on the vehicle). Meanwhile, in the other concept the element facilitating autonomy would be the road infrastructure itself, the safety and efficiency of the system would stem from subordinating vehicles to 'commands' provided by the infrastructure. In simplified terms, in the first case the vehicle is fully autonomous, whereas in the second it is controlled by its surroundings. Naturally, in actual applications we are more likely to be faced with hybrid solutions combining elements of both these approaches, however, it would be very interesting to consider the question, touching upon both technology and cultural studies, of how the cultures of respective countries might influence their preference for the adoption of one or the other of said 'philosophies' . . .

Training, Learning and Simulations

The development of an intelligent solution and its adaptation to work under specific conditions (e.g. on a given production line) requires it to be adequately trained. This creates a niche for a separate class of solutions, i.e.: *training, learning and simulations.*

The first subgroup of solutions falling under this particular category, one that is most intensively exploited by autonomous vehicle developers, includes simulators of road conditions. For safety reasons, it is highly recommendable that algorithms employed in vehicles be pre-trained and tested before being allowed to deal with real-world conditions.

For instance, Cvedia specialises in the supply of carefully labelled (classified), synthetic but highly realistic images and entire scenes depicting road traffic situations, as well as specialised algorithms improving the learning efficiency of image recognition systems used in autonomous vehicles (as already discussed in Chapter 1, in order to operate effectively, such systems need to be highly specialised). The developers of systems intended for the purposes of vehicle control can take advantage of Cvedia solutions to significantly reduce the time and money investment necessary to pre-train algorithms and evaluate their effectiveness in 3D digital simulators. In turn, Playment.io specialises in labelling data obtained from sensors. The algorithms offered by the company have been generated based on a great number of case scenarios, fast and exact identification and localisation of objects, tracking the same with adequate scene understanding, and maps generated on the basis of data obtained from numerous sensors onboard an autonomic vehicle.

In turn, Cognata offers dedicated digital simulators for ADAS (Advanced Driver Assistance Systems) developers. They facilitate the training, evaluation, and analysis of the quality of situational awareness systems. The scenes generated by the simulator are very realistic (varied surroundings, weather and lighting conditions, surfaces, and objects), and the entire system is scalable thanks to the use of cloud technology. A similar solution, offered under the Simulation-as-a-Service formula, is also provided by Metamoto, while Foretelix specialises in modelling and developing scenarios of possible road situations.

However, even the best trained and prepared system has to be eventually taken out of the simulator and tested in the real world. The stage of adapting the robot to work at a given workstation may require tens of hours of tweaking (usually related to movement programming). It is therefore unsurprising that researchers and solution providers are constantly searching for new, more efficient methods in this respect. One of such methods is called *imitation training*, where the task of the system is to imitate the movement of the teacher as accurately as possible (a process not unlike that of teaching a child . . .). One of the providers offering this class of solutions is e.g. Micropsi Industries. The company sells pressure sensors, image recognition systems, and dedicated robot training algorithms. After installing the system, the operator manually guides the arm of the robot, which then attempts to precisely copy the movement of the teacher. After several/a dozen or so iterations, the system is capable of learning even very advanced and nuanced manoeuvres without the need for external programming. The process is both fast (the time needed is shorter than in the case of other methods) and more natural for the operator (no need for programming, simply manually guiding the robot arm). It should be mentioned that imitation training methods are currently subject to intensive and advanced research scrutiny.

Another class of solutions comprises systems capable of continuous self-learning. A very interesting solution in this category is the Brain Builder ^{TM} platform offered by Neurala. It utilises innovative algorithms imitating the inner workings of the human brain to train neural networks, thus facilitating the design and implementation of systems that learn (and consequently self-improve) in a continuous manner based on data received from numerous sensors. Neurala solutions have already been implemented in over 50 million devices, from cameras, phones, and drones to production machinery, where they are used for the purposes of detection, segmentation and recognition of images in the context of industry and robotics. Furthermore, it turns out that continuous learning (including self-learning) is a functionality available not only to individual devices but also entire groups thereof. The solutions that make it possible, commonly referred to as *collective learning*, are offered e.g. by Accelerated Dynamics, a company supplying control systems for drone fleets. With this functionality, drones can not only communicate with each other, but also gather information and experiences in a central database, hence

allowing them to learn in a continuous manner. In turn, Skux utilises a similar solution for the purposes of coordinating the operation of intelligent drone swarms used for agricultural spraying (a systems they describe as a *precision agriculture spraying swarm*), while Autonomous Marine Systems offer fleets of interlinked marine 'drones' used for the purposes of e.g. ocean monitoring. The above are only a few examples of the practical applications of so-called multi-agent technologies – the business potential of such solutions will be discussed in greater detail in the subsequent chapter.

The aforementioned systems providing data for the purposes of simulators and algorithms supporting the processes of continuous learning constitute one of the foundations for the emergence of an entirely new industry: providers of autonomous solutions. This hypothesis will be further explored in the subsequent parts of this book.

Cyber security

Autonomous systems tend to be very complex. For instance, an autonomous vehicle utilises numerous collaborating, electronic subsystems responsible for performing a variety of functions, supplied by different providers, and their communication (both with each other and with their surroundings) in a variety of ways. Such systems are naturally vulnerable to external attacks – it is therefore hardly surprising that another key group of technologies of fundamental importance to the solutions' development comprises a wide range of cybersecurity technologies. The impact of the same on corporate security management will be discussed in the subsequent section. Presented below is only a brief overview of selected providers offering this type of solutions, while one remains aware of the fact that the contemporary cybersecurity industry fully deserves a separate, dedicated analysis.

Autonomous vehicles are particularly susceptible to external attacks. Any attempt to take control over such a vehicle (e.g. while it is travelling at high speed) or any elements of the road infrastructure can pose a lethal threat to the affected road users. This fact creates a niche for companies such as Karamba Security, which not only develops solutions securing various vehicle subsystems against outside interference (Karamba promotes its offering under the slogan of *Self-Defending Cars*), but also aids automotive manufacturers

in identifying security gaps in their software and simplifying testing procedures (in terms of security). Similar solutions are provided by Saferide.io with the added functionality of systems monitoring the vehicle's internal communication, attack simulations, root cause and aftermath analyses for identified incidents (so-called digital forensics) as well as a 24/7 Cyber Defence Operation Centre services (monitoring and attack prevention). In turn, Trilium Secure specialises in securing (including cryptographically) the in-vehicle network (IVN) and its communication with other vehicles (the already mentioned V2X channels), while Regulus focuses on methods of preventing attacks involving disruption of geolocation signals from GNSS and GPS systems.

Digital attacks are a threat faced not only by vehicles but also any industrial systems utilizing e.g. IoT (Internet of Things) infrastructure, or simply corporate ICT systems. Companies such as NXM Labs or Aperio Systems specialise in securing such solutions as well as developing their own products utilizing advanced intelligent systems with varying levels of autonomy. Furthermore, the already mentioned Karamba Security also offers comprehensive security systems to prevent digital attacks against office infrastructure, with a particular focus on the identified weakest link of such systems (that most commonly allows hackers access to internal infrastructures), i.e. multi-purpose devices (e.g. printers).

The presented brief overview illustrates not only the various offerings available in this context but also the abundance of potential threats faced by vehicles as well as other industrial sensor solutions and corporate ICT systems.

Fleet Management and Coordination

The applications of autonomous systems in industry are usually not limited to implementations of individual devices. In many cases, companies utilise entire fleets of more or less autonomous drones, vehicles, robots, or other machines, which in turn necessitates efforts aimed not only at their overall maintenance, but also coordination. To this end, *fleet management* systems are being introduced, which nowadays typically constitute an integral part of the offerings marketed by the suppliers of autonomous solutions.

One such platform is offered by Auterion under the moniker of *The operating system for enterprise drones.* Auterion supports its customers in developing software capable of broadly understood coordination of a variety of autonomous vehicles (currently, mainly drones, although as of March 2020, also supporting ground vehicles) supplied by other manufacturers. In particular, devices operating under the control of such software are able to exchange (between each other as well as with a central control system) a variety of information, jointly perform pre-programmed missions, notify the need to perform service works or charge batteries[10], complete with system-wide attack protection and guaranteed compliance with aviation regulations.

A similar platform, albeit dedicated to robot management, is offered by Fetch Robotics. Adapted to managing logistical robots, the cloud computing-based Fetch Robotics platform facilitates fast (in a matter of hours rather than weeks) implementation of a robot fleet within a given space (warehouse or production line), gathering and sharing information between robots, programming robot behaviour and missions, as well as monitoring their operation. The approach significantly simplifies the implementation of the entire autonomous solution as well as its further development.

Another interesting example of a platform coordinating the operation of various autonomous solutions is the product offered by Unmanned Life. It facilitates the design, coordination, and supervision of missions performed by swarms of autonomous machines, not just drones but also vehicles or robots. The process requires efficient integration and interpretation of data obtained from sensors mounted on numerous devices (*sensor fusion*) as well as coordination of the tasks performed by multiple machines with the view to achieving a predetermined mission goal. Incidentally, the solution described by the company as **Autonomy-as-a-Service** is both network and

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[10] Dedicated platforms for drone charging constitute a separate element of infrastructure, marketed by companies such as e.g. Skysens.co. Clearly, drone managements is gradually developing into a separate industry sector in its own right.

hardware agnostic, it can be implemented both on-premise (client's servers) and in the cloud, it also supports a wide range of communication standards (e.g. Wi-Fi, LTE, 5G).

In the case of air traffic, both safety concerns and applicable air transport regulations necessitate an even greater level of strict coordination. This particular aspect is so interesting that it warrants a somewhat more in-depth consideration.

Unmanned Aircraft Systems (UAS) typically comprise unmanned aerial vehicles (UAV) controlled by operators from ground control stations (GCS), as well as systems facilitating the communication between the control station and the UAVs. A separate class of such systems is composed of solutions capable of controlling the traffic of unmanned vehicles (so-called UAS Traffic Management) which need to be integrated (with due consideration for any applicable legal regulations) with air traffic management systems (ATM). Once we add other market players to the equation, for instance supplementary data & service providers (SDSP) and UAS operators, we quickly end up with a truly complex ecosystem composed of not only a variety of companies and organisations involved but also a wide range of different ICT systems.

It is therefore hardly surprising that the same creates a niche for yet another class of solutions that facilitate the coordination of unmanned aerial traffic with the entire system of air traffic management. AirMap is one of the providers offering this type of solutions. The platform, addressed both to companies and air traffic coordinators, facilitates in particular the management of authorisations and air traffic (in real time), as well as remote identification and authorisation of drones. Additional benefits of the system include improved control, compliance with air traffic regulations, continuous operational capacity of the fleet, mission planning, as well as a package of developer functionalities necessary to ensure effective fleet performance. Interestingly, the AirMap platform takes advantage of the already popular digital-twin technology: a digital map of the air space updated in real time, which is crucial to the entire system's reliable operation.

The solutions described above are just examples of the observed trend wherein the main factor generating value for organisations is no longer its hardware but software, moreover not just that directly controlling a given machine but also that facilitating the coordination of the entire fleet.

Full-stack Development Platform

As illustrated above, the correct, effective and safe operation of an autonomous system involves a multitude of methods and technologies which not only need to be able to collaborate with each other but also provide the adequate level of security. The creators of such complete systems can luckily take advantage of products and services offered by a wide range of specialised providers, not to mention the dynamically growing availability of open-source programmer libraries.

However, the complexity of ecosystems can not only cause a significant headache just to take them in, but also generate a multitude of integration related problems: technologies continue to dynamically evolve, many standards are still in their infancy, not to even mention the constantly changing regulations imposing various security requirements. This boosts the development of the market for solutions facilitating the development of complete, comprehensive systems – which IT terminology often refers to as Full-Stack.

One supplier of such solutions in the context of autonomous car production is Apex.ai. the company's offer combines a practically complete architecture of vehicle software, including in particular:

1. The end product (**user application**)
2. **On-board software** facilitated by algorithms, supporting libraries, and the operating system
3. **Methodologies** necessary for the development of software, particularly in terms of **design, testing and security measures**
4. Creator software, necessary data (e.g. maps) complete with the data processing systems
5. And **hardware**, including sensors and mechanical devices.

An even more comprehensive solution (including sensor systems and algorithms responsible for image recognition, planning, control, and communication) addressed to autonomous vehicle designers is offered by DeepRoute.ai. Situational awareness is facilitated by a **sensor system** (8 cameras, 3 LiDARs, geolocation sensors (GNSS) with a module supporting data integration and telecommunications), **perception technologies**

(algorithms allowing effective image recognition in difficult conditions (storm, night, heavy traffic, roadblock), as well as high-resolution maps. The data is utilised by **planning and traffic control** algorithms, and the package is further complemented by the addition of a **simulator** which facilitates tests under laboratory conditions. Such comprehensive solutions make it possible for car manufacturers to independently develop new vehicles without the need to rely on additional integrators.

In turn, ASI Robots supports the developers of autonomous industrial machines by offering 3 integrated systems: Mobius (managing fleets of vehicles), Nav (supporting navigation, remote control, and autonomy functions), and Vantage (responsible for security). This example is particularly interesting due to the fact that it establishes a basis for an actual business platform supporting the development of multiple autonomous solutions, intended for a variety of industry sectors and marketed by dedicated, specialised providers.

In the sector of autonomous aerial vehicles, a similar comprehensive developer solution is offered by Near Earth Autonomy. The company's product utilises technologies supporting sensor operation, collision detection and prevention, landing zone evaluation, traffic planning, environment mapping, geolocation without the use of GPS systems, and visual pilot support, which jointly allow developers to create dedicated autonomous solutions for the aviation industry, develop custom applications, and ensure the quality and safety of their products.

Companies such as Apex.ai. DeepRoute.ai, Perron Robotics, ASI Robots, or Near Earth Autonomy offer not only complete packages of developer solutions but also systems facilitating the exchange of data generated by the vehicles with the cloud and each other, as well as their subsequent analysis. Consequently, designers can work on specific components of the software within a single, consistent ecosystem.

Applications of self-learning systems in key business processes

Asset Management

One of the main areas of application for the autonomous technologies presented in the previous chapter is asset management. Intelligent systems are used to monitor and secure buildings and machines, but also construction zones, airports, ports, or wind turbines. Robots with a varying degree of autonomy replace or aid humans working on the production lines. Presented below are some of the most interesting solutions considered in terms of their desirable properties, major advantages, as well as the challenges they create.

Autonomous drones equipped with various precision sensors (sometimes referred to as *autonomous flying sensor packages*) are among the most popular methods of monitoring asset levels. Drones offered by Percepto facilitate the generation of exact 3D models of e.g. mines or construction sites, not only for the purposes of creating maps but also tracking environmental changes, which may aid the assessment of construction progress or its consistency with contractual provisions and applicable laws, or simply securing property against damage and theft.

In turn, solutions offered by companies such as Skyspecs, Perceptual Robotics, or Nearlabs are used for automated monitoring of wind turbines. Due to safety concerns and considerable instability of the work environment (wind momentum), this task necessitates the development of dedicated drones and the specifics of the industry require training algorithms capable of recognizing malfunctions characteristic of wind turbines. Automated diagnostic solutions integrated with turbine monitoring systems ensure not only high responsiveness, but also reduction of employee hazard and costs, while also providing the framework for *predictive maintenance* systems.

Analysing large quantities of data gathered by drones is a significant enough challenge to stimulate the development of solutions dedicated specifically to this task (so-called drone

data analytics systems). For instance, 3DR offers a system facilitating the conversion of photographs taken by drones into high resolution orthoimages, point clouds, and 3D meshes. Based on those models, it also allows one to measure volumetrics, compare the same to plans, perform point cloud analyses to calculate distances, heights, angles and areas.

A separate category of systems includes solutions facilitating drone fleet management. For instance, Auterion – a provider of dedicated drone system control software – allows real-time monitoring of individual drones as well as entire fleets, which not only supports system optimisation but also increases its security and ensures compliance. Data concerning the condition and potential defects of the respective devices are aggregated and analysed as a foundation of the predictive maintenance service. In turn, the swarm management system offered by Accelerated Dynamics supports drone autonomous self-repair: devices with suspected defects or running low on energy are automatically returned to the base for maintenance.

Autonomous vehicles can also be used for the purposes of asset management. For instance, robots offered by Clearpath Robotics support remote inspection, monitoring, and infrastructure mapping, particularly in hazardous locations (e.g. condemned buildings), thus contributing to employee safety. Accelerated Dynamics also offers the ability to monitor croplands by using autonomous agricultural machinery equipped with dedicated measuring systems, which, combined with drone-based solutions, provides the foundation for agriculture intelligence systems.

Another noteworthy example of autonomous solutions used in the context of asset management relates to cleaning systems. Cleaning robots offered by companies such as Avidbots or Braincorp operate on production lines, in hospitals, hotels, large shopping malls, airports, and other pubic spaces. Apart from the most evident benefits of automation (such as the ability to work 24/7, elimination of the human factor and increased cleaning quality), when coupled with dedicated analytical systems such solutions support precision efficiency monitoring, which translates to more accurate cost planning and more precise pricing of future contracts (in highly competitive industries where margins are relatively low, pricing errors can have significant consequences – if

the price is too high - the contract may not be awarded, if too low – profits may suffer, even to the point of generating losses).

Autonomous systems are commonly used for the purposes of securing infrastructure. Drones supplied by Percepto, used for port and coast monitoring, facilitate loading operations monitoring (particularly in terms of contractual and legal compliance, as well as identification of unplanned cargo), container monitoring (in particular detection of humans inside the containers, which is an important factor of operational safety), crane inspection, vessel assistance, or simply coastal security monitoring. Meanwhile, in the oil-and-gas industry, such solutions can monitor the temperature of key components of safety-critical infrastructure or identify gas leaks along pipelines.

One should bear in mind that apart from protection and defence, drones may also be used for the purposes of attacking an infrastructure. This fact stimulates the development of systems dedicated to so-called anti-drone protection. Solutions such as those offered by Aeroscope, DJI, or GMP Defence which allow detection and neutralisation of undesired drones are increasingly commonly deployed as part of the protection systems securing airports, critical (e.g. energy) infrastructure, governmental buildings, military facilities, or prisons, as well as mass events. It should be noted at this point that the fast growing availability of drone technologies as well as advanced situational awareness and image recognition algorithms may, in the near future, lead to the emergence of an entirely new type of terrorist threats, and consequently pose a considerable challenge to security services.

Critical infrastructure must be protected not only against physical but also digital threats: the increasing complexity of IT systems deployed in industry (combining transactional systems, sensors of the internet of things, mobile devices and applications, multimedia systems, and many others) adds to their vulnerability to external attacks.

Solutions offered by companies such as Karamba Security or Aperio Systems are used to protect buildings, industrial infrastructures or automated vehicles against cyberattacks. For instance, as follows from the analyses conducted by Karamba Security, one of the weakest points in corporate information security systems are multi-function printers which constitute a relatively easily breachable element of IT infrastructures. While such

a breach most usually results in damage related to information security (theft of sensitive data, employee details, etc.), the ability to take control over a vehicle (e.g. by attacking the multimedia system which also constitutes one of standard weak points) may have far more tragic consequences.

Anomaly detection algorithms are often used as part of attack protection systems: by identifying the typical characteristics of signals reaching given devices, such algorithms automatically block any atypical incoming communication. This is one of the most commonly employed methods, fairly effective in the face of constantly evolving cyberattack strategies. It is noteworthy that apart from the obvious advantages of improved security, a key benefit of increased attack resistance for production companies stems from the ability to minimise machine downtime and the related operational continuity disruptions.

Another possible type of digital attacks against autonomous systems, particularly in the context of transport, involves disruption of GNSS (Global Navigation Satellite System) signals. Regulus is a company specializing in systems dedicated to monitoring and securing GPS receivers against this type of attacks (spoofing). Such protection is critically important not only in the context of autonomous vehicles but also in all forms of air and maritime transport.

Autonomous systems are also used to increase personnel safety. Clearpath Robotics products can replace humans in tasks involving e.g. handling of explosives, while Exyn drones facilitate mapping and monitoring of inaccessible areas, e.g. mines. As suggested by Percepto based on their previous experiences, continuous monitoring of construction zones or areas of planned detonations (e.g. to identify people present in the area), reduces the risk of hazardous incidents by 91%.

Similarly, the already mentioned Nauto system monitoring driver condition and vehicle surroundings significantly increases the driver's safety not only while traveling as such but also in a more general context (thanks to in-depth behavioural and accident analytics coupled with a personalised training system).

Monitoring systems are also employed for the purposes of environmental protection. Information gathered from autonomous drones mat be used to monitor the compliance of hazardous substances production with applicable regulations and contractual provisions (which is particularly important in developing countries where one of the main sources of smog is the illegal use of prohibited fuels for heating), as well as identification of uncontrolled leaks and harmful emissions. In turn, precision agriculture systems offered by e.g. Eco Robotics can optimise the use of spraying and facilitate the reduction (by even 95%) of harmful emissions affecting not just the environment but also in the final food produce. Another solution in the same area is the system provided by AMP Robotics which uses advanced image recognition algorithms to automatize the processes of recycling for the purposes of managing municipal, electronic, or construction waste by improving sorting, lowering costs, and improving employee safety (through elimination of manual sorting) and productivity. Such systems are relatively easy and quick to implement.

Autonomous systems can also be used in quality management. An interesting example comes in the form of the solution offered by AgShift which uses machine learning algorithms to analyse food quality based on appearance, shape, size, and weight. It reduces the duration of inspection, incidence of errors, and helps to standardise the overall quality control process. In turn, inspection drones sold by Exyn aid the assessment of construction completeness and consistency with the design, which facilitates early identification of defects and reduces the costs of remedial works.

Manufacturing

The production line is typically one of the most automated areas in a modern factory. Autonomous (to a lesser or greater extent) robots can perform complex tasks faster and more accurately than humans, thus improving processual efficiency and safety of production. Presented below are several key features of autonomous solutions employed in modern factories, with a particular focus on so-called collaborative robots, i.e. robots cooperating with humans.

Autonomous production systems may replace humans entirely (a fully automated production line) or support their work. An important factor in the context of this distinction is safety: when the full potential of a production robot is unleashed (most importantly in terms of power and speed), significant employee hazards may emerge.

In practical applications, the spectrum of possible forms of collaboration between people and machines in the context of a production line is considerably broader. Kuka distinguishes 6 levels of such interaction:

1. Fixed Safety Fence. In this configuration, the person (operator) is completely separated from the machine, e.g. by a solid, steel fence. Consequently, the robots can work faster, use greater forces or methods that would be potentially dangerous to humans, e.g. welding. This mode is commonly used for a variety of activities performed e.g. in automated vehicle factories.
2. Occasionally shared workspace. A person can safely stop the machine and then enter its workspace to e.g. replace parts, pallets, etc. The machine remains energised but is disabled when the operator approaches it.
3. Virtual separation. The collaboration space is partially separated but in such a way so as not to hinder the operator's contact with the machine (e.g. for loading/unloading). The machine's speed is controlled in a function of its distance from the personnel. Direct contact is only possible when the machine is stopped.
4. Shared space, without virtual separation. The person works together with the machine in a continuous manner, but direct contact is not necessary. The force of movement (e.g. machine arms) is limited so that it poses no danger to humans.
5. The person and the machine must collaborate in order to complete a given planned task in the process of e.g. teaching the machine how to perform precise movement on the production line (so-called imitation learning). The power and speed of the machine are limited, contact with humans is desirable.
6. Autonomous collaborating robots. People move around among the robots in a relatively large space, e.g. in a warehouse environment. The power and speed of the robots are limited. Collaboration takes place in numerous places and is not limited to a single workstation.

It is noteworthy at this point that developing a robot operating in close proximity to humans (levels 4-6 of the classification) poses a considerable challenge. Robots separated from people (levels 1-3) typically function in a relatively controlled, defined and deterministic environment (with the possible exception of logistics robots), and are subject to significantly fewer safety requirements (no contact with humans). Meanwhile, collaborative robots must not only perform often very complex operations within unstable environments, but they must also do so without posing a threat to humans. At the same time, however, robots of this type can significantly improve the flexibility and efficiency of a production line. Hence, companies such as Kuka, Fanuc [@zotero-1571], Universal Robots, Rethink Robotics [@zotero-1573], NextAge [@zotero-1575], Pi4 [@zotero-1577], Onexia, Yaskawa, Nachi, and ABB have been developing and marketing this type of solutions for many years.

For instance, the key features of the Sawyer robot – the flagship product of Rethink Robotics – include ease of implementation (due to software integration with the robot's mechanics, using the methods of imitation learning), flexibility (possible thanks to its advanced visual perception and situational awareness system – Robot Positioning System, and 7 levels of freedom of the mechanic arm), and safety (due to the advanced pressure sensors and torque control systems).

Similarly to Rethink Robotics, Faunc emphasises such qualities of its products as reliability (due to the advanced sensor system), simplicity of learning (understood as the ability to easily learn the ins and outs of a new workstation), ability to recognise various objects, possibility of implementation in a variety of locations and using diverse methods, small size and safety (thanks to pressure sensors and motion controls). Universal Robots makes very similar claims, while also emphasizing the wide spectrum of potential applications including pick-and-place, injection moulding, CNC processing, packaging and palletizing, quality inspection, assembly, polishing, machine tending, screw driving, lab analysis and testing, gluing, dispensing, and even welding.

One should also mention the co-robot offered by Pi4, whose distinctive feature is an animated, highly expressive face projected on a tablet screen, which creates the interesting impression of interacting with an actual, friendly person…

To recap, the key features of modern collaborative robots include:

1. Simplicity of implementation and adaptation to work in a given workstation: easy installation, configuration, and quick learning capabilities;
2. Reliability, not just mechanical but also related to the use of advanced sensor, perception, and situational awareness technologies;
3. Flexibility, accomplished thanks to multiple levels of arm freedom, ease of installation in a variety of places, and relatively small size;
4. Employee safety provided by the systems controlling the power and speed of operation and the ability to handle potentially hazardous materials and tools.

The main advantages of such solutions include increased employee efficiency, higher precision, fewer errors, and improved workplace safety.

Robots can also support humans in spaces other than industrial floors, e.g. at construction sites or in mines. ASI Robotics, one of the suppliers of such solutions, offers its clients a very interesting model of generating value. It facilitates the calculation of additional benefits generated by the subsequent implemented autonomous systems in the context of reduced operating risk, improved productivity, and increased costs (related to the purchase and/or depreciation of equipment). For instance, in the context of the mining industry, such sub-systems include e.g. autonomous drilling, blasting assist, dozer assist, loader assist, and transport, ultimately also automated. Each of said sub-systems comprises numerous components that generate additional layers of value in the aforementioned dimensions.

The primary benefit emphasised by the providers of autonomous solutions supporting manufacturing processes is, quite predictably, enhancement of production efficiency. This pertains not only to production of physical goods but also, increasingly, software, particularly of the intelligent type. Neurala, the provider of a system supporting the creation of image recognition algorithms employed for the purposes of remote inspection,

claims to improve the rate of generating such models by up to 90%. In turn, Cognata, supporting the creators of ASDAS (Advanced Driver Assistance Systems), provides realistic simulation data for the purposes of training specific intelligent systems, including ready-made scenarios of traffic situations and systems for measuring algorithm efficiency, which greatly facilitate the development processes.

Optimisation yields the direct effect of reduced production costs. It is not, however, the only source of savings available: one should also consider the increasingly common, innovative method of investment financing based on the Robot-as-a-Service (RaaS) approach. For instance, Locus Robotics offers its devices under a subscription model, which allows the classification of respective costs to be shifted from investment (CAPEX) to operations (OPEX). In effect, this boils down to minimizing initial costs and therefore precipitating return on investment (ROI), reduces the burden of solution exit barriers, and at the end of the day, shifts focus in automation processes to the problem of hiring a team of adequately qualified employees.

Logistics

Logistics and transport are the areas where the use of autonomous systems is possibly the most widespread. This is the reason why a considerable portion of this book will be devoted specifically to this sphere, presenting the range of applications currently prevalent in the context of warehousing, logistics, manufacturing, transport, and deliveries.

Warehouse and manufacturing logistics

In warehouse and production logistics, autonomous systems are employed primarily with the view to monitoring and handling goods.

In monitoring product stocks and locations, technologies allowing detection and image recognition are paramount. LiDAR technologies have proven themselves to be highly effective in detecting the location of goods within a warehouse. Blickfield mentions product tracking and the related optimisation of the supply chain and reduction of logistic costs facilitated by predictive planning as one of the key applications of their sensors. In

turn, drone systems offered by Enyx support automatic stock taking in large warehouse facilities: drones operating autonomously or as part of a coordinated swarm analyse product images and forward the corresponding data to the analytical systems. In effect, warehouse monitoring becomes cheaper, can be conducted on a continuous basis and provides better data quality.

A similar, drone based solution, albeit dedicated to large commercial spaces, is offered by Pensa Systems. Images provided by dedicated drones are forwarded to the analytical system which continuously analyses product availability on the shelves and recommends, in near real-time, adequate action to be taken by the stock management personnel. As such, the system works to reduce stock shortages (stock-out detection can reach up to 98%), streamline stock planning, analyse the compliance of product positioning with regulations and contractual obligations, and consequently improve profitability.

A more advanced solution utilizing autonomous robots, rather than drones, travelling across large commercial areas, is provided by Simba Robotics. Similarly to the Pensa Systems product, images are forwarded to the analytical centre where they are analysed in detail, with the results of such analyses later used for a variety of purposes and services. Some of the most interesting ones include intelligent inventory management, out-of-stock management, planogram compliance, pricing and promo execution. Intelligent inventory management allows identification of items placed in a wrong place, empty shelves – with indication where the corresponding goods are located and how to efficiently deliver them to their destination, recommendation of optimum stock sizes and identification of the places most susceptible to theft. In turn, the out-of-stock management system informs the staff on the locations of product shortages, recommends replacement goods if the primary product is out of stock, and on top of that provides additional trend analyses and demand forecasts. The planogram compliance function helps to identify the locations where products sell the best, optimise the time of shelf restocking and product priorities. It may serve as the basis for A/B tests conducted e.g. to verify hypotheses regarding optimum product distribution throughout the store. The service is complemented by pricing and promo execution functionalities, which allow precise identification of current product

prices (the robot scans price tags and bar codes) and juxtaposition of that information with the terms of reference adopted under ongoing promotional campaigns.

As follows from the above, Simba Robotics utilises images and localisation data provided by autonomous robots to enable a complex system of services supporting sales outlets in monitoring and optimising their commercial activity. At this point, the system is primarily used by shop owners, but it is easily conceivable that the solution offered by Simba Robotics may serve as an information platform useful in a variety of different applications: e.g. by goods suppliers, market research or security companies. Consequently, the business potential of such monitoring technologies is extensive.

Another task for which autonomous technologies are employed involves handling goods in warehouse or production line environments. The most typical scenarios for the use of autonomous robots in this context (cf. e.g. [@2019]) include Goods-to-Person, Person-to-Goods, and On-Demand models.

In the Goods-to-Person model, logistics robots transport goods from their place of storage to a central commissioning area where employees unpack specific products in accordance with client orders. After the needed goods have ben removed, the robot returns the pallet to its designated location in the warehouse. This approach is employed by companies looking to automatize their warehousing logistics while avoiding the need for massive investments necessary to implement fully automated warehouse solutions (e.g. carousels or shuttle systems). The benefits of this scenario stem primarily from improved volume throughput and the resulting reduction of order lead times – and all that thanks to simply reducing the need for employees to traverse the warehouse space.

In the Person-to-Goods model, employees travel across the warehouse and collect goods necessary to process an order, supported by autonomous robots operating in one of 3 primary modes: Lead me, Follow me, and Swarm me.

In the 'Lead me' mode, the robot receives order details from the Warehouse Management System (WMS) and then guides the employee from one location to another as the device's terminal displays the details of goods that need to be collected from each spot. The employee removes the item from the shelf, places it in the robot and proceeds to the next

location. The main benefit stems from reducing the order processing time (the employee's travel time in the warehouse) by optimising the route and delivering the complete order to the final pickup point (which the robot completes on its own), reducing the employees' workload (no need to carry the goods by themselves), and increasing accuracy (lower risk of missing one of the ordered items).

In the 'Follow me' mode, the robot follows the warehouse employee from one location to another, guided only by its sensors and cameras (therefore, it does not need to be integrated with the WMS, it also does not follow a fixed route). Thee employee assembles the order and sends the robot to the designated pickup point (e.g. commissioning centre). This solution is preferred by companies that do not wish to integrate robots with other logistics systems. The main benefit here stems from reducing the employees' workload and lead times due to the automatization of the final part of the route: once the order has been compiled, the robot independently transports it to the designated place.

The 'Swarm me' mode is usually implemented in particularly large warehouse facilities where respective employees are dedicated to processing strictly designated areas. In the case of orders that require items to be retrieved from a number of such work areas, a robot, having received detailed instructions from the WMS, approaches an employee in a given area, accompany them in gathering the necessary goods, and independently travel to the next area (where it interacts with another employee). Having compiled the order, it then travels do a predesignated delivery point. This solution is employed by companies interested in accelerating order processing activities in large warehousing systems composed of specialised areas (employees work in respective areas while robots travel between the same).

The final mode is referred to as 'On demand'. Here, logistics robots transport pallets from one designated location to another with the transport commands generated either directly by warehouse employees or by the WMS. The principal benefit of the solution, similarly to the previous cases, stems from time savings and reduced employee workloads.

There are numerous vendors marketing systems operating under the aforementioned modes. Autonomous logistics robots offered by Aethon find applications in the health

industry (patient care and board, pharmacies, laboratories, waste removal, or laundry transport), hospitality (management of hotels, mass events, banquets, conferences), warehousing and manufacturing. The benefits extend beyond mere time savings for the employees, including also workload reductions (e.g. handling heavy objects) and safety improvements (particularly in hospitals, where care must be taken to secure various hazardous substances or fragile items). This wide range of applications often necessitates narrow specialisation of the robots used, both in terms of hardware (adapted to the specific requirements of a given industry) and software (dedicated image recognition and situational awareness algorithms).

A similar system of autonomous delivery robots is offered by Savioke. Based on the company's promotional information, the key features of the solution include reliability (ability to operate effectively under a variety of conditions, durability and long battery life) and an advanced analytical system allowing in-depth efficiency measurements. One particularly interesting solution facilitates the delivery of various products (e.g. documents, food, or stationary) to various levels of tall buildings, which can greatly improve the productivity of office workers. The benefits include increased efficiency and quality of the production process, just-in-time delivery, and reduced workload of valuable employees.

Locus Robotics, a company that also specialises in the supply of vertical, autonomous robots supporting employees in handling goods in warehouses and processing stations, describes the main advantages of its solutions by referring to integration with other systems within the organisation (in particularly warehouse management systems), scalability (which is significant in the context of subsequent implementation roll-outs or in the event of business expansion), minimum impact of the investment on other solutions employed in by the company (which reduces implementation barriers), ease of implementation (not only in terms of the infrastructure but also personnel training: Locus Robotics devices are user-friendly and intuitive). In effect, companies employing the system may improve the productivity of their workforce, efficiency of production processes (e.g. by increasing throughput) and reduce operating costs.

In turn, Gideon Bros supply robotic systems take the form of flat platforms allowing autonomous transport of relatively heavy objects (e.g. stacks of pallets) in sometimes difficult conditions (e.g. low temperatures in warehouses or production floors). Such systems rely heavily not only on proper algorithms (recognition of goods or locations) but also highly durable mechanical components.

When choosing the optimum robot for a given warehouse, Gideon Bros recommends referring to the following criteria [@2019a] (although specific to logistics robots, the list could easily be generalised to include other classes of solutions):

1. Navigation
   - How does the navigation work?
   - How are facilities maps created?
   - What if the robot enters a busy area? What happens if there is an obstacle on the route?
   - How does robot avoid crashing into low lying or negative obstacles?
2. Load and Operation
   - What are the maximum dimensions of the load and a maximum load itself?
   - What is the maximum speed? What is the lifting time?
3. Charging
   - How does the charging work? What are the technical requirements for charging stations?
   - What is the charging downtime, and how can it be minimised?
4. Warehouse infrastructure
   - In what sort of warehouses can robots work?
   - What are the environmental requirements, including requirements for the condition of the floor? Which areas must be specially marked or reserved?
   - What is the minimum manoeuvring space?
   - Is there additional infrastructure or equipment that must be installed?
5. IT infrastructure
   - What are the requirements for IT infrastructure?
   - What is the robot interface? How is the robot operated?

6. Safety
    - Are there use restrictions around forklifts and people?
    - How do the safety features protect humans and infrastructure?
7. Use Cases and Efficiency
    - What are the ideal/possible use case scenarios?
    - What efficiency improvements can be expected?
8. Support
    - How is the support organised during deployment?
    - What constitutes normal wear and tear of the robot?
    - What are the options regarding finance and insurance?

It would be prudent to mention at this point the company's recommended model for calculating return on investment for autonomous logistics systems. Among the key benefits listed therein, one could mention:

1. Reduced costs of operator labour. A variable cost calculated on a monthly basis as the product of the number of hours saved and the hourly rate applicable.
2. Reduced turnover of warehouse staff. The shortage of employees willing to do manual labour is a problem increasingly affecting not just by the suppliers but also potential buyers of goods.
3. Elimination of replacement costs. Real savings when buying the solution in the subscription model (RaaS: Robot-as-a-Service)
4. Revenue from selling the equipment replaced by the robots. In the case of the subscription model (as above).
5. Improved efficiency of storage space use. Flat robots are more agile than e.g. traditional forklift trucks and consequently require less space to manoeuvre effectively. This frees up additional storage space, which adds to the overall savings generated.
6. Savings of time and money related to reducing downtimes, both for the equipment and human staff members. For the machines, this means primarily failure downtime, whereas for people: time related to absenteeism (sick leaves, holidays, training, other travel) and turnover.

Logistic platform systems similar to those offered by Gideon Bros are also provided by Otto Motors, who mention additional implementation benefits such as safety improvements and possibility of standardising logistical processes across a number of locations. Standardisation is made viable in particular thanks to the system of robot fleet coordination constituting an integral part of the solution: it allows system configurations to be adapted to a given factory and integrated with other dedicated organisation systems (in particular the warehouse management system), facilitates effective work scheduling and coordination of robot activity. The end result is an autonomous system for transporting goods within the factory, integrated with the customer's other existing subsystems.

Another interesting solution is offered by IAM Robotics. The system features fulfilment solutions dedicated to industries such as e-commerce, pharmacy, or grocery. Autonomous logistics robots aid staff members in the processes of packing orders and forwarding them for pickup by transport services. Similarly to the systems described above, the main benefits of the product include time and money savings as well as enhanced security. What sets this solution apart is the fact that it is specifically dedicated to fulfilment in the context of e-commerce.

Finally, one should also mention the product marketed by iFollow. It employs image recognition technologies (including wide angle stereoscopic cameras), long distance sensors facilitating effective reaction to difficult conditions, and navigational algorithms, which allows the development of collaborative warehouse robots. Similarly to the examples described above, the main advantages of the solution include improvement of work conditions, increased productivity, reduced incidence of errors, and ease of implementation (in particular due to the user-friendly interface).

Why do companies increasingly opt for the implementation of autonomous solutions in logistics? According to Gideon Bros, the most common reasons include:

1. Growing difficulty of recruiting employees now observed not only in the developed but also developing countries. Warehouse work is typically poorly paid, difficult, monotonous, and at times dangerous.
2. High employee turnover due to the above.

3. Forklift trucks can be hazardous, often causing occupational accidents.
4. Customers increasingly expect shorter delivery times, which forces companies to strive for greater logistic efficiency.
5. The limits of warehouse streamlining possible thanks to computerisation (WMS) have already been reached by many companies: consequently, the only way to further improve processes is at the level of operations as such, where human work capacity proves to be the main limiting factor.

Considering the above, it is hardly surprising that the market of autonomous logistics robots is currently on a very dynamic rise: as predicted by the Tractica advisory company [@zotero-1581], the sales in this segment are expected to increase to even 200,000 robots a year by 2018 (corresponding to a revenue of approx. $8.3 bn) and up to a million by 2022 ($30.8 bn in revenue).

It is also noteworthy that from the qualitative perspective, the aforementioned factors (recruitment problems, effectiveness and safety) are relatively universal: they do not affect logistics exclusively but also other areas of business in a contemporary organisation.

Marketing and Sales

A growing number of companies are currently shifting the focus in their communication with present and prospective customers to electronic channels. This is observed in the context of virtually the entire spectrum of possible interactions: from online marketing (using email, social media, or radio/TV), through contact initiation and sales processes as such (telephone, online, teleconferencing), to after-sales and repair services.

A great deal of information is generated in those processes: both in the form of user behaviour logs (e.g. through websites or mobile applications), registered conversations (email, social media or mobile application exchanges, registered telephone calls, etc.), and as various metadata (contextual such as time, place, client ID, responding employee's ID, etc.).

Artificial intelligence algorithms used for the purposes of data analyses in marketing and sales processes are used primarily as means of personalizing the offer, automatizing customer interactions, and forecasting demand. The solutions used in this context include "classic' methods of classification, regression or segmentation, as well as more advanced systems of natural language processing and generation (in customer interactions) or reinforcement learning (optimisation of online communication, programmatic marketing).

Let us begin our analysis of the potential offered by autonomous systems in marketing by approaching the aspect of personalisation. The solutions offered by Reflektion are addressed to companies operating in the e-commerce industry. In the first stage, a product intelligence solution is offered which facilitates offer indexing (with key features extraction efficiency of up to 1 million products per minute). Next, individual shopper insights are created based on their behaviour within the electronic channels. The product intelligence and individual shopper insights elements provide the basis for the subsequent personalisation of the products and services offered: advanced machine learning algorithms generate precise, real-time product recommendations adapted to specific customers. The key benefits of the system include the ability to identify the customers' shopping intentions in real time, precisely personalise individual offers (both in terms of the products included and the presentation as such), and the resulting, tangible business advantages (better customer satisfaction and engagement, increased sales and margin levels).

In turn, Persado utilises advanced sentiment analysis algorithms to optimise marketing communication. Based on a database of over a million classified marketing messages (in this case, the classification entails associating a given phrase with the possible range of emotional responses), it aids its clients in the creative endeavours aimed at triggering a desirable response in recipients of messages. The tools used to this end include natural language generation algorithms as well as advanced self-learning mechanisms: the system continuously tests various iterations, analyses user responses, and updates the relevant models.

Albert.ai offers a comprehensive solution addressed to companies running their own online marketing campaigns. The e-marketing sector can be overwhelmingly complex these days (cf. e.g. https://chiefmartec.com/2019/04/marketing-technology-landscape-supergraphic-2019/ where over 7,000 different companies and products have been listed), which creates a market for solutions that allow simplification, automatization and optimisation of marketing endeavours. The Albert.ai system facilitates planning, launching, optimisation, and evaluation on-line campaigns. Among the benefits advertised to potential clients, one might list considerable cost reductions (cost per action (CPA), cost per lead, cost of customer acquisition, cross-channel costs), increased conversion from social media, and consequently higher return on advertising expenditures (measured by the ROAS (return on ad spent) index). The autonomy of the Albert.ai solution has been achieved by combining over 200 distinct intelligent skills employing a variety of artificial intelligence algorithms.

A relatively new addition to the sales landscape comes in the form of autonomous retail outlets or even entire autonomous stores. Such solutions utilise advanced image recognition, customer behaviour monitoring, sensor fusion, and predictive systems. For instance, the autonomous vending machine offered by Bodega is capable of learning customer preferences in a given location (e.g. an office, fitness club, hotel, or residential building), forecasting the demand for various products, and independently ordering deliveries. This provides customers with a convenient, cashless access to most goods needed on a day-to-day basis.

A similar but larger scale solution is employed by Amazon stores (over 20 locations as at February 2020) advertised under the Amazon Go brand. After entering the store and activating a mobile application, customers can simply pick up products directly from the shelves and the Amazon Go system automatically updates their virtual shopping cart accordingly. Once the customer exits the store, the applicable bill is calculated automatically and charged against the client's registered credit card. A similar solution enhancing the "autonomy' of traditional stores is also offered by Inokyo.

Popularisation of autonomous sales outlets seems to be only a matter of time (unless this development is somehow limited by regulators): the key technologies required already

exist and the costs of sensor implementation are relatively low, which renders such systems easily scalable. The direct consequences of implementation will likely entail significantly reduced employment (especially checkout assistants). One should, however, also factor in another aspect of the situation: suppliers of such solutions will gain access to detailed information regarding the shopping habits of millions of customers, which in the case of Amazon – where this can be coupled with data generated from its massive on-line sales – will further improve their already considerable market advantage. This fact adds a qualitatively new business justification to investment in this type of projects.

Customer Relationship Management

As mentioned above, technologies facilitating natural language processing (NLP), interpretation of large amounts of human-generated messages (so-called conversation intelligence), and natural language generation (NLG) are increasingly used for the purposes of automatizing and optimising marketing, sales and customer service operations. Similarly to perception and image recognition, also in this case advanced language recognition and analysis techniques are employed. However, a new quality in this context stems from the need to 1. account for two distinct modalities (audio and written text), 2. develop text generation technologies (e.g. to reply to questions or verbalise the conclusions reached in the course of analyses), and 3. communicate such content to the user (also within two distinct modalities).

Information regarding user interactions (e.g. registered conversations) can (potentially) have a considerable business value which regrettably remains largely unexploited at this time. There are several reasons why that is the case. Firstly, organisations do not seem to be aware of the value that can be generated by archiving interactions and conversations. Secondly, they are unfamiliar with the technologies that would allow them to do so. And thirdly, even if such technologies are available for the most popular languages (e.g. English, Spanish, Chinese), they usually still do not support many less popular ones. Nonetheless, the dynamics observed in the development of analytic methods used in marketing and sales is so high, and the potential benefits so promising, that it would be prudent to further examine the possible applications of said technologies in marketing and customer relations.

One of the most interesting, comprehensive solutions in this context is currently offered by Chorus.ai and marketed under the name of Conversational Intelligence Platform. Its basis is constituted by specialised speech-to-text algorithms, which allow subsequent tagging of individual conversation fragments. For instance, a telephone interaction lasting several dozen minutes will be transcribed and then key parts of the exchange will be assigned with appropriate key words (e.g. a quote request) as well as designations referring to the type and intensity of the interlocutor's emotions (e.g. client's annoyance). Once the conversation is completed, the data will be integrated with information from the CRM (customer relationship management) system, which will later facilitate identifying a correlation between the specific dynamics and outcome of the interaction. Due to these capabilities, Chorus.ai is increasingly used for the purposes of developing good practices applicable to handling customer exchanges (so-called playbooks), performing fast and efficient meeting reviews, as well as in-depth analyses of sales processes. The key benefits identified by the users include improved effectiveness of sales procedures, automatization of knowledge and good practices sharing, very good sales analytics, and increased productivity.

A similar product offered by Remesh.ai facilitates real-time analysis of conversations held simultaneously with a number of customers. Additionally, the solution supports employees in the course of the exchanges as such by recommending optimum interaction scenarios. This noticeably improves the effectiveness of sales processes, not only thanks to the recommendations provided but also due to the ability to hold several parallel conversations simultaneously. The system finds applications not only in the context of traditional marketing (better insight) but also e.g. politics (voter preference analyses, opinion and voting polls, as well as real-time social dialogue).

In turn, the ability to analyse customer exchanges provides the basis for the development of dialogue systems: autonomous solutions capable of engaging in both textual and audio interactions. A typical architecture of such a system is composed of subsystems (cf. e.g. Conversica.com) responsible for **natural language processing** (with modules such as speech-to-text, **entity** extraction), intent classification and ontology, self-learning, decision and policy management, and natural language generation (including dynamic

response generation, action recommendations, and alerting). Systems responsible for response analysis and generation are integrated with electronic communication channels (e.g. email, text messaging), corporate knowledge management systems (e.g. knowledge bases, training systems, or business intelligence/data science), and transactional systems (e.g. Customer Relationship Management: CRM).

Dialogue systems such as those offered by Conversica, LivePerson, DigitalGenius, or Automat.ai are usually employed to streamline marketing, sales, and aftersales processes. These activities are complex by nature: customer reports require verification of information against data available from numerous subsystems (from technical documentation, through transactional systems, to the history of interactions with the given client), competence in numerous areas, not to mention such aspects as empathy or simply patience. Digital Genius systems are capable of correctly identifying the intentions and needs of customers, quickly searching for and integrating information available from various sources, generating the optimum response complete with reliability assessment, and either providing the response directly or forwarding the recommendation/report to a person operating the helpdesk.

It is noteworthy that dialogue solutions can be employed not only for the purposes of aftersales services, but also – as is the case with Automar.ai – automatization of marketing and sales processes. This creates a niche for an entirely new class of marketing products such as dialogue-based advertising or virtual sales assistants.

The most commonly identified benefits of such solutions include reduced employee workload and the related savings, automatic generation of good practices, in-depth analytics enhanced by the inclusion of machine learning-based analyses, personalised communication and the resulting improvement of customer satisfaction.

Business Intelligence and Cognitive Automation

Autonomous technologies are also employed in the areas of Business Intelligence, human resources and knowledge management. Such solutions employ not only advanced expert systems (to facilitate data-based inference) but also e.g. natural language generation techniques. Presented below are several of the most interesting applications. One has to

bear in mind, however, that due to the limitations of the present format, the real potential of such technologies could hardly be presented in full.

Natural Language Generation is a group of technologies facilitating the creation of natural language narrations based on "raw' data and information. Such techniques are employed not only in automated conversations but also in the context of business intelligence and decision support systems.

For instance, the Quill system offered by Narrative Sciences is capable of generating conclusions by analysing data provided in the form of narrations utilizing the natural language. The user inputs e.g. a spreadsheet containing "raw' data, the system processes and analyses said data, and then communicates its conclusions in the natural language (rather than e.g. in the form of graphs or tables). It is an example of a trend sometimes referred to as human augmentation whereby artificial intelligence boosts the potential of a human user (rather than replacing them). Similar solutions are used by Google in many of its applications (e.g. Google Sheets assistant), or by Grammarly in its systems facilitating stylistic enhancement of English language texts.

A far more advanced solution is offered by Aera Technology. It serves as a perfect illustration for the new class of systems providing decision-making support, so-called Cognitive Automation. Due to its highly innovative character, this product deserves a somewhat more in-depth description.

Aera Technology supports mid-level managers in strategic decision-making. The interface is reminiscent of a typical dialogue system: the user (e.g. a manager) asks a question, and the system answers by providing either information with adequate interpretation, or a recommendation for a subsequent action. The respective groups of skills correspond to the key functional areas of an organisation. In each of such areas, the system is capable of replying to certain questions and recommending certain actions. In order for this to be not only possible but also reliable and effective, Aera Technologies' product utilises tools such as data crawlers (gathering and indexing data from a variety of transactional systems), processing engines (cleaning, classifying and integrating data within a single information model, identifying knowledge structures with the use of methods such as knowledge graphs), analytics engines (analysing data from the

information model with the use of area-specific knowledge and advanced in-memory data processing mechanisms), and co-called skill builders (tools that allow users to develop custom, company-specific system skills).

It is difficult to predict the impact such solutions may have on the way that organisations will be managed in the future. As follows from the above, such systems are able to not only analyse information but also recommend specific decisions. If integrated with transactional systems and provided with adequate competences (i.e. authorisations), this may gradually lead to the emergence of 'self-driven enterprise' solutions. Clearly, the respective elements of the necessary infrastructure are not only already being developed but indeed also commercially marketed.

Recruitment

An interesting application of conversation technologies (specifically chat-bots) can be found in the context of recruitment. Autonomous solutions offered by Impress.ai: 1. integrate with the leading recruitment systems (so-called Applicant Tracking Systems: ATS), social media platforms (e.g. LinkedIn), video-based recruitment tools, and competence or psychometry assessment systems; 2. conduct job interviews providing real-time feedback to applicants; and 3. support candidates in the subsequent stages of the recruitment process. They can simulate various decision-making problems in the course of the conversation, generate applicants' competence and psychological profiles based on the responses provided, and evaluate their suitability for tasks inherent in a given job. The system is not only autonomous but also self-learning: it continuously develops its algorithms based on the feedback received and the decisions made by the recruitment team. Consequently, companies employing Impress.ai solutions are able to increase the scope of their recruitment campaigns even 5-fold, analyse over 100 recruitment interviews in the course of only 15 minutes, reduce the average recruitment time by up to 80%, and overall employee turn-around time by over 70%.

A somewhat different approach to streamlining recruitment processes has been adopted by Riminder. Instead of bots, the company takes advantage of algorithms processing natural language to analyse applicant profiles based on the documents they provide, thus automatizing the painstaking and complex process of analysing CVs and job applications.

The system automatically identifies and classifies text blocks such as personal information, education and experience, compares the same to reference data, and recommends the best candidates to recruitment teams. The benefits include considerable acceleration of recruitment processes, more accurate identification of the best candidates, and reduced overall costs of the entire process.

Information and Communication Technologies

The first section of this chapter discussed the basic technologies employed in autonomous systems. Here, let us focus on the ways in which intelligent solutions may help organisations to develop and manage their ICT infrastructures.

ICT infrastructure: processors, cloud computing and telecommunications

The first example, a processor manufacturer AlphaICs, in a way occupies the middle ground between the two areas discussed above: on the one hand, it supplies highly efficient processors dedicated to artificial intelligence applications, and on the other, it utilises reinforcement learning methods as a core component of the processors it produces. AlphaICs offers highly efficient processors equipped with dedicated compilers, libraries, and developer environments for application creators. This type of processors is commonly employed in data centres (for streaming data analysis, search acceleration, high-efficiency calculations, and business analytics), autonomous systems (vehicles, drones, robots, or industrial automation), as well as edge computing (dialogue systems, extended, virtual and mixed reality applications, surveillance cameras, or intelligent mobile devices). On the other hand, however, AlphaICs engineers elected to base the architecture of their processors on 'agents': basic computer units, understood as autonomous units solving specific computing problems, capable of communication and interaction in varying environments. The agents collaborate with each other (cf. using the Multi-Agent Reinforcement Learning methods described in the previous chapter) in order to jointly achieve the assigned goals. Consequently, we are therefore dealing with a

technology utilizing autonomous systems (MARL) to create solutions designed to facilitate other autonomous products.

Artificial intelligence systems can also be used for the purposes of reducing energy consumption in data centres. As early as in 2016, DeepMind was able to reduce the costs of Google data centres' cooling by 40% [@zotero-1568], and in 2018, to implement an autonomous system of cooling and data centre control [@zotero-1570]. Every 5 minutes, the data centre provides the AI with status updates (based on data collected from thousands of sensors), after which the dedicated neural network forecasts the ways in which various combinations of possible activities may impact energy consumption in the future. Subsequently, the AI system identifies the steps necessary to minimise energy consumption while still observing the limitations imposed by the need to maintain suitable efficiency (e.g. processing capacity) of the centre. The commands are then forwarded to the data computing centre and executed by the local control system.

Autonomous systems also facilitate the management of telecommunications networks. For instance, the solutions offered by Accelerated Dynamics allow the creation of autonomous networks based on Wi-Fi drones. The system designer makes only general indications of the location and network parameters, after which the Accelerated Dynamics application establishes a Wi-Fi network with the use of a swarm of coordinated drones. The solution is easy to control and self-servicing: if a need arises (low battery or failure), individual drones are returned to base for repair. The resulting Wi-Fi network is therefore not only easy to design but also to maintain.

Programming

Autonomous systems also find numerous applications in the context of programming.

Opsani offers support to programming companies in optimising both software and the parameters of IT platforms within which the software is provided to end users. Its algorithms continuously test various configurations of hardware, system and application-specific parameters to ensure maximum efficiency and minimum costs. The resulting improvements may include an even twofold reduction of system response time (so-called

latency), 70% reduction of processing infrastructure costs, and even four-fold reduction of software roll-out time.

As already mentioned above, the high efficiency requirements of autonomous systems necessitate the creation of highly specialised solutions, e.g. dedicated or field programmable gate arrays (FPGA). Silexica supports the programmers of such solutions by offering an environment suitable for software testing, optimisation within a given hardware platform (FGPA system), and identification of possible conflicts or memory overloads. This accelerates the process of software development, as well as improves its overall quality and reliability.

Security

The rapid development of artificial intelligence is not without numerous inherent risks, not least of which is related to digital attacks. The extent of the same is largely dependent on the competence level of persons or organisations responsible for such acts: they are often highly educated, professional computer experts who utilise the newest advances in machine learning for their nefarious purposes. Consequently, it is nowadays increasingly difficult to protect systems against even already known forms of attacks (this logic was developed years ago in the form of antivirus software): new threats and techniques emerge on an almost daily basis, hence the only practically effective form of security entails so-called anomaly detection.

Hunters.ai describes its adopted methodology as 'autonomous threat hunting'. In simplified terms, the method entails analysing correlations between various events, logs and statistical data originating from a variety of IT systems in the protected organisation, and identifying atypical (anomalous) signals within such correlates. In other words, cyber threat hunting comes down to proactively identifying threats or traces of undesirable activity even before (rather than after) an attack takes place. Apart from machine algorithms (e.g. anomaly detection), the system also utilises the results of in-depth analyses of previously identified threats and heuristics provided by cybercrime experts. The benefits for companies using Hunters.ai solutions come in the form of support for cybersecurity teams (Security Operations Centres: SOC) and improved responsiveness to attacks.

Similar solutions are also offered by SecBI. Much like Hunters.ai, the company employs supervised and unsupervised machine learning algorithms for the purposes of behavioural analytics of users or devices operating within protected systems. The solution focuses on prevention (threat detection), and in the case of an attack, incident response systems are triggered followed by a so-called forensic analysis: to identify the attack patterns and update the knowledge base.

Summary

Autonomous systems are employed in practically all functional areas of contemporary organisations: from asset management, through logistics, manufacturing, sales, marketing, and customer service, to human resources management, decision-making support, and ICT infrastructure management.

The key benefits of implementation include:

1. optimisation: in particular by enhancing production processes, improved precision and the resulting reduced incidence of errors, as well as reduced employee workloads
2. streamlining of: employee productiveness, responsiveness (e.g. to failures, customer notifications), maintenance of operational continuity, better quality and speed of decision making
3. reduced operating risks: particularly in terms of improved employee safety (e.g. by replacing them in the performance of hazardous tasks, faster and more efficient failure detection, and consequently reduced downtime or environmental hazards)
4. reduced costs: particularly in terms of labour
5. increased customer satisfaction: thanks to personalisation (offer, communication), better responsiveness and continuous improvement of the customer experience.

The most desirable characteristics of autonomous solutions include broadly understood security, reliability (effective operation even under difficult conditions), simplicity (of installation, adaptation, maintenance and service), flexibility (capacity to work effectively under various, often unexpected circumstances which may occur both in the context of a production line and a customer service centre), ability to communicate in the natural

language (understand and generate messages, be it textual or audio-based), and capacity for self-learning and self-improvement.

Autonomous systems also generate a number of specific challenges. One of the most crucial ones relates to the ability to secure increasingly complex IT systems against external threats. At the operational level, the main challenge is to effectively implement such solutions, which relates to the necessity of digital transformation: not only on the strictly technological but also organisational and social level.

In the global context, one has to remain aware of the risk of digital division of organisations and societies alike: the companies intensively and successfully investing in autonomous technologies continue to improve their competitiveness, which puts increasing pressure on the market and consequently limits the capacity of less digitally mature enterprises/organisations to effectively adopt advanced technologies. Similar phenomena can be expected to also emerge on the labour market: the development of technologies increases the demand for highly qualified employees, stimulating their further intensive development and specialisation, while simultaneously limiting opportunities available to less qualified job seekers.

Potential impact on industry sectors

Intelligent systems change not only the manner in which companies operate in certain key functional areas (e.g. primary and auxiliary processes within internal value chains), but also the way in which said organisations operate on the market. In the previous sections, we focused on the **technologies** constituting the basis for such systems and the **tasks** they can perform in various functional areas. In the section below, the central focus will be on the **impact** thereof on how companies operate in various industry sectors. In many cases (e.g. marketing or sales) 'tasks within the given area' are very difficult to actually distinguish from 'impact on the industry' – for this reason, the **value** generated by technologies was adopted as the key classification criterion, with the respective applications described in terms of revenue stimulation, cost and risk reduction. Notably, the same is not limited to only autonomous and self-learning systems but extended to

include the spectrum of technologies utilising broadly understood machine learning and artificial intelligence.

Manufacturing, Construction and Automotive

The manufacturing industry has been rife with autonomous technologies for years. Data generated by millions of sensors may be used by artificial intelligence algorithms to **drive revenue, cut costs,** and **avoid risks.** Presented below are some of the most interesting applications of intelligent technologies in terms of the above classification.

Drive revenue

The presentation of the impact of artificial intelligence on the manufacturing sector should start with the concept of intelligent, combined products (cf. e.g. [@porter2015]). Sensors installed on end products may gather considerable amounts of data, both on user behaviour (in particular the manner in which products are used) and the context of use (e.g. time, place, ambient temperature, etc.). Such data may be processed directly by the device or transferred to company servers. In both cases, if utilised properly, such data can contribute to improving intelligence of the product itself and the entire ecosystem (e.g. in agriculture), or facilitating the process of personalisation, improvement, and development of new designs. This creates a niche for **intelligent products** which (autonomically) adopt their functions to the needs of a given user and allow the providers to integrate customers' research with the activity of their own development departments.

Another possible use of intelligent technologies is **demand forecasting**. An accurate prognosis of the future demand for existing products allows one not only to ensure the availability of corresponding supply and secure short lead times, but also to simulate the demand for new categories of products and improve overall customer satisfaction. In an effort to improve the quality of such predictions, contemporary systems rely not only on historical data (at the level of both respective products and their categories, but also specific customers or groups thereof) but also contextual data such as the weather or social media sentiment. Example applications include e.g. dynamic allocation of employees to respective commercial areas, recommendations of products that are most

likely needed by respective customers (*next-product-to-buy*), or accurate orders of products from the warehouse, which generates savings (reduced storage costs and waste (e.g. food)) as well as revenues (if product recommendations are sufficiently pertinent).

Competent combination of data from intelligent products and demand forecasting systems may provide the basis for medium and long-term company development plans, and therefore become a significant component of **research and development**. Increasingly short product and service lifetimes as well as their growing diversity are becoming a serious challenge: on the one hand, new products generate a competitive advantage, on the other, any wrong decision made in this context (e.g. bad choice of technology) can be costly and very risky. For this reason, it is necessary to forecast the demand for particular products and services, pinpoint their most desirable properties (e.g. user-friendliness, durability under given conditions, etc.), impact on the environment, cognitive capacity, or health. Properly used machine learning algorithms may help one to identify the optimum directions of development by using data generated by the products themselves, customer reviews and recommendations posted on social media, evaluations of competitor efforts, and automated analyses of market reports or scientific publications.

To sum up the benefits of implementing intelligent solutions in production, the same include **improved efficiency of product design** thanks to in-depth analyses of customer needs and behaviours, **optimised asset allocation** thanks to good quality demand forecasts, and **generation of new business models** based on intelligent, combined products.

Cut costs

It is increasingly common for intelligent technologies to be used in industry with the aim of cutting costs. In this context, systems allowing **predictive manufacturing design** are starting to enjoy particular popularity. Artificial intelligence is used for optimising design processes. Traditional methods of industrial design are often vey cost intensive, both in terms of direct expenditures (numerous prototypes) and time (which generates personnel costs and extends the time needed for a product to be launched). Machine learning algorithms take advantage of data derived from production processes, already

existing products, and databases of material parameters. They aid designers in quickly identifying parameter configurations that best meet given requirements. In extreme cases, an AI may even prepare the solution design itself: the role of the designer is then limited to providing boundary conditions (requirements). Key values generated by such solutions include cost reduction (labour intensity, materials), shorter time frames, and improved design quality.

Artificial intelligence is also increasingly used for **production process optimisation.** Suppliers of systems facilitating simulations of manufacturing processes are keen to equip their solutions with AI optimisation algorithms (cf. e.g. https://www.anylogic.com/features/artificial-intelligence/), especially reinforcement learning techniques. They prove valuable in optimising manufacturing processes, pricing (which in the case of short-series manufacturers can be a considerable challenge), manufacturing parameter control, machine reconfiguration for changing tasks, or simply quality control. The benefits include better pricing for the client, lower material and energy consumptions, shorter lead times and related savings, and better customer satisfaction.

One of the key aspects of optimised manufacturing is **automation** understood here as transferring certain processes previously performed by humans to machines and algorithms. Such solutions have already been discussed in the section on co-bots (robots collaborating with humans) and chat-bots (where conversation systems replace people in providing customer or employee service).

Automation also proves valuable outside traditional industry. Seervision offers solutions facilitating automation of processes involved in the production of movies and television broadcasts. Their automated camera robots employ AI algorithms to precisely guide cameras (their movement in physical space, framing or focus settings) to maximise the accuracy and fluency of footage while minimizing the related costs. With the use of image detection and recognition techniques, the system allows cameras to e.g. follow a moving object (a car, sports person, or lecturer) and select the optimum frame settings. The benefits of the solution include savings generated by replacing a human operator with the machine, improved transmission (avoiding problems related to camera operator's

tiredness), ability to achieve otherwise impossible angles, and increased safety (in situations where filming would be potentially dangerous to a human operator).

One should also mention the use of machine learning algorithms in **optimising transport processes**. For a number of years, intelligent systems have been widely employed for optimising routes – indeed, such solutions are now fairly commonplace. It turns out, however, that even in this context there is still room for new was of utilizing machine learning. For instance, many food manufacturers, particularly in developing countries, are struggling with product wastage in transport. AI algorithms can utilise historical data to precisely simulate the levels of losses foreseeable for the respective methods of transport. This aids producers in selecting the optimum transport methods, defining more effective storage conditions (e.g. freezing), transport routes, or delivery times. Apart from the obvious benefit of savings resulting from minimizing losses, this approach also allows a reduction of overall food waste rates.

By combining intelligent systems' ability to forecast demand, facilitate more efficient production and transport, companies can attempt to **optimise their supply chains**. ML algorithms are helpful in projecting the actual production demand, evaluate raw material needs (and update relevant material requirements), production capacity, as well as supplier reliability. The benefits of this approach include minimisation of production downtimes, reduction of storage costs, and shortening of overall production cycles.

In summary, intelligent systems help to **optimise production** by cutting costs while maintaining the quality levels, **streamline the supply chain** (timeliness, safety), **improve products** and **reduce warranty costs** by employing data generated by intelligent products in design processes, as well as **cut service costs** by projecting and preventing failures.

Avoid risk

Machine learning algorithms can be useful in minimizing operating risks. This usually involves the areas of **quality management**, **safety management** and **production upkeep**.

DataRobot, one of the leading suppliers of Data Science platforms, allows its customers to model product quality based on information from material databases, production lines, and other data gathered from the surroundings. Such models facilitate the monitoring of only selected, highest-risk elements of the production process, thus reducing its overall costs. Additionally, they help to identify previously unknown sources of potential failures, which further reduces costs by minimizing the scope of future warranty services (something of particular significance in the case of e.g. automotive industry, where the costs of pulling a whole batch of vehicles from the market during a servicing campaign can be particularly steep, both financially and goodwill-wise).

The infrastructure monitoring systems described above **improve the security** of the entire manufacturing process, not only on assembly lines but also in construction zones (construction sites, mines), while the ability to **forecast failures** (predictive maintenance) based on historical data (e.g. current strength, vibrations, noise generated by the machine) helps to cut the costs of not just service and repair as such (no unnecessary interventions, reduced inspection costs), but also of downtime resulting from undetected failures.

As a result, quality assurance (QA) processes are optimised, incidence of work related accidents is minimised, downtimes are reduced, and inspection/repair costs are cut (both in the context of production as such and warranty repairs).

Clearly, data generated by things, converted to usable information and processed by intelligent algorithms, may be a source of very tangible value. An interesting example of this kind of transformation (data>information>value) in the automotive context is provided by CCIS (cf. https://www.cccis.com/auto-manufacturers):

1. **Data** are generated by **sensors** mounted in a vehicle, e.g. magnetometer, speedometer, gyroscope, accelerometer, vehicle configuration, no. of occupants, time and date, GPS, diagnostics;
2. They are then used to generate usable **information** such as speed, turns, acceleration, idle time, night driving, hard breaking, and distance;
3. Subsequently, intelligent algorithms use this information to generate **values** such as improved safety, personalisation, or better user experience.

Strict security requirements combined with the availability of the aforementioned data, information, and values can therefore improve the traditional methods of measuring various aspects of machine operation. For instance, measurements of the vehicle's mileage (or kilometrage) can be replaced by measurements of **quality and safety of the drive** (quality of coverage, as part of so-called disengagement reports).

Transport

Transport is yet another area where intelligent technologies can be effectively deployed. Similarly to industry, the values generated thereby will be discussed below with respect to the categories of driving revenues, cutting costs, and avoiding risks.

Drive revenue

One of the most common uses of machine learning algorithms in transport entails so-called dynamic pricing. The same basically entails dynamic adjustment of price levels so that they are better matched to the purchasing power of individual customers (e.g. based on their behavioural profiles, demographic data, or current needs) as well as to the level of demand observed for entire markets or particular segments thereof. The practice is commonly employed by airlines (ticket prices can sometimes fluctuate in a matter of hours), providers of cargo services, and hotel chains. The dynamics of transport pricing can be affected by customer needs and behaviour, as well as external factors such as the weather, fuel prices, security concerns, or even social unrest resulting in e.g. certain groups blocking roads or some means of transport becoming unavailable. At the same time, such costs have a significant bearing on the ultimate pricing of the end product supplied by the manufacturer. Hence, machine learning algorithms facilitate not only increased profitability of shipping companies as such, but also aid manufacturers (due to effective forecasting of future prices) in adopting better projections of future transport costs and more reliably choosing the optimum routes, methods, and e.g. providers of cargo services.

Thanks to their in-depth analyses of user behaviour, AI algorithms allow shipping companies to better identify customer needs and introduce new, tailored offerings.

Cut costs

The use of intelligent systems in transport can also generate savings.

Algorithms offered by Accelerated Dynamics (cf. https://www.accelerateddynamics.co/solutions/logistics/) facilitate automated coordination of a fleet of autonomous vehicles. UAVs connected to the company's system exchange route data and other registered information, which is in turn processed by algorithms responsible for optimising the operation of the entire collective fleet. It is noteworthy in this context that the entire system is capable of learning and self-improving on a continuous basis.

Another interesting application of autonomous systems relates to so-called last mile deliveries (e.g. to the door of the end user). The issue is more complex than it might seem: in many cases reaching the designated residence requires traversing unexpected obstacles that are often not indicated even on most detailed maps. Nuro.ai (a company offering home delivery of everyday shopping) solves the problem by stopping its delivery vehicles in the vicinity of the address and notifying the clients (who then collect the delivery themselves). The vehicles used are similar in size to regular cars. In turn, Robby.io performs its deliveries with the use of considerably smaller autonomous robots which, thanks to their relatively small size, are capable of safely navigating e.g. pedestrian pathways or parks. Both systems aim to replace delivery personnel (e.g. mailmen) while providing a high level of security and considerable savings due to eliminating the human element.

Another interesting approach to the issue of last-mile autonomous delivery is the already mentioned solution offered by Savioke: suitably adapted logistics robots facilitate the transport of various goods (e.g. stationary products, documents, or food) within buildings. Their high level of autonomy allows the devices in particular to: navigate floors with the use of elevators, independently charge their batteries, and secure the deliveries against unauthorised pickups. The key benefit in this case stems from the convenience of the solution and freeing up the time of office staff.

Another category of autonomous systems successfully used in the transport industry involves aerial vegicles. The solutions are used for carrying both small packages (e.g. medicines, medical instruments, small parcels) and larger-scale deliveries. For instance, Elroyair drones facilitate the transport of relatively heavy cargo (up to 225 kg) over long distances (up to 500 km), while maintaining a high level of autonomy (no airport or electric stations required, autonomous cargo loading and unloading, and advanced safety architecture). The solution is usually used in commercial (transport cargo, fast deliveries in heavy traffic conditions, load-balance between warehouses), humanitarian, or military applications. The benefits stem primarily from the ability to quickly deliver items to poorly accessible areas and elimination of risks to human operators (transport in hazardous conditions (bad weather, armed conflict)).

Intelligent systems are also employed in **public transport**. For instance, companies such as Bestmile or Ridecell offer comprehensive solutions for optimising transport services. They allow fast implementation (with minimum investment costs) of autonomous transport technologies, autonomous coordination of vehicle fleets (e.g. identification of the optimum routes at the given time), automatization of overhauls and maintenance, operational monitoring, and continued solution improvement. This can provide the basis for the implementation of services such as Robotaxi (autonomous Taxi on demand), ridehailing services (human-driven door-to-door services for individual travellers), or micro-transit services (similar to ridehailing, but for multi-passenger transport). In this case, machine learning algorithms are used not only for fleet coordination, but also dynamic pricing adjustments and balancing of current supply and demand for transport services. At the same time, such solutions can be integrated with entire ecosystem of public transport, in particular in the context of payment, telematic, or repair services. The key benefits here include improved efficiency of transport fleet use, higher quality of services offered to end users (accessibility from a mobile device, transport on demand, social aspects), and optimised processes of fleet maintenance.

In particular, the aforementioned systems may provide the basis for establishment of **autonomous shuttles**. There are a growing number of companies marketing various components needed for the implementation of autonomous vehicles, from autonomous

buses (Perron Robotics, Aptiv, Optimus Ride) to comprehensive systems facilitating the establishment of e.g. automated transit networks (2getthere). At the current stage, integration of autonomous busses with public transport systems is only possible in controlled environments, hence the first attempts to do so tend to rely on relatively small vehicles (up to 20 passengers) travelling along routes isolated from the public space (automated transit networks) in areas such as university campuses, hospitals, business parks, or airports. The absolute priority in this context is passenger safety and comfort.

Another, more autonomous and personalised iteration of autonomous public transport entails robo-taxi and ride sharing solutions. Waymo is a company supplying one of the most advanced systems of this type in the world: fully autonomous, mid-range (Toyota Prius, Fiat Chrysler Pacifica) and premium (Jaguar I-Pace, Lexus) autonomous cars are connected to a system facilitating rental via a mobile app. In collaboration with Hyundai, Pony.ai has also recently launched a similar service in California.

Notably, cars are not the only viable means of autonomous transport in urban environments. Ehang offers services based on autonomous aerial vehicles used both for cargo and passenger transport. By employing Big Data and numerous advanced machine learning algorithms, the logistical ecosystem provided by Ehang supports automatic route planning and fleet coordination without the need for human input. Example application scenarios include not only passenger transport in crowded urban environments but also e-commerce deliveries (e.g. food products from Yonghui or DHL-Sinotrans parcels).

Autonomous systems are already commonly used in **marine transport**. For instance, Percepto drones facilitate pier surveillance, container monitoring (including detection of human presence and trafficking prevention), crane loading operations monitoring, and transport route supervision.

The final application of AI systems in transport to be discussed in this section relates to the improvement of **operating efficiency**. In this context the greatest benefits are yielded by predictive maintenance systems. AI algorithms provided with data from vehicle sensors and external databases (e.g. fleet management systems, fault logs, and weather services) are able to predict, with considerable accuracy, the incidence of failures with regard to particular elements of the vehicles and their realistic usable lifetime under the

given conditions. In the case of production lines, this allows machines to be repaired before a failure actually occurs, which minimises downtime costs.

To recapitulate, value most commonly generated by intelligent systems in the transport industry includes:

1. **Streamlining and optimisation of transport** thanks to fleet coordination solutions integrated with other transportation systems (e.g. in the airspace)
2. **Increased safety** of flight operators who can be replaced by autonomous systems in hazardous environments
3. **Decreased costs** thanks to automatization and preventive maintenance
4. **Improved user satisfaction** due to e.g. better availability of public transport.

Avoid risk

Specific issues related to security and risk avoidance in the context of autonomous systems used in transport have already been discussed in the section pertaining to technologies. Presented below are several interesting examples of how intelligent systems may facilitate the minimisation of risks inherent in other aspects of transport operations.

The first of the same relates to the already mentioned solution marketed by Nauto. To briefly reiterate the main points, thanks to systems registering images both inside and outside of the vehicle and algorithms capable of real-time recognition thereof (e.g. sudden breaking of the preceding vehicle on the road), Nauto provides the driver with direct alerts (collision warnings) and the fleet operator with in-depth analyses regarding driving styles or incidence of accidents. The immediate benefits include improved driver safety (thanks to alerts and personalised driver training programs) and the resulting savings (less downtime, lower costs of insurance) as well as acceleration of certain processes (e.g. claims settlement).

Another safety requirement, this time related to the transported goods, is fulfilled by Mujin. The company's advanced logistics robot control systems support the **transport of fragile goods**: from depalletization, through warehouse handling, to packing. This is possible thanks to dedicated image recognition systems and precise gripping and

handling mechanisms. In turn, systems offered by Blickfield (a provider of Lidar sensors) and Exyn (drone software) facilitate exact detection of obstacles emerging in the way of autonomous vehicles (e.g. agricultural machines or drones), thus preventing **collisions**. Yet another aspect of safety is addressed by Clearpath Robotics: the company's robots are used for the handling of hazardous materials (e.g. explosives) thus naturally increasing the safety of operators.

The main benefits include reduced costs stemming from damage to the transported goods, minimisation of the risk of autonomous vehicles' collisions and improved safety of human personnel.

Agriculture

An interesting approach that may contribute to increasing the revenues generated in the marine industry entails the use of autonomous sailing drones deployed to identify the most promising fisheries. For instance, the solution offered by Saildrone in the Mission-as-a-Service formula allows the collection of reliable data from marine regions, which may be used to identify potent fisheries (fish biomass, sustainable fishery management), track the movements of animals, or forecast weather. Such data can be useful when making sort or long-term fishing decisions, and thus naturally contribute to increased revenues.

However, a vast majority of autonomous systems used in agriculture are deployed with the view of cutting costs (due to automatization).

The first class of such systems covers **autonomous agricultural machinery.** Asi Robots is a company offering one of the most advanced solutions of this type. Asi Robots machines are capable of operating completely independently or in a tandem with a human-operated machine. The provided obstacle detection systems allow safe manoeuvring, while advanced algorithms plan the optimum routes depending on the task being performed (e.g. sowing or harvest), taking into account such factors and specifics of the field surface, type of tasks performed, type and size of the vehicle, turning angles, etc. When coupled with dedicated agricultural navigation systems provided by Swiftnav, the solution is capable of significantly increasing the efficiency and precision of cultivations.

Autonomous systems are increasingly used **for sowing and sowing monitoring**. The solutions offered by Clearpath Robotics are designed to allow the equipment of existing machinery with autonomy-facilitating components (e.g. phenotyping, remote sensing of crop health, seeding and top dressing). The Accelerated Dynamics platform allows integration of data from moisture sensors and irrigation devices to instantly monitor crops, while autonomous drones sold by Percepto can gather information about water consumption, crop health, soil composition, and weed growth – significantly contributing to the reduction of inspection costs. In turn, Tevatronic offers **autonomous irrigation** systems capable of independently controlling and maintaining the level of watering in particular areas.

Yet another category of solutions includes spraying and weed control systems. The providers of such products utilise specially adapted agricultural machinery (Accelerated Dynamics), dedicated robots (Gussag, Ecorobotix), drone swarms (Skyx), or electric vehicles (Flypyka). Typically, the first phase of operation entails an inspection of the given area, followed by planning and application of spraying, and finally monitoring the results. The main benefits stem from automatization of the entire process, reduced workload and improved safety of human workers, acceleration of the respective processes, minimised incidence of errors, reduced downtimes, and ultimately cost savings stemming from all of the above. Notably, such systems also facilitate a **significant reduction of herbicide use** (according to Ecorobotix by even 95%), which is beneficial to the environment as well as the quality of food produce. The same can be categorised within the emerging trend of so-called **precision agriculture.**

In summary, autonomous systems are employed in agriculture primarily with the view to automatizing farm work (e.g. sowing or harvest) and improving its precision (e.g. irrigation or spraying). The benefits include lower costs, increased efficiency (e.g. due to better monitoring), and a significant reduction of the environmental footprint.

Mining and energy industry

The mining and energy sectors are yet another area where intelligent systems are currently intensively introduced. Similarly to the previous sections, the below analysis will focus on the benefits stimulating growth, cost and risk reduction.

Drive revenue

One of the most cost-intensive aspects of mining operations is the search for new viable deposits. Any errors made in this context can potentially lead not only to increased direct costs but also a deterioration of overall productivity. Machine learning algorithms take advantage of the large quantities of data available from already completed explorations to facilitate more accurate evaluation of the mining capacity of already exploited deposits as well as new potential sources of raw materials. They allow mining companies to better estimate the value of deposits (e.g. forecast total recoverable reserve volumes), limit the risk of new geological explorations, and forecast future prices of raw materials. Such solutions can also provide valuable information when deciding on the purchase or acquisition of other enterprises.

In the energy industry, information about the profiles of energy consumption by respective clients (both individual and corporate) may facilitate the development of new offers tailored to specific customer needs. The ability to precisely predict trends in energy consumption (e.g. related to ecology) and production, both in medium and long term, may provide e.g. companies in the fuel industry with information needed to effectively make strategic decisions regarding e.g. modifying the forms of energy produced or developing the existing distribution networks.

Yet another, somewhat exotic but undoubtedly interesting example of the ways in which AI can be employed to increase business efficiency in the energy sector entails the use of systems analysing sentiments expressed in press releases to predict future decisions of market regulators, and consequently better prepare corporate strategies for imminent negotiations with the government.

Cut costs

Intelligent systems are also helpful in reducing costs at the stages of drilling and completions, production, gathering and transportation, **as well as** processing and refining maintenance.

In the first of the same (drilling and completion) AI is used for the purposes of well design, drilling and completion execution. In particular, algorithms are used for optimising the position and distribution of wells (to maximise output) and projecting risks related to subsurface phenomena, while autonomous drones are utilised for detailed mapping of deposits or mines. At the stage of production, machine learning can help to calculate parameters such as speed or pressure (with the view to maximising deposit output), optimise frac design, model and simulate various proppant and fluid loading options, while anomaly detection can aid operators in predicting failures. The mined produce can be transported with the use of autonomous vehicles (cf. e.g. https://clearpathrobotics.com/queens-university-autonomous-mining/). If the need arises to shutdown a refinery, intelligent systems can aid engineers in shutdown planning by recommending the best possible scope of works. At each of the aforementioned stages, predictive maintenance systems facilitate timely failure prediction and prevention, while systems of supply chain optimisation help to ensure their security and minimise the related costs.

As for the energy sector, one of the most common applications (apart from the 'ubiquitous' predictive maintenance) relates to energy supply and demand forecasting, which allows analysts to identify periods of peak network loads and plan adequate preventive action to minimise the risk of overload (peak maintenance).

The AsiRobot website (cf. https://www.asirobots.com/mining/#Mobius-For-Mining) provides an interesting value calculator allowing users to estimate the benefits of implementing the company's autonomous systems. The tool illustrates how the implementation of the subsequent components of the solution (from autonomous drills, through blasting assist, dozer assist, loader assist, and haulage assist, to autonomous haulage) can contribute to reduced operating risks and improved overall productivity relative to the relevant investment costs.

Avoid risk

Equally interesting solutions allow the application of intelligent systems for the purposes of risk avoidance.

The most popular method entails the use of autonomous solutions in the context of various inspections. SMP Robotics offers fully automated inspection robots specializing in e.g. detecting electrical damage, gas leaks (both underground (mines) and on the surface), as well as oil leaks. Dedicated algorithms prevent collisions with objects, other vehicles and people. Others facilitate autonomous battery charging. An additional quality of the devices, important when working in difficult condition, allows them to operate in ambient temperatures from -25 to +55 degrees Celsius.

Drone systems are often used for purposes related to minimizing operating risks. For instance, Percepto drones allow generation of thermal maps, monitoring of infrastructural elements prone to component overheating, and detection of harmful leaks of e.g. gas (cf. https://percepto.co/oil-gas-drones/). Exyn offers dedicated autonomous solutions for mapping out inaccessible and hazardous areas (e.g. underground). In turn, the system marketed by Skyspecs is designed for wind farm monitoring: it not only takes photographs but also analyses them (to identify potential failures) and forecasts potential future breakdowns.

A separate category of systems designed to mitigate operating risks includes solutions detecting energy theft. Based on characteristics of user behaviour, in particular energy profile anomalies, such systems identify points with the highest risk of theft, supporting the work of respective control services.

In summary, the key benefits generated by intelligent systems in the area of risk avoidance include cost reduction, increased efficiency of sensitive and critical infrastructure monitoring, and related safety improvement (both of the infrastructure and the supervising personnel), decreased costs of repairs due to effective preventive maintenance, as well as reduced theft-related losses (thanks to better detection).

Marketing

Given the large amounts of data generated through various signals and contact channels utilised, marketing and sales can benefit from the use of intelligent systems in a variety of ways. Let us first have a closer look at such applications in the context of marketing, to be followed by commerce in the subsequent section.

Drive revenue

Intelligent systems can help to increase revenues in the marketing industry by optimising **advertising**, **improving customer experience** and aiding **market analyses**. Due to the considerable complexity of this subject mater and the very wide spectrum of the possible ways in which machine learning can be employed in advertising, the below deliberations will be limited only to some selected aspects of the same.

Firstly, based on analyses of user behaviour across digital channels as well as, increasingly, more traditional ones, machine learning algorithms allow marketers to identify individuals most likely to be interested in purchasing a given product or service (or, analogically, products which the given client is most likely to look into). Furthermore, conversation intelligence systems (e.g. Chorus.ai) are capable of analysing thousands of sales interactions to identify optimum conversation scenarios, thus broadening the spectrum of customised marketing messaging to include direct client interactions. This allows for effective personalisation of advertising: a customer is provided with advertisements tailored to his or her expectations while the company can limit the costs of ineffectual advertising (both financial and image-related (SPAM)).

A separate category of systems comprises solutions facilitating the assessment of marketing effect on sales in particular channels (so-called **Multichannel Marketing Attribution**). This allows companies to fairly accurately calculate the return on investment (ROI) in respective marketing activities, which then helps them in relocating resources to the areas where the same prove most effective, thus improving sales and limiting campaign costs.

Another area where intelligent systems can yield tangible benefits relates to **improvement of customer experience**. Artificial intelligence algorithms facilitate precise micro-segmentation of the customer base based on behavioural data, purchase histories, demographics, as well as information obtained from professional third-data providers. The conducted analyses allow one to identify small groups (segments) of customers based on certain specific attributes (e.g. shared characteristics, shopping intentions or interests). Based on the same, marketers are able to tailor their messages and offers to the specific segments, which leads to improved campaign effectiveness (higher conversion rates and revenues, lower cost of misplaced advertising).

AI algorithms can also be employed to identify products (or services) which the customer may be interested in buying in the near future (so-called **Next Best Offer**). Studies indicate that offers which are too broad can negatively impact customers' purchase decisions: clients tend to prefer fewer but more accurate recommendations. It therefore becomes critically important to be able to identify those products included in often very extensive and fast changing catalogues that the given customer is most likely to be interested in. Intelligent systems (so-called recommender engines) offer a solution to this problem by analysing large datasets pertaining to the shopping behaviours of individual customers or entire groups (e.g. microsegments) thereof.

Yet another challenge for marketers relates to the need to personalise channels and methods of interaction with the given users (so-called **Next Best Action**). It has been observed that every customer journey is unique: activities that are effective on some customers can have a completely different (including negative) effect on others. This depends not only on the product but also the customer's position in the overall sales process, the communication channel used, and even local context (e.g. travel, health condition, etc.). AI systems allow one to identify the optimum channel at a given time (e.g. e-mail, phone call or text message) as well style and content of the message to maximise the chances of eliciting a purchase decision (or the customer's conversion to the next

stage of the sales process).[11] At the higher level, AI helps to identify customer behaviour patterns not only in the context of the sales process itself, but the entire customer journey (from the first contact with the brand to the after-sales service and social media interactions).

The benefits generated from the aforementioned applications of IA include **improved customer satisfaction** and the resulting **greater inclination to buy** (due to the increasing personalisation of offers, these days possible even at the individual level (rather than at the level of (micro)segments), reduced time necessary to successfully **launch a new product** (thanks to early identification of trends and needs, e.g. using social media), improved **business scalability** (digital algorithms are often capable of identifying market needs faster than human analysts, which limits the need to hire local experts), and **better understanding of customer behaviour** (through the use of so-called self-eXplainable AI (XAI) algorithms).

Cut costs

The complexity of the online advertising market (cf. e.g. over 7,000 companies listed here: https://chiefmartec.com/2019/04/marketing-technology-landscape-supergraphic-2019/) naturally creates a niche for services related to simplifying the management of marketing campaigns conducted online. Although the entry barrier for systems facilitating the creation of campaigns through e.g. Google (Google AdWords) is relatively low (in terms of simplicity of use), one quickly discovers that actual optimisation of such campaigns requires a great deal of expert knowledge.

Albert.ai works to utilise this particular niche by developing advanced AI algorithms capable of **optimising online marketing campaigns**. Over 200 different AI 'skills'

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[11] Notably, such personalisation may be applied not only to individual messages but also entire marketing campaigns: e.g. Persado supports the preparation of marketing campaigns oriented to eliciting a certain response in specific groups of customers.

utilised in the solution provide users with the tools to plan, create, optimise, and monitor the effectiveness of their projects. The key benefits identified by the company's customers include reduced costs of customer acquisition, improved conversion of Facebook users' activity, lower cost per action (CPA) cost per load (CPL), and consequently improved Return on Ad Spend (ROAS).

In turn, DataRobot (supplier of the Data Science platform) offers a solution supporting buyers during real-time bidding (RTB) advertising auctions. Based on previous marketing activity, it allows one to create models of the online advertising market and uses the same to **identify the optimum bidding prices** (low enough not to overbid but high enough to win the auction). Naturally, this can go a long way towards improving the profitability of advertising spending.

Another use of AI with the aim of generating savings in the sales process entails so-called **Lead Scoring**. Machine learning algorithms can assess the likelihood of success relative to sales efforts aimed at potential buyers and identify the sales contacts with the highest promise (and, analogically, those with low chances of success). Additionally, the algorithms can identify factors that condition the actual sales potential of the given contact. With effective prioritisation, this can significantly improve the effectiveness of sales efforts and, in time, overall customer satisfaction (thanks to fewer futile interactions).

Avoid risk

Intelligent systems can also help companies in minimizing the risk of so-called customer churn (a customer choosing a competing offer). By using historical data, in particular past behaviour patterns, the algorithms are able to single out the clients who are most likely to stop buying (or using) a given product/service and recommend action that may prevent the churn. And given that customer retention is cheaper than customer acquisition, this approach can significantly reduce the risk of profit loss (due to losing an existing customer).

Retail

Drive revenue

Nowadays we observe, on the one hand, a growing number and variety of products and services, accompanied by their increasingly short lifetimes, and on the other, a progressing diversification of customer groups (in other words, nothing is as simple as it used to be . . .). Consequently, the experience and intuition of even the most seasoned salespeople is beginning to prove insufficient, which creates a natural niche for algorithms capable of better handling certain complexities of the contemporary world.

The first notable area where intelligent systems can prove valuable is a detailed study of customer behaviours and needs (Know Your Customer: KYC). By conducting in-depth analyses of the so-called customer journey, advanced analytical systems facilitate a better understanding of behavioural patterns, identification or discovery of new needs, and optimisation of the offered product combinations (e.g. by way of so-called basket analysis).

One particular form of such efforts entails **geospatial analyses** which allow a better insight into correlations between customer behaviours observed in brick-and-mortar shops and those registered online. The data used in such analyses increasingly includes not only online or mobile app statistics, but also behavioural data gathered in physical outlets. For instance, Pilot.ai offers systems utilizing advanced image detection and recognition algorithms which can be implemented with relatively low infrastructural requirements (e.g. using simple in-store CCTV) to provide retailers with so-called in-store analytics. The same systems can provide governmental or security services with a simple method of monitoring and analysing the behaviour of people in public spaces.

Additionally, geospatial analysis can support retail in planning optimum locations for new outlets as well as identifying stores that should be closed down for a variety of reasons. In combination with in-store analytics (analysis of customer behaviour) and demand forecasting systems, the solution can be used to continuously monitor and plan both the optimum staffing levels in respective outlets (with daily or even hourly accuracy), as well

as the necessary personnel qualifications (e.g. language fluency in places frequented by foreign tourists).

Information generated by KYC systems can be used in particular for the purposes of **offer personalisation**. Utilizing data reflecting particular customers' online activity (e.g. time and frequency of visits, level of interest in a given class of products, social media interactions, purchase history, and geolocation), information on their behaviour in brick-and-mortar outlets (e.g. time spent inside particular stores or even in front of particular shelves, as well as other behavioural data), and demographics (age, sex, education, even income levels or sexual orientation), algorithms are capable of identifying their current needs, which can be used as a basis for subsequent advertising and product recommendations. If done effectively, this can significantly improve sales levels and profitability as well as customer satisfaction (by limiting the amount of persistent advertising offering unattractive products and being able to better match recommendations to the customer's actual expectations).

It is noteworthy that personalisation pertains not only to the product offer as such, but also the format of messages, communication channels utilised, as well as the price. **Price optimisation**, in particular so-called **dynamic pricing**, is yet another area where intelligent technologies can prove invaluable. They facilitate parallel pricing and promotional offerings management across numerous sales channels and geolocations taking into account competitor prices and activities with the view to identifying the best possible price point for a given product relative to the current needs of particular customers, thus maximising revenues and margin levels.

The adaptation of the product (including price) to the current needs of a given buyer constitutes an element of a broader strategy whose aim is to retain the customer and maximise the so-called customer lifetime value (CLV). Machine learning algorithms assist companies in **optimising their loyalty programmes**: natural language processing, image analyses, and semantic reasoning techniques aid marketers in better understanding the customer-to-brand engagement, and consequently in improving their loyalty programs. Such solutions can also be used in the development of new products, e.g. new tea or coffee flavours in café chains.

At the end of the day, the reason for implementing all of the described methods is to improve customer satisfaction. Contemporary consumers are characterised by increasingly low issue tolerance with regard to products and services alike. Being mobile themselves, they expect fast, personalised response and delivery at all times. Purchasing priorities are now increasingly oriented towards health, wellness and fitness. Intelligent systems aid the retail industry in forecasting the next customer state as well as the time and character of the next customer purchase. They help to identify offerings that are most likely to successfully influence future shopping decisions (next-product-to-buy), perform in-depth analyses of customer needs e.g. based on surveys or data generated directly by the products themselves. They also facilitate better assessment of marketing campaigns and their impact on customer conversions, relative to different communication channels.

To recapitulate, the key benefits of artificial intelligence in this area include hyper-personalisation, product mix and pricing optimisation, more effective campaigns and loyalty programs, and, ultimately, consumer empowerment.

Cut costs

The progressing complexity of product offers and the related services, combined with shortening product lifetimes and growing expectations of customers in terms of services procured over a variety of channels, be it brick-and-mortar, online, social networking, or mobile, continues to generate new challenges directly affecting bottom lines.

One of the machine learning applications that helps companies to optimise their business operations entails **product assortment and supply optimisation**. Retailers rely on algorithms to recommend products that should be made available at a given time and location, determine optimum assortment and price levels (so-called AI-driven Demand Forecasting by store, date and SKU), as well as even plan optimum product distribution in retail outlets (cf. e.g. the already discussed Simba Robotics systems). To that end, forecasts are generated based not only on historical sales data for the given outlet, but also by taking into account customer behaviour online, information on ongoing marketing campaigns, as well as other contextual data regarding e.g. local holidays or weather conditions.

Furthermore, intelligent systems can be helpful in **identifying the most cost-efficient suppliers** and conducting negotiations with the same. Although seemingly simple, the process is in fact anything but, largely due to the complex discount mechanisms employed in commercial contracts. Such recommendations, complemented by projections of producer supply levels and transactional histories, are also the basis for **scoring systems** – which nowadays, constitute an important factor when deciding whether to engage in a given collaboration. The benefits here stem from improved customer satisfaction due to the availability of most desirable products, optimisation of supply chains, elimination of non-profitable products, reduced food losses (fewer products have to be utilised due to expiry of shelf-life), and automatization of the processes related to ordering products from suppliers, which results in reducing the costs thereof.

In turn, predictive systems can be used to **estimate the likelihood of returns for particular products purchased via different distribution channels**. This is particularly important for companies operating on the e-commerce market where the rate of returns can reach up to several dozen percent, which naturally generates very significant costs. Additionally, a more in-depth insight into the underlying causes of such behaviour may help to identify and, where possible, eliminate the key factors responsible.

Avoid risk

Intelligent systems also support companies in minimizing operating risks. They let them **identify consumer behaviour patterns indicative of potential theft or fraud** (e.g. unwarranted returns), as well as **improve product quality** by identifying and predicting the most common defects. Furthermore, artificial intelligence can be utilised to improve recruitment processes with the aim of **optimising personnel selection for respective positions**. Thanks to the above, companies can significantly reduce their operating risks.

Insurance, finance and banking

The ability to generate good quality data for the purposes of risk assessment has long been one of the primary foundations of effective operation in the insurance and finance

sectors. It is therefore hardly surprising that AI systems find numerous applications in this context to facilitate higher revenues, lower costs, and minimum risks.

Drive revenue

One of the most popular uses of machine learning in the finance sector relates to investment portfolio management and is commonly employed by large organisations and individual investors alike (wealth management). Algorithmic trading systems are capable of not only supporting brokers in short and mid-term decision-making, but also by effecting independent investment decisions and completing transactions, often in extremely short time frames (e.g. High-Frequency Trading - HFT). Companies such as World Quant utilise AI algorithms to optimise the portfolios of e.g. insurance funds with the aid of Delta Science consultants scattered all over the world. Suppliers of investment systems described as robo-advisors (for instance Wealthfront, M1Finance, Betterment, or Personal Capital) market such algorithms also to private investors.

Another possible application of AI algorithms in the financial sector facilitates in-depth assessment of customer behaviour. The results of such analyses are then used to anticipate new needs, develop new products and services, identify clients who are most likely to be interested in them, precisely target offers, and evaluate the price consciousness of individual customers.

In turn, one of the most popular uses of machine learning algorithms in the insurance sector relates to **pricing optimisation**. The traditional approach to determining the price level of e.g. a vehicle insurance relies on a number of basic, relatively simple indicators including year manufactured, model, or specification. What it fails to account for, however, is a number of important factors relevant to individual risk assessment applicable to a given driver, e.g. driving style and temperament, place and manner of using the vehicle (e.g. typical routes, projected mileage), or typical weather conditions. Machine learning algorithms, in particular their prognostic, segmentation and anomaly detection capabilities, allow insurers to generate in-depth behavioural profiles of individual drivers e.g. based on the frequency and dynamics of their use of the brake pedal. This way, individual insurance pricing can be accomplished without the need to use 'sensitive' information such as short-terms speed, or accounting for the aforementioned factors

(weather conditions, typical places and times of day when the vehicle is used, etc.). Notably, the applications of this approach are not exclusive to vehicle insurance. For instance, John Hancock offers health insurance discounts to individuals willing to share information about their behaviour (mainly movement patterns) generated with the use of wearables (e.g. smart watches). Beam.dental, a company selling dental insurance, offers similar discounts to customers who agree to use intelligent electrical toothbrushes supplied by the company. On the one hand, this practice serves to generally improve sales (lower-risk customers are offered better prices), and on the other, it helps to minimise insurance risk.

Machine learning algorithms can also be used for the purposes of **personalizing insurance products**, in particular by combining several services in consistent packages. The diversity of options available even from a single insurer, with different types of insurance sometimes offered by separate business units within the same insurance institution (e.g. life insurance, property insurance, transport insurance), can lead to misunderstandings and general dissatisfaction of the end customer (e.g. after being contacted by several branches of the same company). AI systems can aid insurers in integrating data pertaining to a given customer gathered through various divisions of the organisation, and preparing a dedicated product package that not only benefits the customer (lower price, single point of contact, service tailored to specific needs) but also creates upsell opportunities when new insurance products are introduced. Artificial intelligence aids insurers in **evaluating the likelihood of a given customer accepting an offer within a particular price range** (so-called conversation modelling). It allows companies to optimise their promotional campaigns and maximise margins while limiting the risk of the customer ultimately deciding not to buy or choosing a competitive offer.

Cut costs

AI algorithms utilise historical data as well as other information obtained from a variety of external sources to facilitate automatization, cost reduction, and quality improvement in underwriting processes. This is particularly important in the context of property and health insurance policies. In both cases, the cost is often the decisive factor for customers, forcing insurers to engage in price competition, which may affect the bottom-line and

generate losses if the cost of potential claims and probability of risk emergence are wrongly assessed. On the other hand, if the insurer were to play it too safe, excessive price levels may deter potential customers. For this reason, precise and efficient algorithms using data gathered from a variety of sources can create a potential competitive advantage for contemporary insurance companies.

Intelligent systems also begin to play an increasing role in **automated claims settlement**. For a claimant who e.g. took part in a car accident any simplification of the processes involved in reporting and settling the claim is a highly welcome development. This can include both matters related to filling out various documents or evidencing the event, as well as to damage evaluation or settlement oversight. On the other hand, any systems used to simplify such processes may create potential vulnerabilities, e.g. due to fraudulent claims. AI algorithms help to identify such anomalies and score them in terms of likelihood of illicit behaviour. This way, 'honest' claims are forwarded for automated processing and event automated settlement, whereas risky cases are submitted for verification by an experienced agent. Furthermore, algorithms may aid agents in the assessment of claims by assigning so-called reason codes to help with their subsequent classification and further processing of the case.

A good description and characteristics of the damage may, at the subsequent stage of processing, provide the basis for **automated valuation**. It is often the case that the preliminary estimate is substantially different from the actual cost of the repair. Precise prediction of such discrepancies may significantly affect the financial standing of insurance companies, particularly in terms of necessary reserve allowances. Predictive algorithms coupled with so-called claim development modelling help to considerably reduce such inconsistencies between cost estimates and actual repair costs, thus minimizing the financial risk of insurers and contributing to the pricing of future insurance offerings.

Autonomous technologies are also used to **automatize interactions with clients**. The bot (named Amelia) offered by IPSoft provides support to bank customers in the process of opening an account by education them on various available product lines, processing of personal data, checking account balances, credit card payments, or preparing loan

applications. When applied in the context of insurance, it supports fast and user-friendly claim registration, verifies policy details, and helps to estimate the amounts obtainable. It also facilitates the processes of amending insurance policies e.g. by updating personal information, as well as sales e.g. by recommending optimum products for a given customer. Whenever the bot is unable to provide the requested information or perform a specific action, the customer is redirected to a competent consultant. The solution helps to reduce the time-consumption and costs of customer service.

Avoid risk

One of the most popular applications of intelligent systems in the context of financial services is **credit risk scoring** assistance. Credit risk profiles are generated based on information such as the current level of income, employment opportunities, credit history, and demographics. It has been observed that overly 'secure' algorithms can lead to losses in potentially profitable segments of the market. For instance, students of good universities with a chance for relatively high earnings in the future, freelancers without a fixed source of income but with a portfolio of regularly paying customers, or people with low income but high aversion to legal conflict with a large bank may ultimately prove to be safer borrowers than others, seemingly preferable on the basis of standard criteria. Hence, advanced machine learning algorithms are able to not only protect banks against bad credit but also identify potentially profitable and relatively secure customer groups.

Intelligent systems support banks and insurers in **fraud detection and mitigation**. This problem poses a very serious challenge for many firms in the sector: indeed, significant enough for many companies to establish dedicated teams focusing solely on this particular issue. Basic AI algorithms help to identify suspicious credit card transactions or claims by detecting anomalies or based on predefined patterns of fraudulent behaviour. More advanced systems are capable of identifying new illicit patterns, substantiating (reason codes) their suspicions, and assessing the risk associated with particular claims, which may also be used as the basis for prioritisation of in-depth analyses. As a result, insurance companies are able to better allocate the time of their experts who can focus on the most risky claims, and minimise losses related to this form of theft. In turn, predictive algorithms and anomaly detection systems help banks to

identify money laundering mechanisms by tagging suspect transactions with the corresponding risk level and providing justification for the assessment (reason codes). This allows controllers to focus their attention on the most suspicious activity, thus significantly improving the efficiency of their work.

A particularly important functionality for banks and insurance companies alike is offered by customer retention and churn prediction systems. The cost of customer acquisition is relatively high in this sector, while the most valuable clients tend to take advantage of a wide range of products. It is therefore natural that their loss is considered to be particularly costly, which creates a niche for churn prediction solutions. Their task is to identify customers who may, in the near future, decide to discontinue using the given offer, but also to indicate the possible reasons for the situation, which may be a good basis for preventive action. If well implemented, customer retention solutions can minimise the risk of customer churn and increase the customer life-time value.

Another interesting application of autonomous systems in the insurance sector entails the use of drones for the assessment of damage occurring e.g. due to natural disasters. Devices equipped with advanced image recognition algorithms, capable of autonomically reaching places inaccessible to humans (e.g. buildings consumed by fire), can quickly and effectively provide information on the scope of damage and potential cost of repair.

The impact of intelligent systems is particularly evident in the context of automotive insurance. As already mentioned above, algorithms can support the processes of damage evaluation and adjustment. New interesting trends also emerge, for instance usage-based insurance. It will be particularly interesting to observe how the development of autonomous vehicles may impact the future market of automotive insurance. On the one hand, autonomous cars will certainly cause fewer collisions than human drivers, which is likely to reduce both insurance prices and revenues of insurance companies. On the other hand, however, insurers will have access to substantially more information about the risks generated by such vehicles, which in a sector so heavily reliant on good datasets and well trained algorithms, may create grounds for an initial price increase. Nonetheless in time, as autonomous vehicles become more popular, the progressively generated

historical data and continuously improved algorithms can be expected to eventually lead to a considerable insurance price decrease, despite the initial inflation.

Healthcare

Healthcare is yet another area where the availability of large amounts of good quality data creates ample room for the use of AI solutions. The sector itself is highly specific: one would be hard pressed to find or identify in its context activities aimed solely at driving revenues (or, even if they do exist, they tend not to be directly verbalised), while goals associated with minimizing costs and risks tend to be closely correlated. For this reason, the below analysis will depart from the categorisation of values into those related to *driving revenue, cutting costs, and avoiding risks*, focusing instead on the respective entities for whom value is generated: *pharmaceuticals* , *healthcare vendors and providers,* and *payers.*

Pharmaceutical manufacturers

The most promising area of AI applications in medical production relates to the development of pharmaceuticals. It is estimated (2019) that the development of a single innovative drug in the USA requires an investment of approx. USD 1 billion. For that reason, companies operating in the BigPharma market are intensively searching for methods that would allow them to reduce the cost and time investment (a need that became ever so pressing with the emergence of the COVID-19 virus) necessary to conduct the pharmaceutical development phase, while maintaining the adequate levels of quality and safety. For instance, researchers working at Phizer (cf. https://medium.com/datadriveninvestor/interview-peter-henstock-machine-learning-ai-technical-lead-pfizer-e6400b56c225) utilise natural language processing algorithms to analyse millions of abstracts, medical publications, and patents in search of information potentially relevant to the development of new active compounds.

An increasing number of pharmaceutical manufacturers are now turning towards artificial intelligence in their research efforts (cf. over 40 listed at https://blog.benchsci.com/pharma-companies-using-artificial-intelligence-in-drug-

discovery), and there are over 200 start-ups currently specialising in providing solutions dedicated to that end (cf. e.g. https://blog.benchsci.com/startups-using-artificial-intelligence-in-drug-discovery). This is possible due to the availability of large quantities of data generated by contemporary biochemical techniques, which provides a good foundation for teaching AI algorithms able to effectively project the processes of synthesis and compound bioactivity, support molecular designs, and perform visual analyses. This allows them to identify desirable molecules more quickly, asses the risk of complications and toxicity, as well as suggest new uses for existing drugs.

Intelligent technologies have also been successfully employed in **drug trials**. AI algorithms support the selection of patients for clinical trials, individual identification with the use of natural language processing algorithms, in-depth analyses of patient records (e.g. to find those with specific symptoms) as well as geographical data. Advanced analytics facilitate faster and more effective clinical trials thanks to e.g. the ability to integrate data from numerous studies.

AI is also used by pharmaceutical manufacturers to support operational and strategic decision-making in areas such as market potential analyses in the context of developing product portfolios, revenue optimisation, or product life cycle forecasting.

Healthcare vendors and providers

Machine learning algorithms are primarily used in the healthcare sector in the context of direct patient care, for instance in AI supported diagnostics. Companies such as Zebra Medical, Enlitic, or Caption Health (ultrasound diagnostics) offer systems capable of utilising image recognition algorithms to analyse medical imaging results. In many countries, the increasing availability of e.g. magnetic resonance imaging technologies has long outpaced the availability of trained radiologists – specialists competent in a number of diverse areas: pathomorphology, anatomy, even physics. Consequently, there has been a growing demand for expert systems capable of effectively and safely improving the efficiency of their work and possibly event replacing them in the near future.

Dedicated algorithms, trained to recognise medical images based on millions of correctly diagnosed cases, are already able to identify e.g. certain types of cancer more effectively

and 'safely' (lower false-negative rates) than humans. Currently used as support for human radiologists, they are increasingly recognised as viable tools by governmental agencies certifying medical technologies (cf. e.g. the list of algorithms cleared by the FDA https://www.acrdsi.org/DSI-Services/FDA-Cleared-AI-Algorithms) – we can therefore expect to see radiologists being gradually replaced by algorithms. Notably, the implementation strategy commonly adopted in this context can be described as 'skill-by-skill': products launched on the market are dedicated algorithms capable of detecting very specific conditions.

Diagnostic applications also include algorithms recommending the set of tests optimum for a given patient. The current potential for monitoring various aspects of human health is vast, hence, selecting the right set of tests to administer can often pose a considerable challenge. AI algorithms use patient data as well as databases made available by pharmaceutical companies to aid physicians in selecting the optimum scope of diagnostics, which not only serves to reduce patient load but also to accelerate the process and reduce its costs.

Intelligent technologies are also used with the aim of **improving patient safety**. Models offered by DataRobot facilitate the identification of patients who are likely to suffer from a particular disease even before they are referred to hospital, which means that adequate preventive measures can be taken beforehand. A separate class of algorithms utilise patient data, history of previous illnesses and periods of hospitalisation to project the likelihood of hospital acquired infections (e.g. during blood transfusions). Equipped with such information, medical personnel can more effectively monitor high risk patients and prevent infections, with due consideration for the most probable sources of infection and the applicable good practices (identified by the so-called reason codes generated by the algorithms). A very specific type of infection whose prediction is often facilitated by this type of solutions is **sepsis**: based on patients' vital signs and metabolic levels, the algorithms identify those with the highest risk of developing this life-threatening condition.

Intelligent systems also help to identify patients who ought to be hospitalised at intensive care units (ICUs) even before they crash. Patients admitted to the ICU early enough have

a significantly better chance of recovery – in this sense, algorithms facilitating early detection of serious health symptoms literally help to save lives.

The applications of intelligent systems presented above can be jointly described as a 'human predictive system'. Early identification of threats is a way not only to reduce patient risks but also overall costs of treatment. Notably, the path adopted in the development of such solutions is not unlike that pertinent to industrial technologies. One can observe increasing prevalence of e.g. the term digital twins being used with reference to humans – i.e. digital models of vital organs (e.g. the heart, cf. the solution offered by Siemens: https://www.siemens-healthineers.com/fi/press-room/press-media-gallery/im-20190412001shs.html) which allow physicians to e.g. simulate the effects of various drugs or therapies on the organism of a given patient. It is undoubtedly one of the more interesting directions in the development of medical technologies.

AI systems are also useful in **predicting and reducing hospital readmissions.** Determining the optimum time of patient discharge can pose quite a challenge: if done too early, a discharge can run the risk of relapse and exacerbating health problems. On the other hand, late discharge will generate unnecessary financial and human costs, e.g. prolonged lack of contact with family members. Predictive algorithms analyse the individual patient's situation and medical history to aid healthcare personnel in assessing the risk of discharge at a given point in time and recommend the optimum time frame, complete with the relevant justification and indication of possible consequences of discharging the patient ahead of schedule. This helps to limit the incidence of readmissions, improve the overall quality of care, and reduce treatment costs.

Intelligent systems also assist hospitals and healthcare centres in optimising their operating performance. Firstly, based on the available patient data, environmental (e.g. climatic) statistics, and event-specific information (epidemics, natural disasters, etc.), such systems help hospitals to forecast occupancy in both regular wards and ICUs. This allows them to prepare in advance by securing necessary resources, both in terms of infrastructure and staffing, which serves to reduce patient risks and improve the quality of service. In turn, solutions offered by companies such as Qventus or LeanTaas take advantage of machine learning to improve clinical workflow. Predictive and optimisation

algorithms facilitate better processual efficiency of infusion centres or operating rooms, as well as conduct in-depth analyses and re-engineer other processes performed at the hospital. The benefits include faster access to medical services due to optimised patient flow (e.g. by managing admission room waiting lists, processing standard test cycles, or predicting the incidence of no-show patients). Other benefits stem from shortening the duration of medical processes and procedures, reducing their costs, and assisting tactical and operational decision-making by implementing advanced systems for monitoring the efficiency of key processes.

An important factor affecting the operating efficiency of healthcare institutions is the ability to **automatize simple activities**. Autonomous logistics robots dedicated to healthcare applications offered by companies such as Aethon or Savioke can handle transport of a variety of items: from cleaning products, through food and medicines, to hazardous substances. They help to streamline processes such as nursing, medicine administration, feeding, cleaning, or transporting dirty laundry. They are capable of independently navigating hospitals (including elevators) without creating hazards to people or other equipment, delivering sensitive items (e.g. medicines, medical records, hazardous materials) to designated recipients ('in person'), and independently charging their batteries when necessary. In turn, cleaning robots supplied by Avidbots help to maintain interiors and control the required level of hygiene in particular areas. The benefits here include in particular reduced human workload but also improved security of transported goods (e.g. reduced risk of loss) and overall safety of both the personnel and patients (secure transport of hazardous substances). The growing popularity of financing options based on monthly leasing (Robot-as-a-Service) helps to limit the entry (e.g. investment cost) and exit barriers of such solutions, as well as to calculate the foreseeable return on investment (ROI).

Communication with patients can also be automatized, at least partially. The already mentioned Amelia bot offered by IPSoft can aid both hospitals and patients in booking and scheduling appointments and procedures, providing pre-consultations, distributing questionnaires, exchanging information, and managing tests. It can make appointments, track relevant test results, and help patients to better understand medical procedures. It

provides support in home treatment, particularly to patients using medical devices, transfers the results of relevant measurements to a physician, and offers advice on simple measures that can facilitate faster recovery. In turn, empathic robots offered by Consequential Robotics in the form of cute rabbit-like animatronics are designed to boost emotional recovery: equipped with six senses, the devices are capable of recognising emotions and reacting accordingly, which may constitute a valuable therapeutic aid particularly in the case of patients living alone. The solution helps to reduce patient discomfort as well as the workload of the medical personnel.

Payers

Intelligent systems can also support payers, i.e. organisations responsible for healthcare financing. In this context, the main applications of machine learning focus on the **identification and minimisation of fraud, waste and abuse.** Similarly to insurance, fraudulent healthcare claims constitute a significant budgetary burden for many countries (US Justice Department estimates that 3% of healthcare claims in the United States, worth almost a hundred billion dollars, are fraudulent). Models based on machine learning principles use previously identified patterns to facilitate automatization of claim assessment processes in terms of potential fraud and flagging of the most suspicious operations.

More advanced systems are also able of identifying new fraudulent patterns and methods, as well as provide controllers with reasons for the classification (reason codes). Early detection of fraud before a payment is even made greatly reduces losses and allows any suspicious transactions to be submitted for legal evaluation.

Public sector

Similarly to healthcare, the mission of the public sector is not focussed on generating profit but rather ensuring comfort and security. For this reason, the below analysis of benefits offered by AI systems will not follow the *drive revenue, cut costs* and *avoid risk* pattern.

One of the main uses of autonomous technologies in the public sector relates to a variety of **security measures**. Most commonly, the same entails not only surveillance and tracking in public spaces, but also projecting a variety of potential risks.

The most popular systems in this area offer the ability to analyse images obtained from CCTV and other perception systems, e.g. using LiDARs. For instance, Blickfield algorithms can track persons, groups of people, and objects (e.g. vehicles) to detect suspicious behaviour. If a potentially dangerous situation is detected, an alert is sent to the appropriate authorities.

A more mobile solution is provided by SMP Robotics. Its autonomous robots are designed to patrol critical infrastructure (e.g. factories, power plants, or water intakes), limited access areas (gated housing estates, golf clubs, resorts, etc.), university campuses, airports, governmental buildings, warehouses, parking lots, or data centres. The devices equipped with cameras, position sensors, and advanced image recognition systems offer not only the capacity for 24/7 patrolling but also such functionalities as e.g. facial recognition (up to 50 m) or thermal surveillance. Their autonomy includes the ability to monitor the designated area, avoid obstacles (including people), charge batteries when necessary, but also coordinate with other devices (swarm surveillance). Notably, the robots can effectively operate under difficult conditions (in temperatures from -25 to +55 degrees Celsius, high humidity and dustiness).

Such systems are typically used by security companies and public institutions with the view to limiting the costs of monitoring (replacing human guards), increasing efficiency (image recognition, advanced situational awareness, alerting), and improving the safety of security personnel (reaching otherwise inaccessible areas, lesser risk in the event of a conflict).

A similar solution - autonomous security robots – is supplied by Knightscope. The company's products offer similar functionalities to those provided by SMP Robotics and can be deployed both in open (e.g. parks) and closed spaces (shopping malls, conference centres).

Some of the most noteworthy benefits of such solutions include reduced assault and break-in rates, faster identification of fires, assistance in identifying insurance fraud attempts (e.g. thanks to footage recorded at the site), identification of hit-and-run incidents, reduced incidence of car and bicycle theft, acts of vandalism, and even sexual harassment. The measurable effects of implementation are mainly seen in the lower incident and crime rates in the monitored area, as well as reduced costs of the monitoring itself.

In turn, Nightingale Security employs a fleet of autonomous drones to monitor public spaces. The system can operate during the day and at night as well as in difficult weather conditions (including rain and snow), stream video in real time, coordinate its operation (swarm intelligence), independently charge batteries and signal the need for maintenance. The most common application scenarios include **autonomous threat response** (the system automatically dispatches a drone in an emergency and streams a video feed to the security team), **scheduled autonomous patrols** (with pre-programmed routes, elevation and camera positioning), and **manual surveillance missions** (performed e.g. during emergency evets such as chemical leaks or other industrial failures). The key benefits in this context include cost savings and fast response capabilities – the drones can accelerate to 100 km/h in a matter of seconds.

Another interesting application of intelligent systems in public security entails **crime prediction**. Based on historical data on crime incidence (mainly place, time, and type) as well as additional information (e.g. weather, surroundings, internet usage), predictive algorithms are increasingly able to asses the likelihood of a crime being committed at a given place and time. This allows security and police services to proactively prevent such events. Interestingly, the algorithms are capable of not only indicating the type, place and time of a crime, but also people who might attempt to commit it. Consequently, a natural application for such algorithms is in **counterterrorism –** one of the key aspects of preventing terrorist acts is currently related to analysing the online activity of suspects (or entire groups of people).

However, security is not the only area of AI applications in the public sector. Intelligent systems can identify, analyse, and monitor the behaviour of individuals and entire groups.

When combined with traffic flow analysis, such solutions may be used to optimise municipal transport. In the long term, this may facilitate better planning of city transport systems, e.g. in terms of constructing new roads, simulating the impact of various municipal transport solutions (e.g. transfer nodes) on congestion or the environment, as well as simply identifying available parking spaces in real time, which would be particularly useful in heavily congested cities. Furthermore, companies such as Lumebot supply municipalities with fleets of autonomous, collaborating street cleaning robots capable of working in diverse weather conditions (e.g. able to clear out snow).

Summary

The analyses presented above paint a rather compelling picture. Autonomous systems support companies in key areas of their operations: from design, through logistics, production, marketing, and sales, all the way to customer service. Companies able to wisely utilise the potential of such solutions gain a considerable competitive edge, which naturally influences the sectors in which they operate. This, in turn, determines not only the demand for intelligent systems, but also the growing expectations towards the same.

There is much to also suggest that we are now seeing an entirely new sector of autonomous solutions emerge. Similarly to the case of e.g. the automotive industry, where there is a very complex supply chain ecosystem in place to connect the manufacturer and the end user, the producers of e.g. autonomous robots or vehicles take advantage of advanced components supplied by specialised providers. One can therefore expect market of autonomous solutions to grow exponentially irrespective (to a reasonable extent of course) of their specific industry applications – this trend in itself can constitute an interesting subject of future research.

Now that we have discussed the key groups of autonomous technologies, their impact of functional areas and industries, we can proceed to the analysis of the best practices applicable to their implementation in organisations – which will be the focus of the subsequent Chapter.

Autonomous Systems in Value Generation

Practical aspects and good practices in autonomous systems implementation

In [@saleh2019] Saleh et al describe good practices applicable to the implementation of AI projects in large organisations. The publication identifies top management support as one of the key factors of success, with one of the major threats stemming from perceiving such projects as solutions that can not only be purchased as ready for deployment but also expected to generate quick returns (the approach is described as 'AI as plug and play'). Consequently, decisionmakers make large investments in pilot implementations only to face subsequent difficulties in the transition to ordered solutions closely interrelated with the wider program policy.

In turn, the report published by Kaggle.com [@zotero-1586] from a study conducted on over 16,000 respondents identified the following key challenges related to data science projects: bad data quality, difficulty in acquiring data, shortage of data science talents, lack of support from top management, lack of clear questions to answer, or the fact that the implemented systems are not used by decision-makers in everyday practice.

It seems clear that the practice of AI implementation is similar to that observed in the case of large-scale IT systems: the main danger stems from the misconception that technologies are capable of generating business value on their own, while in order to achieve one's goals, one must realize that success is conditional upon also, if not above all, proper implementation of change – not merely in terms of preparing employees but also duly modifying operating procedures, organizational structures and culture.

Presented below are some of the good practices applicable to the implementation of intelligent solutions, compiled based on both relevant expert reports and supplier recommendations. The same will provide the framework for the original approach described in the subsequent section under the working title of AIR Management.

Initiation

The first stage of a typical IT implementation process is referred to as **initiation**. It usually leads to the building a business case duly approved by the decision-makers. Presented below are several recommendations and examples to illustrate the possible methods of approaching this stage.

Preliminary assessment of possible goals and the role of AI in the organisation

As discussed in the previous chapter, **intelligent systems can offer a fairly extensive range of "skills"**. They are capable of environmental perception, in particular image and behaviour recognition, language comprehension, as well as broadly understood situational awareness. They can detect anomalies in in a variety of contexts, from internet traffic, through data from IoT sensors or credit card transactions, to pathological lesions in medical imaging. They provide forecasts for the future by evaluating such diverse factors as demand, weather conditions, value of financial assets, etc. They are also capable of operating autonomously, be it individually or in collaboration with other similar devices (intelligent swarm).

Those skills allow such solutions to perform a wide spectrum of **tasks** with varying levels of autonomy – e.g. on-demand human assistance, human collaboration, or human replacement. The tasks can also be performed in **a variety of places**, understood both as 'physical' locations (e.g. a production line, construction zone, travel route) and 'logical' places (a component of an intelligent product, an activity in a business process, or outside communication – e.g. with a client). As such, they serve certain specific **roles** in the context of an organisation and/or product. In order to perform their tasks, the solutions need access to specific **resources**: space, infrastructure, human support, integration with other systems, or simply data. A particular form of said human support entails preparing the machine to perform certain designated activities ('training', onboarding).

Tasks constitute the source of **value**. Intelligent systems assist and/or replace people, improve safety, facilitate faster response, often improve quality. The end results can

include cost reduction, productivity enhancement, or higher responsiveness. This, in turn, translates to tangible financial advantages and better customer satisfaction. As observed by Saleh et al. [@saleh2019], AI implementation can also affect **the way in which a given work is performed, the decision-making methodology,** as well as **organisational culture.** In particular, work within organisational silos needs to be replaced by work within interdisciplinary teams, decisions previously made by leaders based on professional experience have to be replaced by decisions made by line managers based on data, while a stiff organisational culture oriented towards risk minimisation needs to transform in a more flexible culture open to experimentation.

In light of the above, one must first consider the best possible **use for the intelligent system within the organisation**: be it associated with a given business process, location, or product. It is beneficial to already have a relatively good data culture in place: implementations of AI systems are more likely to succeed if they are to incrementally improve processes already based on high quality data, rather than in processes where such good practices are absent. The next step is to take advantage of expert interviews, opinions of business process owners, and information provided in supplier offers to determine the **tasks** that can be performed by intelligent systems in the context at hand. At this stage, it is particularly important to 'involve' (e.g. by inviting them to partake in the formulation of the business case) persons actively engaged in the given functional area – this helps to reduce the risk of later problems in the course of change implementation or adoption. Finally, one should preliminarily make sure that the **environment** in which the relevant tasks are performed is capable of accepting the introduction of autonomous solutions: a quick analysis may identify e.g. legal, safety related or infrastructural barriers to implementation. This initial analysis provides the basis for initiation of the subsequent stage: building the business case.

Building the business case

In considerably simplified terms, the object of the business case is to describe the change that will result from the implementation of the given solution and provide a cost and benefit analysis thereof as the basis for the decision whether or not to set the project in motion.

K. Veljko [@veljko2019] presented an interesting procedural model that accounts for the specificity of AI implementations. As suggested in this method, the first question one should ask pertains to the way in which AI is to be utilised by the organisation. The author recommends four options in this respect: support for **decision-making processes, AI as an integral part of an existing product, automatization of business process activities,** or **AI as a product in itself.** If the first option is selected (decision-making support), the subsequent questions will pertain to the exact type of decisions that are to be supported as well as the current way of making the same. In the second option (AI as a part of a product), K. Veljko recommends considering whether the product is to be fully (or only partially) automated and what tasks the AI is intended to perform (as well as under what conditions). If the ultimate goal is to automatize activities in a process, the subsequent questions will pertain to the current characteristics of the process, the expected changes (after the implementation), as well as the actual feasibility of the same. Finally, if AI is to constitute a product in itself, one should consider the benefits that it would provide to the potential customers and follow up with questions analogous to the ones posed for the other options. In summary: the first step is to roughly characterise the intelligent system and then identify its intended tasks and the environment wherein they are to be performed.

An interesting method (recommended to suppliers of robotic systems) of building a strong business case has been described by S. Kalouche in [@kalouche2019]. Firstly, one should reflect on the price of the solution that will be offered to the end user: the same should correspond to a portion of the total cost of employment applicable to the personnel that is to be replaced by robots, calculated relative to a certain, reasonable period of time (not longer than two years). Consequently, the suppliers of robotic solutions will need to be able to accurately identify the relevant structure of employment costs (remuneration and its derivatives, but also the costs of absenteeism, employee turnover, and related downtimes) in the industry to which the solution is to be marketed. This, in turn, assumes continuous collaboration with e.g. financial auditors specializing in issues related to the given sector or functional area. Alternatively, offers can be addressed to sectors wherein the workforce demand significantly outstrips the availability of skilled workers (as is now increasingly the case in e.g. logistics), which would take advantage of a market where

customers are desperately searching for operating solutions alternative to traditional employment.

Another important factor that ought to be taken into account when considering the business case for the implementation of a robotic solution is **workstation adaptation**. As observed by Kalouche, attempts to adapt robots to workstations designed to be operated by humans prove ineffective in the long-term perspective. On the one hand, one has to account for numerous greatly limiting safety concerns (e.g. movement speed or power), and on the other, designs should not be limited by the specifics of human labour. The fact that a given task is currently performed in a certain manner often does not mean that a more efficient alternative cannot be found. Hence, consideration should be given to the possibility of reorganising workstations and methods in such a way, so as to fully utilise the potential of robots – naturally, this is bound to affect the role of human employees, excluding them from certain tasks but also creating new ones, which may prove profitable in a longer-term perspective. Following this train of thought, one might reasonably expect the emergence in the near future of a trend related to reengineering work methodologies (and adequately adapting workspace organisation and human roles therein) with the view to maximising the return on investment in autonomous solutions (in short: designing work environments around robot capabilities).

One should bear in mind that industrial customers tend to be more inclined to purchase comprehensive robotic solutions rather than individual systems collaborating with other systems. In particular, this pertains to the requirement for a given activity to be performed in a practically maintenance-free manner with 99.9% reliability. The level of 95% that is commonly reached in laboratory conditions is therefore considered insufficient – such robots require nearly continuous supervision, which significantly limits heir capacity to generate tangible benefits. At the same time, practical examples seem to indicate that wile reaching the 95% threshold is relatively easy, any improvement beyond that point by even a single percentage point (or a fraction thereof) may indeed prove relatively cost intensive. Given the above, the author recommends focusing on the design of near perfect and narrowly specialised solutions rather than more universal but less reliable ones.

An interesting method of building the business case for the implementation of logistics robots has been proposed by Gideon Bros (por. [@2019a]). The company recommends that the first step should entail becoming acquainted with the basic concepts and methodologies of robotics as well as the most commonly employed models of robot application in logistics (e.g. the *Goods-to-Person, Person-to-Goods,* or *On-demand* modes discussed in the previous chapter). The next step ought to focus on determining the most pressing need that the robot fleet is to address: be it solving the problem of workforce availability (employment, employees quitting, absenteeism, etc.), improving work conditions and safety (which directly affects workforce availability), growing demands of customers in terms of shorter lead times, or exploiting the full potential of already existing warehouse management solutions.

Once the need has been identified, the next stage should involve an analysis of offers provided by suppliers of relevant solutions. The strategies for securing the resources necessary to implement an AI project will be described in detail in the subsequent section. At this point, let us only indicate that the recommended approach would be to firstly check the availability of ready-made solutions, secondly consider the possibility of adapting the existing systems, and only then, if the market is found to lack viable options – attempt to design and implement a dedicated solution. Hence, an in-depth analysis of the supplier market, e.g. in the case of Gideon Bros – manufacturers of warehouse robots, constitutes a very important next step in the efforts aimed at formulating a strong business case[12].

The need identified in the preceding step will serve as the basis for estimating the applicable return on investment (cf. [@2019b]). The authors of the study identify the following costs and savings in this context, which can be fairly easily generalised to include other autonomous solutions:

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[12] The criteria proposed by Gideon Bros in the context of selecting suitable warehouse robots, which can be fairly easily generalised to include other autonomous systems, were presented in the subsequent chapter.

1. Costs
   - Initial investment: one-off or, in the case of leasing, a monthly fee
   - Costs of delivery (one-off)
   - Costs of installation (one-off)
   - Costs of extended warranty: (one-off, not applicable to subscription models, e.g. Robot-as-a-Service)
   - Costs of employee training (one-off, with the option of follow-up, on-demand training)
   - Costs of energy (settled on a monthly basis)
   - Costs of other materials consumed, operation, maintenance and repair (settled on a monthly basis, may not apply to subscription models)
   - Costs of decommissioning and disposal (one-off, not applicable to subscription models)
   - Costs of downtime (robot unavailability) for reasons such as installation, employee training, maintenance and repair, or adaptation to new tasks
2. Savings
   - Reduced operator costs. Preliminary estimates can be obtained from the solution's supplier or verified at the stage of pilot implementation. Monthly savings can be calculated as the product of man-hours performed by the robots per month and the total cost of one man-hour to the employer (established in collaboration with the finance controlling department)
   - Reduced costs of employee turnover. Can be estimated based on the intra-month rate thereof, the overall number of employees, and the approximate cost of a new hire (including recruitment, administrative, medical, training, and onboarding costs)
   - Revenue from the sale of equipment replaced by robots
   - Improved space use efficiency. In many cases, the implementation of a robotic solution frees up considerable usable space, e.g. due to greater mobility (compared to the existing systems) or establishing a warehouse space dedicated exclusively to robot use, which means some of the safety requirements no longer need to be followed (e.g. in terms of shelf spacing).

The related savings can be calculated by applying the estimated percentage of freed space to the total warehouse floor area and the monthly costs of maintaining the same (e.g. lease cost)

- Savings due to eliminating the downtime experienced with currently used solutions. Analogically to the cost of downtime of the robots to be implemented, the downtimes of the current solution also generate a determinable cost (failures, maintenance and repair, replacement, etc.)

3. Once the relevant costs and savings have been estimated, the expected return on investment can be calculated fairly easily, e.g. by dividing the difference between costs and savings by total costs.

It is noteworthy that suppliers of autonomous systems tend to follow the current trend of switching from investment costs (CAPEX) to operating costs (OPEX). This is noticeable not only in the context of offerings available to companies (all the X-as-a-Service models, where X can stand form anything from computing power, through infrastructure, software, or robots, to intelligence or autonomy), but also those addressed to individual consumers (see e.g. the new practice of car manufacturers (e.g. BMW) entailing a shift from selling vehicles to models of providing mobility as a service (e.g. car-sharing or Mobility-as-a-Service).

In the context of autonomous solutions, suppliers commonly refer to the already discussed concept of Robot-as-a-Service. There are numerous benefits that this model brigs to the table from the customer's perspective: it eliminates the need for a high initial investment as well as the need to separately cover the costs of maintenance, repair or additional warranty, facilitates a relatively easily calculable and verifiable business case and, a factor whose significance should not be underestimated, a low exit barrier (quick disassembly by the supplier, which means no costs of decommissioning, scrapping or disposal).

Preparing the organisation for project implementation

Once the business case has been established, the next step in the initiation stage entails preparing the organisation for the potential change. As already mentioned above, Saleh et al. [@saleh2019] posited that AI implementation can affect **the ways in which a given**

work is performed, decision-making methodology, as well as **organisational culture.** Therefore, before setting such projects in motion, efforts should be made to ensure acceptance of said processes and engagement of change leaders. In particular, it is necessary to establish interdisciplinary teams so that the priorities of various groups can be duly addressed in the course of the implementation (business-oriented, technical, analytical) and new operational challenges identified (changes to processes, work methodologies, etc.). One should also take into account the possibility of changes to decision-making processes, including increased decentralisation and greater significance of data and/or recommendations generated by intelligent systems – in some cases greater flexibility, willingness to accept risks and engage in informed experimentation would also be recommendable. Moreover, the **potential barriers need to be identified** that might affect the changes resulting from the implementation of intelligent systems (anxiety?, organisational culture?, silo-specific processes?, shortage of talent?, unavailability of good data?).

It is also important to ensure early and **clear communication of the underlying reasons behind the project**. This helps to limit the anxiety and resistance of employees whose work will be affected by autonomous systems. The aforementioned Gideon Bros recommends the following stages of preparing internal communication strategies for the purposes of imminent implementation:

1. **Develop a communication plan.** It should constitute an integral part of the overall project planning, equally significant as the business case or implementation plans.
2. **Explain the underlying reasons.** It is worth referring to the aspects of improved quality and safety, as well as the possibility of enhancing personnel capacity (by reducing workloads). Oftentimes, the 'existential argument' may prove the most effective: *if we do not implement this solution, we will only benefit our competition and we may not survive.* In this case it tends to be helpful to reflect on four simple questions: *what will happen if we implement this change? what will happen if we do not? what will not happen if we implement the solution?* and *what will not happen if we do not?*

3. **Allow time to reflect**. After conducting a meeting with the staff during which the reasons for the planned changes have been be explained, it is a good idea to give them some time to reflect and analyse (approx. two weeks, although extreme changes may require more time). This allows for the first wave of emotional response to subside, the most ardent opponents as well as supporters of the change to be identified, and the groundwork for the subsequent negotiations (if necessary) to be laid.
4. **Focus on the opponents**. In the next stage, one ought to identify the most active opponents of the change and try to bring them onboard. Their position has a high likelihood of influencing those still on the fence about the project – hence it is certainly worth expending the effort to gain their support before proceeding with the implementation. In doing so, one should bear in mind that the change will affect every person on an individual level, and as such will often be met with an emotional response (emotion outweighs rationality). It is therefore important to start by gaining a good understanding of the effect that the intelligent system will have on the organisation's operations. Experiences of other companies shared e.g. by the supplier may prove valuable in this context.
5. The communication should also account for **generational differences:** the reception of the change may vary significantly depending on the employee's age but similarly, the style and content of a given message can have a different affect on different people.

As evident from the above, the described stages constitute fairly typical efforts undertaken by organisations with the view to managing any change, increasing the likelihood of project success, and providing a solid framework for the subsequent implementation.

Implementation

After goals have been defined and the organisational stage has been preliminarily set, the approved solution needs to be duly prepared and implemented. Presented below is a brief overview of the necessary resources and ways of securing them, methods of

implementing intelligent systems as well as preparing employees to be able to take full advantage thereof.

Resources

As already mentioned, the implementation of systems utilising AI is, in many aspects, similar to that of other complex IT systems. It is therefore hardly surprising that the resources needed in this context are also largely similar.

One thing that significantly sets e.g. advanced machine learning algorithms apart is the inherently high reliance on human contribution, particularly with regard to data science experts (commonly referred to as data scientists). In a sense, such persons serve as intermediaries between the decision-makers initiating the project and the IT or analytical departments that operate the infrastructure necessary for the implementation e.g. in terms of providing good quality data required by the algorithms. Their tasks in a typical data science project (in accordance with e.g. the CRISP-DM methodology, cf. e.g. [@chapman2004]) include in particular converting a given business need into a problem compatible with machine learning, identifying organisational problems that need to be solved (particularly in terms of data), gathering data, preparing the same for modelling (cleansing, integration, enhancement, and transformation), developing and evaluating models, and communicating the final results to decision-makers in a form they can understand.

Those tasks require competences in a wide spectrum of areas, in particular **mathematics and statistics** (algebra, differential calculus, probability, machine learning, modelling, experiment design, Bayesian inference, supervised, unsupervised and reinforcement learning, optimisation, etc.), **programming and database operation** (e.g. script languages such as Python, DataScience libraries, relational and non-relational databases, parallel computing and BigData processing, cloud computing), **domain-specific knowledge and soft skills** (understanding business processes in the given functional area, data curiosity, approach oriented towards solving (rather than just generating) problems, capacity for objectivism, and **communicativeness** (including in terms of data visualisation, digital story-telling, explaining the business consequences of algorithms at a given quality level, and recommending specific action based on analytical conclusions).

It is therefore unsurprising that even given the wide availability of very good coursebooks and training opportunities (both traditional and online), the market struggles with a limited supply of experts who, even if available, expect to be very well compensated for their work. On the other hand, companies are increasingly able to gain access to good quality data, they are also becoming more aware of the benefits that can be potentially gained by implementing intelligent systems. Consequently, the demand for expert services continues to significantly outstrip the supply.

There are several possible solutions to this problem. The first entails **minimisation of the entry barrier by developing simple solutions requiring only rudimentary competence in IT and statistics**. This niche has been explored by such suppliers of data science platforms as Dataiku, RapidMiner, DataRobot, Knime, or H2O (cf. also Gartner report discussing the landscape of this class of solutions [@zotero-1600]). Their software facilitates the performance of the entire data science process, including creation and evaluation of advanced machine learning models, practically without the need for any experience in programming or statistics. This allows the companies' customers to greatly accelerate the adoption of AI in their organisations by reducing entry barriers, and to significantly increase the potential of employees already working with data (e.g. business analysts).

Another solution could entail **recruiting employees without all the necessary competences but showing a high potential for development**. A particularly promising group in this context would be that of young scientists (e.g. in the process or shortly after earning their PhDs) whose research heavily relies on quantitative methods (e.g. in social sciences, life sciences, or mathematics). Their financial expectations are often significantly lower than those of professional IT specialists, while at the same time they bring to the table previous experience in working with data and many desirable soft skills such as research curiosity, ability to formulate and verify hypotheses, as well as plan experiments.

Employees could also be recruited through **online platforms organising data science competitions**. Kaggle.com is the current leader in this market, allowing various organisations (public and corporate alike) to submit problems that need to be solved, and

then inviting data science teams from around the world to compete in solving them. The incentives here go beyond eliminating a problem or winning a reward, as one of the key purposes of such competitions is, on the one hand, to identify talents, and on the other, to promote one's personal brand and secure interesting commissions/job offers.

Finally, yet another source of potential talents might be found in **institutions and universities offering expert training programmes in the areas of AI, machine learning and data science.** These include prestigious universities (operating via e.g. e-learning platforms such as Coursera or EdX), as well as companies specialising in exactly this type of activity (e.g. Udacity, DataCamp, or DataQuest) as their offer often goes beyond training itself to facilitate contacts between business and the most promising graduates.

Effective data science teams are able to not only independently develop AI solutions, but also support organisations in the process of searching for and negotiating with potential suppliers. Teams composed of ambitious programmers and machine learning experts may be tempted to develop a solution themselves – particularly since most of the libraries necessary to attempt such development (currently (2020), the most popular of which include Sklearn, PyTorch, and TensorFlow) are available under open licencing (i.e. practically for free). However, the complexity of AI projects strongly supports the viability of the approach long established in the IT sector: 'buy ready-made if available, adapt existing solutions if not, develop it yourself as the last resort'.

The first of the three options is particularly recommendable: to perform an in-depth analysis of solutions already existing on the market and determine the feasibility of their deployment in the given organisation. This stems from a number of reasons: the possibility of quickly verifying the viability of the business case (often before the expiry of the trial period), shorter implementation time-frame, high chances (albeit not certainty: many of the suppliers are startups whose future remains uncertain) of continued maintenance and development of the solution, less burdensome entry and exit barriers, etc. One should consider the possibility that the AI solutions required by the organisation, even if unknown to it, may in fact be readily available on the market. It is also often the case that the suppliers of solutions already used by the organisation (e.g.

specialist devices or production machinery) may update the software in their products by implementing certain AI components that will facilitate a significant improvement of their effectiveness. A good example of the same is provided by image recognition algorithms used to evaluate the quality of welds in welding machines, or even advanced software expansions offering image processing vapabilities. It may therefore be the case that intelligent solutions are withing arm's reach and the barriers to implementation are in fact negligible as the personnel is already familiar with the relevant systems.

At the same time, adaptation of a third party supplier's intelligent system to the requirements of a given organisation may prove to be a considerable challenge. A typical problem in this type of implementations relates to being able to build a viable business case without engaging in an in-depth relationship with the supplier, which may result in analyses that fail to account for certain factors significant to project's ultimate success. A good example here would be the purchase of a predictive maintenance system intended for industrial applications: the quality of algorithms offered by suppliers is often strictly related to their experiences rooted in a specific industry sector (e.g. energy), but not necessarily the requirements of a new client. Consequently, achieving the desired reliability of e.g. damage detection systems will require such algorithms to be trained in the course of the implementation, in particular by providing them with good quality data. However, after purchasing the solution and attempting to implement it, the customer might discover that they simply do not have such data at their disposal – which may not only lead to frustration but ultimately to failure of the entire project.

As already mentioned above, the alternative of independently developing the system should only be considered as the last resort. In the context of machine learning, the first steps of training usually yield very significant quality improvements, but with each subsequent iteration the learning curve tends to plateau – afterwards, each subsequent percentage point of improvement comes at a significant cost. For this reason, this particular approach should be adopted only if faced with a 'higher necessity', e.g. if the market is unable to provide a suitable solution or the same is justified by valid legal (e.g. regulatory) or safety concerns.

LivePerson – a supplier of intelligent chat-bots, offers an iterative methodology for implementing a ready-made AI solution consistent with the above considerations. Firstly, the company recommends using preconfigured bots developed on the basis of data obtained from numerous clients. LivePerson also develops chat-bots dedicated to specific industries. They are practically ready for implementation out of the box: they are compete with templates for addressing scenarios typical for the particular sector (including intentions and dialogue flows), support integration with domain systems (e.g. CRM), and are able to communicate across typical interaction channels. However, if the above skills prove insufficient, the company also offers a service under the name of Conversation Builder: the solution allows even a person without any programming skills whatsoever to develop a bot from the ground up. In turn, the Macen AI component allows the system to continuously learn and improve itself by using natural language understanding algorithms to recognise intentions and evaluate the effectiveness of the interaction.

Methodologies

As intelligent systems continue to be developed and implemented, certain implementation methodologies specific to this type of projects are being developed. Some have now been formalised and commonly accepted by expert circles (e.g. the mentioned CRISP-DM methodology), some are more of an ordered collection of good practices recommended by suppliers, while others can be directly inferred from case studies or based on the observed development of intelligent product ecosystems. Presented below are some of the most interesting methods employed in the development and implementation of AI systems, as identified in the course of the author's research.

The design, creation, and implementation of systems utilising artificial intelligence is still not without certain inherent risks, particularly when attempting to develop a system from the ground up. In this type of projects, realised by interdisciplinary teams with varying competences, experiences and workstyles, where adequate coordination is of vital importance, it may prove difficult to clearly define a common goal, settle for a consistent choice of methods and technologies, and therefore identify the necessary resources and agree on a viable time-frame. For this reason, many researchers and practitioners (cf. e.g. [@saleh2019]) lean towards so-called **agile methodologies** (e.g. SCRUM) in this context.

Agile methodologies, popular among startups and IT professionals, are particularly applicable to situations characterised by low certainty in terms of scopes, resources, and timeframes. Projects implemented in accordance with these principles are divided into short stages (one week to one month long) and expected to yield specifically defined results – for instance subsequent iterations of prototypes. The decision whether to continue the project is made at the end of each stage depending on the success thereof, projected future progress, and the sponsor's patience.

In the case of particularly complex projects (such as AI implementations and especially AI design from the ground up), the measure of success may be e.g. the achieved improvement of an algorithm (e.g. regression or classification) relative to its previous iteration and/or the related increase of business value (the problem of measuring the effectiveness of AI will be discussed in greater detail in a further section). The main benefits stemming from such an approach include better control and minimisation of business risks (the risk being limited to the cost of e.g. monthly of quarterly stages), but also the equally important comfort of the project team. High levels of project uncertainty render it difficult to prognose its scope and time-frame – under such circumstances excessive financial or time pressure may prove to be detrimental rather than beneficial. Moreover, excessive pressure can discourage key personnel, often scientists with low risk tolerance, from becoming involved in similar projects in the future, which , given the short supply of talent on the market, may prevent the organisation from proceeding with its other AI-related initiatives.

The methodology employed by IAM Robotics, a supplier of logistics robots, may serve as a good example of the iterative approach to implementing a ready-made solution (cf. [@zotero-1602]). In the initial stage (**feasibility and planning**), the company prepares a feasibility study with the aim of performing an in-depth analysis of operational needs and challenges, defining expectations, developing the action plan, and formulating the business case (including a return on investment (ROI) assessment). The subsequent stage (**build and integrate**) entails the performance of a pilot implementation (the solution is created and tested) at one of the designated locations. Once stage two is successfully completed, the company recommends proceeding to stage three (**site deployment**)

wherein the system is adapted to the applicable safety procedures and methods of measuring performance. Successful completion of this stage allows the process to continue to stage four (**full-scale development**) during which the system is implemented at the customer's remaining sites.

Another implementation methodology viable in the context of intelligent solutions, in a sense an expansion on the idea of prototyping adopted in agile methodologies, entails gradual **skill-by-skill** deployment. This approach has been successfully employed by Amazon the development of its intelligent assistant – Alexa (in February, the number of its skills exceeded 100 thousand items [@zotero-1606,@2020] classified under such categories as Business & Finance, Communication, Education & Reference, Food & Drink, Games & Trivia, Health & Fitness, Sports, Travel & Transportation, Home Service, and even Connected Car).

A similar approach was also adopted by the already mentioned Aera Technologies. By integrating its solution with domain-specific organisation systems, and taking advantage of advanced data processing and analytical technologies, the company has been launching subsequent skills ('skill-by-skill') under such categories as Demand Management, Manufacturing Performance, Supply Chain 360, Revenue Maximization, Margin Optimization, Predictive Sourcing, Delivery Optimization, Market Share Growth, or Working Capital. Similarly, Albert.ai, a company offering optimisation systems for online campaigns, has been developing its product by implementing individual skills in the areas of planning, creation, monitoring, and optimisation of campaigns. IBM has also been consistently adding new cognitive skills to its Watson.

Notably, what distinguishes companies such as Aera Technologies and Albert.ai from Amazon and IBM, apart from the obvious aspect of scope and target group for the intelligent services, is the composition of the companies' development teams: in the former case (Aera Technologies and Albert.ai), solutions are designed by internal teams, while Amazon and IBM also encourage their partners (independent contractors) to contribute to the development, thus creating an entire ecosystem around their core cognitive services. Ultimately, this may provide the basis for structures such as cognitive networks which will be discussed in a further section of this book. Nonetheless, all of the

above methodologies remain consistent with the already mentioned recommendation by Kalouche: it is better to offer a single solution that works perfectly than to offer many less than perfect ones.

Apart from the aforementioned agile and 'skill-by-skill' methodologies, one could mention at least two other methods employed in the implementation of intelligent solutions. The first relies on gradual development of increasing **autonomy levels**. This approach has been successfully used by the already mentioned ASI Robotics whose recommendation can be boiled down to the following principle: 'Start with basic automation; upgrade when ready'.

As the first step, the company recommends implementing the simplest remote control solutions. The operator remains nearby but at a safe distance from the device, controlling it with a dedicated handheld remote. This allows employees to relatively easily and quickly become acquainted with the new system. The subsequent stage entails transition to operation from a more distant point (teleoperation). In this case, operators control devices from a separate room, overseeing their work with via dedicated camera systems. Only after this stage is it recommended to move on to automated operation that utilises the full potential of the autonomous solution. This approach is also reflected in calculators of value generated by autonomous systems. For instance in the mining industry (cf. [@zotero-1608]), the company recommends starting the implementation with work assistance devices in areas such drilling, blasting, loading, or slag heap formation, and only then gradually transposing to fully autonomous solutions.

The last methodology to be discussed here aims to minimise the risk of implementing autonomous solutions by introducing gradual levels of confidence in AI recommendations – for this reason, it is often described as the '**by-confidence-level**' method. It is employed by Digital Genius in the context of its chat-bot solutions. For instance, a bot utilised by one of the airlines categorises questions into those where answers can be provided with high confidence (confidence level > 95%, e.g. questions regarding the allowable hand baggage limits) – which it answers independently, those with confidence under 95% (e.g. 80-95%) – where it recommends the possible answer to the human employee, and those with low confidence levels (e.g. under 80%) – where the question is transferred directly for human

processing. On the one hand, this approach ('by-confidence-level') facilitates automatic processing of many simple queries (thus reducing the workload of customer service staff), and on the other, minimises the risk of wrong answers and the resulting potential costs (e.g. complaints). The principle of limited confidence in autonomous solutions can paradoxically increase their adoption capacity and, with the available self-learning mechanisms, allow companies to gradually improve their effectiveness. It is noteworthy that this method is, in a way, an alternative to the 'skill-by-skill' strategy as it facilitates fast implementation of less than perfect solutions whose operation continues to be supervised by humans.

As follows from the above, the specificity of intelligent systems forces providers to develop new, dedicated implementation methodologies. In a sense, each of the same is a variation on the so-called agile approach: be it 'skill-by-skill', 'by-autonomy', or 'by-confidence level'. Each of the methodologies is heavily reliant on prototyping – the overall development of the production capacity relative to the respective implementation methods applicable to autonomous systems will undoubtedly constitute an interesting area to be considered in the context of management.

Preparing employees and the organisation

The next natural step of implementation, characteristic also of any IT system introduction, entails preparing employees to be able to take full advantage of the potential offered by intelligent solutions. In [@saleh2019] Saleh et al. provide the following recommendations:

1. **Define clear roles in the AI project.** Make sure you understand the key competences required and tasks/areas of responsibility involved.
2. **Establish 'interpreter' teams** – persons who will act as intermediaries between business and data science or technology teams. Their role will be to take part in the building the business case, communicate the relevant needs to IT/AI departments, and support the adoption of the solution in particular business units.
3. **Integrate analytical and business teams.** Organise the project in a way that facilitates rather than hinders cooperation.

4. **Wisely prepare and perform data organisation and cleansing projects.** Organise them around the most compeling business case without spending too much time or money on comprehensive data governance.
5. **Organise trainings for key process participants and end users.** Take advantage of various available formats (F2F classes, online courses, workshops, on-the-job training, study visits, etc.). Prepare the leaders (top-tier management with adequate of how the AI works, able to identify and prioritise the capacities of such solutions, necessary culture changes, etc.), the data science teams (hard and soft skills necessary for the design and development of intelligent systems), as well as the end users: decision-makers, managers, first-line specialists (prepare them for the practical use of the solutions).

As we can see, this step in a way complements the organisational preparation steps taken during the project initiation stage. The goal here is to consolidate the change resulting from the implementation of intelligent systems, as discussed in the subsequent section of this book.

Consolidating change

In order for the change introduced with the implementation of an intelligent system to find a permanent place within the organisation, people engaged in the processes affected by the same need to see and understand its effects, not only from the technological but also business perspective. For this reason, it is vital for the owners of AI initiatives to be staff members of business units rather than analytical or IT departments. It is they, naturally with adequate support from experts well versed in AI, who should identify business goals, manage scopes of responsibility (ideally in the entire project) and monitor the business-wide impact. The latter requires developing a suitable performance management framework (cf. e.g. [@saleh2019]) and continuous monitoring of the relevant indicators.

Such combination of technical (e.g. algorithm quality or efficiency) and business-centric measures constitutes a new, interesting area for scientific research. For several years now, Provost and Fawcett [@provost2013] have been using the concept of a so-called

profit curve to determine the correlation between such distinct measures, with the technical values corresponding to the cause variable (axis of abscissae (X)) and the business values as the response variable (axis of ordinates (Y)). With this approach, already at the stage of formulating the business case, one is able to determine the economically viable values of technical parameters (e.g. quality of classification algorithms), which may significantly facilitate definition of the first or subsequent iterations in a data science projects (the first stage under the CRISP-DM methodology). Notably, the practical application of this approach (and other similar variants) does not require advanced knowledge of machine learning methods and helps to significantly reduce the risks inherent in AI implementations, as well as monitor the benefits derived therefrom, thus consolidating the change in the context of the organisation.

Another very important factor is the active engagement in and support for the project from the relevant leaders. Typical good practices in this area (cf. [@saleh2019]) include **walk-the-talk** management, active training participation, continuous informative and inspirational communication with teams involved in the project, asking about the effects of the implementation (*what works? what could be improved?*), and encouraging experimentation. It is also highly advisable to urge or even oblige managers to periodically (e.g. on a quarterly basis) recommend possible applications of AI methods in facilitating better performance in their respective areas of business activity, e.g. in the form of short presentations or reports.

Summary

As follows from the above, similarly to IT systems, the 'installation' of an intelligent technology as such is far from sufficient for effective implementation and consolidation of change, i.e. the ultimate goal of the entire project. It is necessary to activate additional initiatives (communication on change, training, performance monitoring, etc.) and to ensure active support from the top management.

Dataiku, the supplier of one of the most successful data science platforms, identifies the following good implementation practices, so-called Proof of Concept (PoC), in data science

projects (cf. [@zotero-1613]), which may be quoted here as a good recapitulation of the deliberations presented above:

1. **Choose a realistic, specific application scenario.** Focus on solutions to real, significant problems. Analyse existing processes and determine how and to what extent machine learning algorithms could be helpful in improving the same.
2. **Adopt a reasonable time frame.** In most cases, 60 days should be enough to formulate the PoC, on the one hand, allowing comfortable evaluation of its potential, and on he other, preventing the investment of excessive resources.
3. **Clearly define the expected project outcome.** In particular the methods and measures to be used in determining the success rate of the PoC.
4. **Engage the right people.** Data scientists and analysts, IT teams, and managers (where possible).
5. **Collaborate with the solution's beneficiaries**: end users, domain experts, and process owners. Their experience, particularly understanding of the limitations inherent in a production implementation, will be invaluable even in the early stages.
6. **Be flexible, but focused.** Excessive flexibility may sometimes hinder the achievement of the adopted goal, discourage greater effort or create distractions (too many potential options). Hence, it is important to strike the adequate balance between agility and focus the ultimate objective.

Benefits of implementing intelligent systems

One of the main aims of this book is to propose a method for identifying values that may be generated by autonomous systems, building a business case based on the same, and developing strategies for adopting such solutions in organisations. The issues have been subject to research scrutiny by both the academia [cf. e.g. @porter2015, @saleh2019] and consulting companies [cf. e.g. @bughin]. Therefore, when developing the methodology of generating value for self-learning, autonomous systems which constitute a natural sub-group of a larger class of AI solutions, it would be prudent to consider the previously proposed concepts as a certain theoretical framework.

The deliberations presented below rely on the model proposed in [@wodecki2018]. Firstly, the factographic material presented in Chapter 2 was used to identify the particular values generated by autonomous systems at the level of an organisation. Next, a model was proposed to allow the adoption of such solutions (under the working name of AIR Management) based on analogies with good practices applicable to talent management. For the sake of simplicity, the analyses presented below will be limited to the values generated by autonomous systems, while the risks related thereto will only be mentioned indirectly in the *avoid risk* section.

Chapter 2 presented numerous case studies under the categories corresponding to the technologies constituting the basis of autonomous systems, as well as their impact on key functional areas in organisations and industry sectors. The latter two sections considered not only the possible applications but also the potential benefits of implementing the same. What follows below is a synthesis of the values themselves, which will prove helpful when formulating the business case for an AI project, as discussed in a further part of the analysis.

The below indications for introducing intelligent systems have been formulated based on an analysis of several hundred case studies, in particular derived from offers presented by suppliers of sector-specific solutions and data science platforms. There are several viable ways in which the same could be organised, however, in the present considerations the categorisation criteria rely on the distinction between *revenue generation, cost reduction, and risk avoidance*. Such a classification is not only relatively simple and understandable (particularly to decision-makers), but also justified by the results of studies on the effectiveness of AI implementations in real-life organisations: it has been discovered that most of the successful projects were those developed around 'positive' (i.e. growth-oriented rather than targeted at cutting costs) business cases.

In the below presentation of the respective benefits it will be difficult to avoid some repetition of information already presented in Chapter 2, particularly in the section focusing on the impact of intelligent solutions on the key areas of operational activity. With this in mind, the number of illustrative application scenarios (which took up a considerable part of the previous chapter) was kept to a minimum and the main focus was

shifted instead to the actual benefits that can be derived from the same. The values are presented below in a synthetic form, steering clear, wherever possible, from referring to specific functional areas or industries.

Drive revenue

The analysis of benefits that may stem from the introduction of autonomous systems starts with the presentation of possibilities created by the same in the context of organisational growth, in particular market expansion and revenue increase.

Intelligent, integrated products equipped with sensors that allow tracking the ways and context of their end use (including place, time, even weather) offer a great potential for creating new, even better solutions, as well as entirely new business models (cf. e.g. @porter2015). They replace the traditional methods of evaluating user satisfaction with cheap, fast and reliable technologies capable of providing adequately processed measurement data directly to project divisions (rather than via customer service or marketing structures) – which has the added effect of improving the overall information flow. In-depth analyses of the thus obtained data constitute one of the means of identifying behaviour patterns and (often unrealised) user needs. Consequently, **intelligent systems facilitate the processes of streamlining the existing products and services as well as developing new ones.**

Data received from sensors installed in devices are one of many available sources of information about customers. Clients leave a so-called digital trail from all their internet and social media interactions, information regarding their activity is also increasingly available from various video monitoring systems and other sensors present in the public space. Such data, processed by machine learning algorithms, facilitate a new, better **understanding of customers** (know-your-customer – KYC). The resulting benefits include the possibility of very precise **segmentation** (so-called micro-segmentation, in extreme cases even down to the individual level) based on key customer characteristics. This allows a considerably deeper insight into their **behavioural patterns, habits, opinions**, and **shopping preferences**, facilitates prediction of the **most likely time and object of their next purchase** (next-to-buy), as well as the **next stage of their purchase**

funnel (next-action). Equipped with such information (in a sense already knowledge) about the customer, companies are able to identify the optimum offer at any given time, which constitutes a sound basis for next-level **personalisation of products and services**.

Machine learning algorithms facilitate not only tailored products but also communication. Contemporary customers expect access to multichannel communication, preferably fast and compatible to their current needs. AI can support companies in identifying not only the optimum means of such communication, but also the actual form and style thereof. This allows organisations to **continue expanding the scope of their omnichannel communication, optimise marketing messages,** and **generally improve their customer service quality**.

Additional benefits include **better effectiveness of marketing campaigns** and **development of deeper relationships with customers** by **anticipating their needs** as soon as they emerge (and sometimes even beforehand – cf. next-to-buy), identifying the customers most likely to purchase the promoted product (and therefore engaging in more targeted campaigns), and ensuring that customers receive the support then need, thus boosting **sales** (also through effective up-sell and cross-sell strategies). At the end of the day, apart from direct business benefits, companies achieve better **satisfaction** by providing cutomers with exactly what they need, in a form that is attractive to them (form and channel of communication), at the right time, while simultaneously not losing the element of positive surprise.

Intelligent systems also facilitate optimum pricing that dynamically accounts for not only current market demand but also the spending potential of a given customer. Such methods of *dynamic pricing* account for the financial standing and price sensitivity of a given customer, as well as various geographical factors. This allows sellers to **maximise margins without significantly huting the demand** and to **identify the best business solutions**, in particular the so-called product-market fit (which is especially important for start-ups that need to very quickly find an acceptable equilibrium between the complimentary and paid facets of their offerings, identify the payer, and establish optimum price levels).

AI algorithms also facilitate **demand and sales forecasting**. This helps not only to better design **promotional campaigns**, but also to **optimise supplier purchases** (and consequently stock levels) and **reduce delivery times**. Additionally, a detailed analysis of sales (especially shopping carts) allows vendors to design **optimum product combinations** (so-called product-mix), which, combined with all the aforementioned benefits facilitates **fast (short time-to-market)** and effective (sensitive to current needs) **adaptation to existing market conditions.**

A synthetic summary of the values that can be generated by intelligent systems in the area of *drive* revenue has been presented in the table below.

Product	Development	Enhance product design & support
		Build intelligent product
	Pricing	Optimise pricing to price sensitivity by geography and individual clients
		Improve pricing and capture the best business opportunities
	Sales	Forecast product sales more accurately
	Quality	Improve product and service quality
	Product mix	Ensure the right product mix
		Find powerful product combinations
Customer Understanding and Experience	Better customer understanding	Understand customer segments
		Understand customers needs
		Understand a customer's spending preferences
		Assess the long-term value of each customer to personalise service
	Better reach and communication	Expand multichannel strategies
		Build more effective patient/customer messaging
	Better marketing campaigns	Optimise marketing campaigns, messaging, and call center operations
		Boost the results of every campaign
	Personalisation	Anticipate next customer CRM state (e.g. activation, regular, high value, decliner, dormant, churn) to inform the strategy of future marketing comms

		Hyper-personalise and identify the most effective products to present to each individual customer based on historical data
		Predict time to next purchase order based upon prior purchase history and customer segmentation
	Better customer satisfaction	Anticipate client needs and identify new needs as they arise
		Determine which client is likely to need which product or service
		Expand customer relationships through targeted up-sell & cross-sell
		Precisely target offers
		Ensure clients get the support they need when they need it
		Use analytics to understand client price sensitivity and preferences
Market	Trust	Be a leader in quality rankings
	Agility	Adjust to market conditions
		Decrease time to market

Table 2. Possible values that can be generated by AI systems in terms of organisational expansion and driving revenues. Source: own elaboration.

Cut costs

Cost reduction is the currently dominant meta-category of benefits generated by intelligent systems. The main source of savings stems from broadly understood **automation** entailing, to a varied extent, replacement of humans in the performance of various tasks (cf. the different definitions of autonomy discussed in Chapter 1). Automatization can pertain not only to manual tasks (e.g. in logistics or production), but also to work traditionally performed by experts (e.g. medical diagnoses or analyses of bank transactions to identify potential fraud) or customer service (conversation bots), as well as even the processes of knowledge extraction (cf. *conversational intelligence*).

Particularly intensive automatization is currently observed in the context of **asset management** processes. Systems of visual surveillance utilising sensors (optical, laser and thermal) installed in fixed locations or on mobile drones enable **monitoring** of office, industrial, and public spaces. Their capacity to recognise, track and count objects can

generate benefits such as **optimisation of warehouse space use** and product distribution on shelves, **optimisation of staffing for respective workstations** (depending on the current demand, e.g. in retail outlets), **improved communication** (thanks to coordination and better planning of such systems in the context of e.g. cities, but also in enclosed spaces such as factories of university campuses), as well as **early identification of threats** (e.g. crop blights). Automated monitoring can also provide data for the purposes of evaluating the performance of both people and machines (so-called accountability) which, when correctly employed, results in **improved productivity**. Moreover, collecting data on the traffic taking place over ICT networks can lead to significant **security** improvements.

Automation can also be applied in the context of scheduled monitoring activities, i.e. **inspections**. Autonomous systems, both ground and aerial vehicles, facilitate **remote inspections** conducted over large geographical areas e.g. construction sites, airports, or large structures such as bridges or high voltage lines. Data obtained from such sources can be processed by people (in which case automation is not complete) or the system itself (with operators receiving only selected, already processed data about e.g. a potential failure, threat, fraud attempt, or deviation from adopted investment plans). Dedicated systems are capable of processing data to e.g. generate high-resolution maps, perform automated dimensional calculations (distances, areas, volumes) , as well as identify human and machine behaviours.

The main benefits generated by this class of solutions include **reduced inspection costs, time savings** (a drone can reach the inspected machine faster than a person), **operating continuity** (systems can perform their tasks virtually nonstop, often in very difficult conditions (e.g. low visibility or bad weather)), **improved quality** (sensor data are more accurate and their interpretation by a machine is often more accurate than by what can be reasonably expected of humans), and **improved employee safety**, both active (operators do not need to personally conduct inspections in potentially hazardous areas, e.g. in mines or at high altitudes) and passive (machines can detect people present in hazardous areas, e.g. before commencing with demolition). Autonomous inspection systems can also help to **reduce losses due to illicit activities** (e.g. theft at construction

sites, smuggling in ports, as well as e.g. suspicious credit card transactions, insurance fraud, or money laundering), and **limit the risk of damage and the costs of resulting repairs** (the generated data can be fed to *predictive maintenance* systems). Additionally, data provided by such systems can constitute the basis for the creation of digital models of devices or areas, which in turn translates to **improved quality of planning and development of the monitored devices/spaces**. It is noteworthy at this point that in the context of such solutions, automation means in particular **maintenance-free operation**: if necessary, the need for maintenance can be reported automatically, the devices are capable of charging their own batteries when needed and, where missions are performed jointly by a number of machines, independently coordinating their work.

Automation can also affect the sphere of infrastructure **maintenance**. A good example would be the use of cleaning robots in places such as airports, hospitals, schools, shopping malls, or production lines, even parks or streets. The typical benefits include **time savings for employees, reduced workloads, higher quality of cleaning** (and automatic supervision of compliance with applicable standards, e.g. in hospitals), **increased corporate resilience to risks related to absenteeism or employee turnover**, and finally simply **improved productivity** and **cost efficiency**. Furthermore, some implementations also serve to **increase employee safety** (e.g. cleaning objects in the presence of hazardous compounds) and more reliable reporting facilitating **quick profitability assessment** for such solutions.

Another interesting example of streamlining customer services is the use of conversation bots for processing helpdesk requests in large organisations. Such bots are able to provide automatic responses to many standard questions, and help users in solving the most common problems. The resulting benefits include in particular **improved responsiveness of call centres** (measured by **reduced average time needed to process a query** by up to 90%), the related **savings** (reduced costs of repair services), and **better employee productivity** (shorter downtimes due to system unavailability).

Similar benefits can be achieved by implementing conversation bots in processes related to general customer service. In this case, additional value stems from **automated analyses of customer interactions across a variety of channels** (so-called

conversation intelligence) and the related benefits such as **more comprehensive knowledge based on behavioural patterns and good marketing practices, reduced time frame and improved quality of playbook development processes, effective adaptation of such guidelines to quickly evolving products and varied, also evolving customer segments,** as well as the resulting **improved efficiency of sales processes**.

Furthermore, many key corporate processes can also be subject to automatization. On production floors and in logistic centres, robots replace or assist people in the performance of many manual activities (cf. scenarios of production applications) as well as in monitoring and quality control tasks. The benefits include often significantly **increased performance (higher quality, shorter lead times), reduced operating risks** thanks to operational continuity (shorter downtimes due to e.g. human errors or absenteeism), **improved safety, boosted productivity, and lowered costs**. In the case of medical diagnostics (imaging analyses in e.g. magnetic resonance) the potential benefit (should such solutions become more widespread) would entail **better availability** (increased efficiency, shorter analysis time) **and reduced cost of medical services**, as well as, most likely, **improved accuracy** of diagnoses as such.

In yet another context, automatization of online advertising is fast becoming the go to solution to an increasingly common problem: the overwhelming complexity and dynamic growth of the e-marketing segment, as well as the resulting shortage of experts capable of not only effectively running such campaigns with the use of advanced tools, but even simply keeping up with the rapid advances made in this area. Automated systems dedicated to online campaigns facilitate a significant simplification of such processes and consequently allow such projects to be managed by people without in-depth business or technical knowledge. The main value here stems from the **availability of advanced e-marketing methods** to companies that would otherwise have a hard time using them, as well as reduced costs and increased effectiveness of campaigns.

A parallel problem, in this case in the areas of data science and machine learning, is addressed by DataScience platforms and so-called AutoML (auto machine learning) methods. Similarly to e-marketing, the scope of competences required to effectively utilise such technologies is so wide that even large companies capable of securing good

quality data, considerable budgets, and highly determined top management can struggle to recruit suitable teams. DataScience platforms help to **reduce the entry threshold** of such solutions thus increasing the availability of methods and tools, lowering the costs of processes, and giving many organisations a **viable chance of generating value from the data and information they possess.**

Another interesting area in which autonomous systems can offer considerable advantages is improvement of back-office productivity. Classification algorithms are increasingly effective in automating many activities performed with the use of IT systems, e.g. automatic classification of accounting documents (by analysing electronic documents or photographs of hardcopy invoices), distribution of incoming mail, or workflow management. Such systems, marketed as *Robotic Process Automation (RPA)*, are continuously developed – so much so in fact that a new subcategory of solutions is already emerging, described as *Cognitive RPA* – i.e. systems taking advantage of more advanced cognitive functions and dedicated to automatizing processes in expert systems (e.g. management of medical documentation). Such products are employed not only by large organisations, but also their shared services centres and companies operating in the BPO (Business Process Outsourcing) market. The benefits generated by such systems include primarily **reduced employee workload** (or elimination of employee involvement in simpler cases) and the resulting **savings, efficiency** and **productivity improvements**. Moreover, it is noteworthy that automatization of office work does not have to be limited to information (or documentation) management: significant productivity improvements can be achieved by supporting employees even in simple, everyday office tasks such as e.g. distribution of documents, food or coffee – a functionality offered by the already discussed office logistics robots.

Another source of value resulting from implementations of intelligent solutions stems from **optimisation**. What sets it apart from automatization is that is does not need to replace a human employee: in this context the performance improvement usually stems from identifying a set of parameters for a given process or system to maximise a particular target function (e.g. cost reduction, acceleration, or quality improvement). The ultimate effects may be similar to those discussed above (automation) but their genesis is

somewhat different. It its noteworthy that under real-world circumstances, one would be hard pressed to separate automation from optimisation – they are strongly intertwined, in fact optimisation is often one of the effects of automation, and automatic systems typically include built-in optimisation mechanisms.

As already mentioned, optimisation can be applied to product development processes. Designers can use artificial intelligence by providing it with only certain general conditions (e.g. shape, durability, maximum weight, or acceptable material use) and allowing the intelligent system to analyse the often vast range of possible parameters to identify a solution best suited to a given application. This process can sometimes yield constructions that humans would have a hard time conceiving independently (interestingly, often not unlike structures found in living organisms). The main benefits here include **increased effectiveness of the development process** and **production savings** due to the identification of the most cost-effective solutions (e.g. cheaper or fewer materials). Similar methods are also employed in the development of new medicines (active ingredients) – with the main purpose being a **reduction of development costs** (which in the context of pharmaceutics can be quite overwhelming).

An important source of value can also be found in **simplification**. The implementation of autonomous solutions, particularly in production or logistics, often entails considerable investment expenditures: necessary to not only adapt workstations to accommodate machines but oftentimes also significantly overhaul the entire factory. For this reason, one of the more interesting trends relates to so-called **on-demand automation**, where solutions are specifically designed in such a way, so as to minimise, or better yet eliminate the entry barrier. For instance, suppliers of logistics robots design their products to allow customer implementation in literally a single day, without the need to create dedicated routes, integrate anything with existing IT systems, adapt robots to particular workstations, or conduct advanced employee trainings. Naturally, this approach is not without some considerable challenges but may prove highly profitable in the log run: such a **low entry barrier (nearly no cost of investment)** is bound to encourage potential customers to test the solution and will work well in the **robot-as-a-service model**, consequently limiting the costs of system deployment to **operating costs**, which in turn

makes them easier to calculate and monitor, as well as facilitates a relatively **fast return on investment**. Additional benefits may include **easy scalability** e.g. to include other facilities, **no repair and maintenance costs or costs of additional insurance**, as well as **negligible exit barrier** (the supplier can collect the robots from their place of deployment within a single day). It is noteworthy that simplification can be achieved not only in the context of logistics or manufacturing, but also e.g. with respect to establishing Wi-Fi networks (e.g. with the use of drones serving as relay stations), as well as many other activities of varying complexity.

Also the learning processes employed in autonomous solutions themselves may be subject to optimisation. The same can be accomplished at even relatively 'low' IT levels (e.g. with the use of dedicated processors or adequately optimised machine learning algorithms), as well as the 'macro' level, e.g. with the use of advanced methods of on-the-job robot training (cf. the already discusses *imitation learning* methods). In fact, the autonomous vehicles industry now accommodates an entire group of providers offering systems facilitating faster training of various machine learning algorithms used in such vehicles, as well as situational generators and simulators that can be used both for the purposes of training and testing vehicle software. The main benefits of such solutions include **accelerated training of intelligent systems, lowered costs** and **improved safety** (simulators are safer, faster, and more cost-effective than testing in real-world conditions).

Intelligent systems are also used to **optimise resource management**. For instance in the mining industry, the same can boil down to optimising the drilling locations, forecasting the availability of natural resources, or managing mining machinery – so that, at the end of the day, **margins** generated from the entire process can be **maximised**. A particular role in this context is played by predictive algorithms: they facilitate not only fault prediction (which can **lower the costs of repairs**) but also forecast e.g. the number of customers visiting stores or number of helpdesk calls to determine the **optimum staffing at a given time and place**, which in turn **reduces employment costs** (no overemployment) **with minimum risks of negatively affecting quality.** Furthermore, robots scanning shelves in commercial outlets and recommending, in real time,

restocking orders or changes to be implemented by employees help to **minimise the loss of profits** (due to unavailability of products) and **legal costs resulting from breaching supplier agreements**.

Machine learning algorithms also aid optimisation of supply chains. With precise forecasts of the availability of specific good from specific suppliers, companies can predict potential delays, consequently **reducing their operating risk** and **maintaining operational continuity**. Additionally, supplier scoring systems aid representatives in negotiating commercial terms and optimising purchasing strategies, which facilitates **cost reductions** and **margin maximalisation.**

Production processes are yet another area that has long been subject to consistent optimisation efforts. AI algorithms help energy providers to better forecast supply and demand levels (e.g. in the case of prosumer clusters), which allows them to project peak consumption periods and take adequate preventive action (e.g. by offering limited consumption discounts). This not only serves to **limit operating risks** (grid overload) but also aids them in **developing entirely new products** (e.g. based on data generated from intelligent networks and sensors (smart metering)). Demand forecasts are also very valuable for traditional production companies as they allow them to **limit** overstocking and **the related costs** (frozen assets, costs of storage, loss of profits). First-line production managers are also supported by production process simulators, which allow them to utilise e.g. reinforcement learning algorithms to optimise the same (cf. solutions offered by companies such as AnyLogic [@zotero-1615] or Cognata).

Intelligent systems are also employed for the purposes of **optimising financial management**. In particular, predictive algorithms provide revenue projections, assist managers in managing financial flows, automatize the processes of financial controlling (by classifying transactions and identifying potential anomalies), and even recommend optimisation possibilities in various functional areas (cf. Aera Technologies solutions). In this sense, they become an important component of decision-making processes: the manager's main focus can shift from tasks related to data analyses (the area of *descriptive analytics*), identification of cause and effect relationships (*diagnostics*), and forecasting, to evaluation of recommendations provided by the system (including analyses of their

potential consequences in the context of identified market trends). The main benefit here stems from **limiting the extent of managerial tasks** and allowing leaders to focus on more strategic areas.

Autonomous systems and AI algorithms are also increasingly used to **optimise sales**. Apart from the already mentioned benefits stemming from more effective pricing (with the use of *dynamic pricing* methodologies) and marketing campaigns, such systems also facilitate distribution of products in commercial outlets (cf. e.g. Simbe Robotics), identification of the best possible locations for new outlets, as well as locations that ought to be closed down for a variety of reasons. A somewhat exotic but also viable use of such algorithms entails optimising the distribution of various objects inside casinos – one can only guess what the benefits are to be expected here . . .

One of the values most often emphasised by suppliers is the **increase of employee potential**. This means not only reducing workloads (as it may be broadly interpreted as human replacement – which, at least for responsible employers, continues to be a negative rather than positive aspect), but also, possibly above all, enhancing their abilities. A good analogy can be found in the context of manual labour: a crane or digger naturally 'enhances the abilities' of an employee: without it, one would simply be unable to perform certain tasks. It is interesting to consider how this process, which literature sometimes refers to as *human augmentation,* has evolved with the introduction of intelligent technologies. Interestingly, while *augmented reality* solutions have surprisingly quickly found a niche in industrial contexts (accelerating employee training, facilitating repair and maintenance processes, or adding an entirely new facet to design work), 'analytical assistants' (such as e.g. the convenient analytical system available in the Google Spreadsheet package) still remain at the stage of testing and verification by more advanced users. Other notable solutions that could, in a sense, be placed in this category include e.g. the already mentioned Cognitive Automation system offered by Aera Technologies (with its constantly expanding range of skills), as well as the more and more popular telepresence solutions (autonomous robots with a display projecting the user's face and capable of moving around in e.g. conference hall or hospital environments). The key benefits boil down to **broadly understood enhancement of human abilities**, albeit

no longer exclusively mechanical (as in the case of the digger) but also analytical or related to telepresence.

Before concluding the part focused on optimisation, one should also mention that intelligent systems are capable of optimising their 'environment' (e.g. processes) but also themselves (they are not only autonomous but also self-learning). The theoretical basis for these mechanisms has been presented in Chapter 1 and the example applications thereof in Chapter 2, in this section we will only focus on identifying their inherent benefits. A good illustration of self-learning systems is provided by those utilised in the context of medical diagnostics. The operational improvement of such solutions can be achieved in at least two distinct ways: the first is for the algorithms to be developed by experts in laboratory conditions and made available to end users (physicians) only upon successful testing (and often certification). The second option is to enable updates based on feedback received from trusted users (similarly to reinforcement learning): an expert evaluates the system's recommendation (e.g. confirms or rejects a diagnosis), after which a package of such feedback assessments is used to update, in near real time, the 'central' algorithm which is later periodically distributed to end users. The undeniable advantage of the first approach is the complete control over quality and therefore higher safety, whereas in the second case the benefits include accelerated learning and faster updates. Depending on the character of a given application (particularly in terms of safety requirements), the developers of autonomous systems tend to lean, to a varying extent, towards one or the other of the two presented extremes.

As follows from the above, there are many ways in which tangible value can be generated by intelligent, self-learning, autonomous solutions – the Table below presents a summary of the main points discussed.

Automation	Asset management	Monitoring	
		Inspection	
		Maintenance	
	Manufacturing	Process automation	
		Quality assurance automation	
	Logistics	Process automation	
Process optimisation and better productivity	Design	Analyse service patterns to improve product design & cut warranty costs	
	Faster implementation	On-demand automation	
		Faster system adoption and learning	
	Operational efficiency	Resources	Allocate resources most profitably
			Predict maintenance needs
			Improve on-shelf availability
			Improve scheduling of support resources
			Accurately forecast staffing needs

		Supply chain	Identify and negotiate with the most cost-effective manufacturing and delivery partners
			Enhance predictions of delivery times
			Optimise procurement strategies for proppant, water, and other consumables
		Production	Optimise production while keeping costs low and product quality high
			Accurately predict production and consumption to
			Optimise pipeline scheduling for product flows
			mining
			building AI models
			ADAS developers
		Delivery	Minimise time to deliver the shipment
			Improve on-shelf availability by each individual SKU and store and
		Back-office	Optimise cleaning
			Enhance back-office and invoicing/billing processes
		Finance	Optimise revenue cycle management and revenue prediction
			Optimise commodity trading and hedging strategies
			Automate financial controls for high-volume transactions

			Robots as a Service
	Sales and Marketing	Price	Use Price Optimisation and elasticity to identify optimal price points influenced by multiple factors such as item, brand, category, and more
		Product	Product Assortment and Supply
			Predict sales of product innovation and new product launches before they reach the market
		Promotions	Identifying the best SKUs and best promotion strategy (e.g. rebate, discount, BOGO, etc.) to achieve targeted revenue or volume
			Maximise profits through targeted offers
			Achieve marketing ROI across digital and traditional channels
			Make loyalty programs more effective
		Place	Identify best sites to open, expand, reduce, or close stores based on strategic goals without cannibalisation of existing store sales
			Optimise the placement of equipment on the casino floor
	Customer Service	Responsiveness	Faster notification response
			Shorter lead-times
		Communication	More effective customer messaging
		Claims Management	Identify claims for straight-through or manual processing

			Flag potentially fraudulent claims
			Identify subrogation opportunities
			Predict claim severities and large loss potentials
			Improve adjuster performance with outcome-based assignments
		Reduce warranty and service costs	Reduce repair costs
	Cyber-security	Better threat identification	
		Shorter response time	
	Human augmentation, employee potential	Simplification, better user-friendliness	
		Human augmentation, enhanced employee potential	
		Reduced workload of human employees	
Better planning and coordination	Predictive planning		
	Mission planning and execution		
	Automated vehicle controls		
	Swarm coordination		

Table 3. Possible values generated by AI systems in terms of cutting costs. Source: own elaboration.

Avoid risk

Intelligent systems are very often employed with the view to minimising widely understood risks. In many cases, this is made possible by their enhanced, compared to humans, perception capacity, speed and precision of operation, or ability to process large amounts of information, as well as the simple fact that they are machines – capable of performing a variety of hazardous missions without endangering human personnel. Similarly to the facets of *driving revenue* and *cutting costs*, the below presentation will synthetically identify the respective values that can be generated in this particular area.

Intelligent systems are often deployed as elements of infrastructural security. As already mentioned on several occasions, IT systems can generate both benefits and threats: their high complexity and significant exposure to outside interference (sensors, communication) render them susceptible to external attacks, primarily digital but sometimes also more traditional in nature. Given their increasingly critical role in many organisations and processes, the adequate protection of such systems becomes paramount. It is therefore hardly surprising that there are already many companies on the market whose chief offering constitutes solutions enhancing the security of critical infrastructure.

The first class of the same is dedicated to securing sensor systems and, in more general terms, advanced IT systems employed in autonomous devices (e.g. vehicles), industrial machinery (IoT sensors), as well as entire smart buildings. Thanks to mechanisms facilitating ultrafast detection of a potential attack (e.g. by identifying anomalies in signals transmitted by or to sensors), it is possible to **prevent** attacks and therefore **minimise their adverse consequences**. In turn, so-called *forensic analysis* aims to **gain better understanding of new forms of attack**, which in turn leads to **improvement of security measures and systems' defences against a growing range of potential threats**. The real-time support of Security Operation Centres (SOC) allows experts employed there to **focus on the most significant threats** and **quickly identify the same**, which naturally **reduces response times** and **contributes to better system security**. Another important value, similarly to DataScience platforms or automated online campaigns,

stems from **reduced overall costs of security measures** (even taking into account any applicable licensing fees).

Another area of AI implementations oriented towards minimising operating risks relates to monitoring and securing industrial infrastructures. Systems supporting thermal mapping, leak detection and inspection, emergency response or anti-drone protection facilitate **faster identification of potential threats** (e.g. attacks or hazardous leaks) **and failures** (e.g. of power grids), which in turn allows **faster response** and consequently effective **threat prevention** or **minimisation of adverse effects**. When a threat occurs (e.g. an explosion in a refinery or oil leak at sea), autonomous systems provide the needed situational awareness to support decision-makers and emergency response units in **coordinating and effectively executing their response**, and afterwards to **quickly and accurately assess the damage caused by the attack or disaster** (e.g. for the insurance purposes).

In turn, monitoring of border zones, border crossings, ports, and airports helps competent authorities to detect unlawful crossing, smuggling, or attempts to carry potentially hazardous substances on board. In such instances, the detection capabilities offered by machines exceed those of human security staff (e.g. thanks to advanced laser sensors, thermo-visual cameras, or infrared based sensors detecting specific substances), **thus expanding the range of tools available to controllers** and **allowing them to focus their attention on high priority incidents** (rather than statistical surveillance).

Autonomous systems are also used for the purposes of minimising operating risks in the context of key business processes. Intelligent sensors help to identify production line anomalies contributing to **product quality assurance** and, due to early identification of threats, **minimising the risk of failures** and related downtimes, which facilitates better **operating continuity**. Automated quality control systems (e.g. using dedicated image recognition solutions calibrated to detect particular problems) improve the overall monitoring of product quality (be it fruit or laser-welded elements). Furthermore, advanced logistical systems are able to gently transport goods in a way minimising the risk of damage, which **adds to product and goods safety in a variety of processes**.

Finally, systems facilitating prediction of deliveries from respective suppliers and, in more general terms, their automated scoring, help to **reduce supply chain related risks.**

Intelligent systems can also be effectively used to limit risks related to certain products themselves. The first category of the same includes various more or less autonomous vehicles – not only those present in the public space but also a wide range of industrial vehicles operating within the confines of warehouses, production floors, construction zones, or mines. The systems installed in such vehicles must provide adequate **situational awareness** (by interpreting both external circumstances and e.g. condition of the driver) in order to **minimise the risk of collision**. They **protect the vehicles** against external attacks (both mechanical (which is fairly common with drones) and digital), as well as manage their overall condition (maintenance, battery charging, etc.). The large amounts of data generated by autonomous vehicles, in particular by their dedicated security subsystems, can also be useful in the development of new insurance products: both in terms of adjusting the insurance cost depending on driving styles or actual usage (in e.g. vehicles with low levels of autonomy), and dedicating products specifically to autonomous vehicles (indeed, this market is already beginning to gradually emerge). As shown above, security systems can also yield certain benefits related to the **development of new products** or lines of business.

The direct beneficiary of such solutions is, quite naturally, the operator (e.g. driver) and road user (or person employed by a company utilising robots), while the indirect beneficiaries include fleet operators (**higher security and better efficiency** of the entire system), suppliers (**goodwill, reduced costs of potential claims**), as well as entities operating in their immediate business milieu (e.g. regulators or insurers), be it in the context of **compliance monitoring** or **development of new products**.

Intelligent drones constitute yet another interesting group of secured 'products'. Similarly to vehicles, they are vulnerable to various failures and attacks, including data theft or hijacking attempts. Attackers can target sensors (e.g. motion detectors or CCTV cameras) but also actuators (e.g. garage door controllers or bank locking mechanisms) and central operating systems (such as e.g. Amazon Alexa or Google Home). Securing such systems generates benefits similar to those already discussed above in the context of industry,

although in the case of smart homes, one also has to consider the additional aspect of **protecting privacy**. Data from cameras or microphones installed both inside and outside of buildings are often very sensitive in nature. The risks here go beyond simple intimacy, including also e.g. the ability to identify certain daily patterns (e.g. leaving the house), which can be exploited by criminals or companies trading in so-called behavioural pattern data. It is noteworthy that similar threats are also faced by users of most mobile devices, in particular the increasingly popular wearables – utility electronics capable of gathering staggering amounts of data on user behaviour. Systems assisting the suppliers of smart home solutions have the added benefit of improving the **sense of security, privacy, and intimacy** of their users.

Autonomous systems are also often used for the purposes of increasing personnel safety. **They can replace people in the performance of hazardous operations** (e.g. transport of hazardous chemicals, work on high-voltage power lines or in disaster areas) as well as **ensure that no humans are present in high-risk zones**. The solutions assist e.g. drivers (**warning** them when detecting distractedness or when a the situation on the road suddenly changes), as well as production crews (using e.g. enhanced reality devices (glasses, tablets)) by **identifying potentially hazardous objects** (e.g. hot machine elements). At the same time, they also reduce the risks faced by persons responsible for finance or compliance by e.g. identifying suspicious transactions or non-compliant behaviour of other employees. The key benefits in this context include, quite naturally, **better employee safety** but also **reduced operating risks** and **financial costs** (e.g. related to damages and/or litigation).

A particularly interesting area of intelligent systems' application entails **limiting risks related to customer behaviour**. AI algorithms facilitate early detection and prevention of fraud (finance and insurance). Advanced analytics of credit histories aid the assessment of creditworthiness and insolvency risks, both at the level of individual customers and entire asset portfolios, and allow organisations to take adequate preventive action if the risk of insolvency is detected. The obvious benefits include **reduced financial risks and limited likelihood of loss**, but also **ability to develop new financial products**

(addressed e.g. to client groups normally perceived as risky but in fact showing high capacity for debt repayment) as well as **optimisation of product and client portfolios**.

Intelligent systems help to identify clients showing high likelihood of churn and recommend adequate, personalised preventive measures. This is particularly important in contexts where the costs of acquiring new clients are particularly steep: in order to maintain profitability, companies must foster good relationships to ensure continued interest in their offer. The benefits of churn prevention systems include **minimisation of profits lost** but also **increased customer satisfaction and loyalty**.

One sector where the use of intelligent systems for safety related reasons is particularly interesting is healthcare. Notable examples in this context include solutions reducing the risks of hospital readmission due to premature release, contracting new infections during hospitalisation, as well as sudden deterioration of patient health (with indication for transferring the patient to an intensive care unit). The evident benefits include **improved patient safety** as well as **reduced treatment costs** (in line with the principle of prevention over treatment).

The final area where the safety of 'customers' is of key importance is the public sector – in this case, the 'customers' are simply e.g. city residents or participants in mass events (sports, music, political, conference, etc.). Autonomous robots are being increasingly deployed for the purpose of securing such places or events. The resulting benefits include primarily **improved surveillance capacity** (by assisting and sometimes replacing traditional patrols) both space and time-wise (robots are able to operate non-stop), **improved quality** (similarly to inspection robots, such solutions offer enhanced perception and situational awareness capabilities – e.g. the ability to detect unconscious or fighting people), as well as the resulting **improvement of surveillance efficiency and responsiveness (reduced response times)**. Indirectly, the use of such systems (not only in urban areas but also at e.g. airports) facilitates **better safety** (including prevention of terrorist attacks) **of both citizens and security personnel**, as well as **reduced surveillance costs**. One the other hand, one should also consider the potential risk related to the possibility of authorities abusing the data gathered by such systems.

Autonomous systems are also used for the purposes of ensuring legal safety. For instance, financial institutions take advantage of such systems to prevent money laundering, while companies managing fleets of autonomous aerial vehicles (e.g. drones) use them to ensure continuous compliance with the applicable air traffic regulations. The results include **reduced costs of supervision, better compliance, reduced risk of unlawful behaviour** and **reduced costs that may stem therefrom (legal, financial, and goodwill related).**

Finally, one should mention the role of autonomous solutions in environmental protection. This is facilitated by the aforementioned monitoring and inspection systems, but also solutions used in precision agriculture (capable of reducing pesticide use by over 90%) or recycling automatization. The benefits include **reduced environmental pollution** but also **reduced costs of waste processing**, which is often subject to regulatory requirements.

Similarly to the previous section, a brief summary of the presented benefits resulting from autonomous implementations in the sphere of risk avoidance has been presented in the table below.

Infrastructure	Data and Sensors
	Industrial infrastructure
	Digital systems and networks
Processes	Reduce operational risk
	Reduce supply chain risk
	Improve cybersecurity by prioritising attack investigation and remediation efforts with AI
	Improve QA and minimise liability exposure
Product	Vehicles
	Energy
	House
	Finance
Employees	Work methods
	Workplace safety
	Human replacement in hazardous conditions
Customers	Default risk
	Churn
	Medical safety
	Public safety
Compliance	Assure compliance
Environment	Improve environmental safety

Table 4. Potential values generated by AI systems in terms of risk avoidance. Source: own elaboration.

Value Generation Model

As mentioned above, intelligent systems rely on a variety of machine learning **methods and technologies** (e.g. image recognition, segmentation, or time series forecasting) utilising data obtained from both external and internal systems (e.g. sensors). In turn, the technologies are the basis for various **skills** which allow the systems to perform their designated **tasks** and, at a somewhat higher level, also certain **roles** within organisations. Said tasks **influence** diverse areas of activity by generating specific **value**, which may eventually completely **change the rules of market competition.** Presented below are details of the value generation method previously described in [@wodecki2018], developed in the light of the above deliberations – in the subsequent part of this study, it will be used as the basis for analysing autonomous systems.

Value generated by AI systems

Intelligent systems can impact various areas of an organisation – it would be prudent, therefore, to begin by identifying the key foci of such influence.

AI naturally impacts the operations of an **organisation** as such. This influence may pertain to **infrastructural resources** (production lines, machinery, transport, storage or commercial areas) as well as **information** (data, experience and knowledge bases). Ai can also have an impact on **employees**: the ways in which they perform their work, the necessary competences, decision-making or roles within the organisation, as well as **business processes** (primary and auxiliary). Technologies have the capacity to reshape entire **markets**: change competition rulesets, generate customer or partner advantages, or create niches for entirely new roles (e.g. market coordinators or information proxies). This, in turn, may necessitate adequate adaptation of **business models**.

Once the areas of influence have been identify, the next question should pertain to its character: **how** can AI affect the given area?

Intelligent systems can alter **modes of operation**: how we **perform** tasks, **group them into processes**, the **competences** necessary to do the same as well as the work **environment** as such. The **place** of performing respective actions can also change (e.g. research, testing, maintenance) as can the **time** (for instance, products were once configured before being made available to the end user, nowadays it is more and more common for the users to configure devices during initial startup) and **duration** thereof.

Systems employing AI can significantly affect the processes of management and decision-making. In particular, the constantly improving effectiveness of predictive systems leads to a rapidly increasing popularity of *prescriptive intelligence* solutions, capable of providing management boards with in-depth recommendations for future actions, along with the underlying justification. This means that the **place** of decision-making processes as well as the **role of humans** in the same are changing.

The next stage of change is of **qualitative** nature. AI facilitates **new methods of product development** (where the role of the designer can be reduced to providing certain limiting values, while the design process as such entails a vast number of AI generated simulations), **new insights** into data (e.g. thanks to segmentation, simulation, forecasting, or self-explainable systems (XAI), **development of qualitatively new products** (e.g. identification of new market needs based on social media analyses), as well as **new communication methods** (e.g. V2X) and **new methods of threat identification** (e.g. anomaly detection in response to cyberattacks).

Intelligent systems also influence the forms and methods of generating, gathering, and disseminating knowledge within organisations, be it with regard to **customers** (their needs, ways of using services, or customer journey), **own products** (and their potential), **risk factors**, or **competition**. Finally, the **methods of observation** can also change: including the success indicators and tools used to measure them.

The **scope** and **manner** in which AI influences organisations generates certain **values** (**but also threats**). By analysing several hundred cases, it was possible to group said values under the following categories:

1. **Automation**: performance of activities, management of assets and other processes
2. Optimisation and productivity improvement in various processes and tasks
3. **Increased speed, responsiveness and flexibility** of processes and activities, and consequently shorter response times (to customer queries, new competitive products, or external threats)
4. Improved product quality and credibility (in the eyes of customers and partners)
5. **Improved safety** of customers, employees, infrastructure, and the environment
6. **Increased satisfaction** of customers, employees, and partners
7. **Financial benefits**: increased revenues, reduced costs and greater profitability.

The above list of possible areas of influence, character of changes and generated values may be used as the starting point when formulating a business case, e.g. by following the below steps:

1. **Identify the area**, where AI can be expected to generate the greatest value. To do so, one may refer to the possible applications of AI systems described in Chapter 2 in the context of different functional areas and/or industry sectors, other available expert analyses (e.g. provided by established advisory companies), or simply review the offers of potential suppliers.
2. **Identify the benefit** that can be derived from the intelligent system
3. Specify **tasks** that the intelligent system would be expected to perform in order to facilitate the identified benefit

4. Identify the methods, tools or third-party products needed to perform said tasks
5. **Plan the implementation** following e.g. the good practices discussed at the beginning of this chapter. In particular, decide on the project management approach, identify the necessary resources, mile stones, and methods of measuring success at the end of each stage
6. Perform the implementation and evaluate its results.

It is noteworthy that the aforementioned procedure can be used when initiating a single project but would have limited application in the context of planning an entire portfolio of projects and their alignment with the corporate strategy. Nonetheless, it provides a solid basis for developing the methodology whose first iteration is presented in the subsequent part of this chapter.

AIR Management: method of formulating business cases and strategies for adopting autonomous systems

Building a business case and developing a strategy for implementing a complex system in an organisation is only the first step of the way. One must not forget that the ultimate goal is not the installation or commissioning of the new technology, but initiating processes of change which, at the end of the day, are expected to permanently and sustainably aid the organisation in creating value and increasing its competitive advantage or, in the case of most public organisations, capacity to perform their statutory missions.

If the implementation of change is to be successful, it is necessary to involve many actors (stakeholders): from leaders, through sponsors (e.g. owners or investors), managers, specialists and first-line personnel members, to partners and customers. Securing their willingness to participate, however, is hardly ever easy: the underlying technologies are

difficult to understand, the consequences of implementation can be unpredictable, there are still no reliable good practices on the market, not to mention that the media are nowadays rife with near-apocalyptic visions. Hence, a viable method of formulating a good case and then successfully adopting it may be highly desirable but certainly not be easy to put together.

The proposal presented below relies on a **metaphor**: the assumption being that a reasonable analogy to well known processes and contexts may help change leaders to better understand the new processes and consequently, be more effective when the time comes to design and implement the relevant solution. Based on numerous conducted analyses as well as the author's own implementation experiences, it was decided that the best metaphor for the adoption of autonomous systems would be a comparison thereof to **human resources and talent management**. Presented below are the details of the concept which, one hopes, may aid managers in developing effective solutions in their own organisations. Firstly, certain analogies between AI systems and human cognitive processes will be highlighted, followed by a short consideration of the key functions of processes observed in the context of organisational human resources management (HR management). Finally, the proposed method of managing AI 'resources' (AI Resources Management: AIR Management) will be presented.

Let us begin by referring to the concept of *Cognitive Computing.* It reflects specific methods of developing AI technology based on the analogy between intelligent systems (their operation and architecture) and the inner workings of a human brain, particularly its cognitive areas. Similarly to humans, intelligent systems posses certain **sensory** capabilities: they can receive signals from their surroundings by using senses in a variety of modalities (e.g. image, sound, or touch). Data provided by such senses, once properly processed, becomes the foundation of **perception** and creates the ability to recognise things and situations. In order to effectively function in its environment, an AI also needs to have access to **memory**, both **short-term** (including working memory) and **long-term**, as well as the capacity to **learn**. Successful performance of tasks also requires access to

higher cognitive functions such as the ability to determine (or interpret) **the objective** and functional **planning** (in particular to decompose the primary objective by identifying particular goals and activities necessary to accomplish the same, as well as establish a timeline – which in turn requires the ability to recognise cause and effect relationships). Finally, intelligent systems are increasingly expected to show **empathetic** and **social skills** (e.g. in the case of intelligent swarms). Said abilities, currently intensively studied and developed by many of the most esteemed academic and corporate research teams, are not only modelled after the inner workings of the human brain but more and more often allow one to explain many unresolved cerebral phenomena – the knowledge transfer is evidently beginning to work both ways.

Equipped with such skills, intelligent systems can perform a variety of **tasks** and serve a variety of **roles** in organisations. The tasks are performed in the context of a certain **work environment** (which ought to be properly prepared and secured beforehand), both **independently** and in **collaboration** with people, other machines, or the organisation's IT systems. In order to be effective, however, such intelligent systems must first be **adapted** to the requirements of given tasks: **retrained or onboarded**, and in a longer-term perspective further developed.

The tasks performed by intelligent systems are the primary source of their **value**. When 'hiring' an AI at a given workstation, it is necessary to define not only the scope of applicable work but also the goals (benefits) and **methods of measuring their achievement**. Monitoring the value generated by autonomous systems is, in turn, the basis for verification of the implementation's business case, **in particular the relevant ROI estimates.**

Even the brief analysis presented above reveals many possible analogies between managing AI 'resources' and personnel. In considerably simplified terms, the key goals of managing human resources and talents in organisations include:

1. **Recruitment**, i.e. ensuring the availability of adequately qualified people to perform tasks related to workstations or projects. It is usually preceded by an **analysis of staffing needs** based on competence requirements adopted at the level of respective workstations and/or roles within the organisation, or requirements notified by managers running respective projects. Recruitment may be conducted **continuously** (e.g. in hight turnover positions) or on a **case-by-case** basis. Employees can be sought both **outside** the organisation (e.g. searched for 'passively' or by analysing candidate applications) and **internally** (internal recruitment). The role of the HR department most commonly entails candidate **selection** (identification, preliminary incentive, and competence assessment) and presentation of the most promising candidates to the interested managers (e.g. process owners or business unit heads). After the recruitment, new staff members go through introductory processing: so-called **onboarding** (formal organisation (e.g. medical examinations or H&S training), as well as initial orientation, on-the-job training, providing access to company systems, etc.)
2. **Measurement of competences, performance and potential** of employees, candidates, or subcontractors. Competences are broadly understood as knowledge and skills (both 'hard' and 'soft'), experience and education, which can be measured in a variety of ways: from tests and exams, through interviews, to individual performance reviews. In many cases, the role of HR departments includes **performance** management and evaluation of employee **development potential**.
3. **Management of the work environment, knowledge and change.** Creating optimum work conditions (ergonomics, safety, adequate tools for communication and groupwork, etc.), supporting knowledge management processes (generation, storing, organisation, and dissemination of knowledge), as well as supporting various transformation processes (e.g. in the context of mergers and acquisitions) and managing change.
4. **Workforce development.** Identifying developmental and training needs, identifying and implementing of optimum development programmes (e.g. training, workshops, study visits, coaching or mentoring), as well as managing talents (e.g.

identifying roles within the organisation that ensure the fullest utilisation of employee potential, planning professional development and career paths, etc.).

5. **Motivation.** Identifying factors most effective incentivising employees, atmosphere within the company, taking adequate steps to motivate staff members, develop remuneration schemes to maximise engagement, proactively identify and retain key employees.
6. **Correlating human resources development with corporate strategies.** In particular, accounting for medium and long-term plans of the organisation as well as the current and prognosed market conditions when formulating HR policies.

Based on the analogies between AI systems and human cognitive functions, as well as the identification of key functions of HE departments in contemporary organisations, we now have the basis for the formulation of the AIR Management model.

The gist of the proposed approach lies in the **treatment of broadly understood AI resources as analogous to human resources**. As already mentioned, this metaphor will facilitate building the business case (in particular implementing technology that immediately starts yielding visible effects, rather than technology for technology's sake), envisioning the scope of AI projects, simple and effective communication about the project within the organisation (particularly to people without technical knowledge), and consequently more effective adoption of the solution.

The presentation of our metaphor ought to begin with **recruitment**. In the context of technological solutions, the process entails acquiring and supplying a given solution. Similarly to HR, the first step is to analyse needs: process owners an/or end users identify the **range of tasks** with which the AI systems are to assist, at times also certain **quality parameters** applicable thereto (e.g. speed, precision, performance standards, etc.). However, at the end of the day, similarly to tasks performed by a newly employed staff member, the introduction of intelligent systems should also constitute a source of

concrete benefits – which can be identified by conducting the analysis discussed in previous sections.

One of the most common mistakes when identifying needs stems from the fact that persons attempting to define them are not aware of the possibilities offered by modern technologies, including the ways in which they can facilitate more efficient processes. At times, the optimum approach is for the customer to formulate a problem and/or need that the system is expected to address and allow the solution provider to propose the adequate configuration – in line with the old adage that 'business should determine *what* is to be provided, but technologists should be responsible for the *how*'.

The next step entails '**searching for employees**', which in the case of AI systems boils down to **identifying the available suppliers of solutions and/or technologies**. Similarly to HR, the 'recruitment' can be conducted both externally and internally. External robot sourcing can take the form of leasing (X-as-a-Service, where X can stand for a Robot, Autonomy, Insight, Intelligence, or AI), which corresponds to the *body leasing* formula popular in IT services; purchase of a ready-made solution (in a sense equivalent to hiring an expert); or adapting an existing solution (which in HR terms would correspond to hiring an inexperienced worker and providing them with training). 'Internal recruitment' can entail the development of a solution by the company's internal teams – in this case, trying to find an HR analogy could prove somewhat risky . . .

Once the potential suppliers have been identified, it is time to proceed to **selection**. Again similarly to HR Management, the offers submitted by respective suppliers are evaluated, after which the best offer (in a perfect world at least . . .) is selected. In HR, the best candidate is hired and **brought onboard**. In the case of autonomous systems, the same corresponds to the process of **implementation**, with the respective stages thereof focusing on **installation, configuration, additional training** (adaptation to specific tasks, e.g. in the case of robots), **integration** with other systems operating in the company

(ERP, domain systems or other machines), and **on-the-job training** – a process almost perfectly analogous to corresponding HR activities.

The implemented system starts to perform its tasks. Similarly to a human employee, its work is monitored and evaluated. The advantage of intelligent systems stems from the fact that a lot of the data they generate (their own status and working environment, task performance, etc.) can facilitate very in-depth **performance analyses** conducted in a variety of dimensions and in near real time – something that would be considerably more challenging in the context of human *performance management*. The **development potential** can be estimated, on the one hand, by analysing the given supplier's market position (the better the position, the higher the potential), and in the case of independent development with the use of open source technologies – based on the popularity of the particular technology among programmers measured with the use of statistical tools available on websites such as GitHub or Stack Overflow.

Similarly to a human employee, an autonomous system performs its tasks within a certain **work environment**. Adequately adapting such an environment can sometimes pose a considerable challenge (e.g. in the case of large production lines), although trends such as *on-demand automation* aim to minimise the efforts required. Nonetheless, companies implementing AI solutions must take into account the possible need to modify their working environments (e.g. by creating maps, marking out routes, modifying digital systems, etc.) and duly prepare their employees. A particularly important factor is that of **ensuring safety** of an 'AI employee', be it **physical** (mainly safety of people working in the vicinity of machines), **digital** (e.g. protection against cyberattacks), or **legal**: AI systems must comply with all applicable regulations, not only those pertaining to e.g. air traffic but also e.g. data protection (GDPR) – a particularly important consideration in the context of monitoring or medial diagnostics where the processed data is often of sensitive nature.

As already mentioned, a common method entails adapting the machine to a given workstation with minimum interference with the workstation itself, the established method of performing the given activity, or the related business process. This seems to be the most natural approach when attempting the first implementation of autonomous systems without having much experience to lean on. However, it is worth considering somewhat more revolutionary paths possible at this stage: paradoxically, it often turns out that it is ultimately easier, cheaper, faster, and decidedly more profitable to modify the place and manner of performing particular activities, which allows one to utilise the machine's potential to a considerably greater extent. This is reminiscent of the mass computerisation of companies observed several decades ago (in the days of MRP I, MRP II, and ERP systems): when faced with the dilemma whether to adapt the IT system to current business processes or vice versa, most opted for the former – while it was the latter option that could often generate considerably greater benefits.

HR systems also play a significant role in **organisational knowledge management** – the same processes are equally important in the context of intelligent systems. The case studies presented in Chapter 2 (cf. also [@wodecki2018]) show that autonomous solutions take advantage of large amounts of data (Big Data), often provide qualitatively new information, and are able to automatically generate knowledge (cf. e.g. *conversational intelligence* solutions taking advantage of *natural language generation* technologies, or Business Intelligence systems). The same constitutes a **new source of knowledge**, significantly enhances its **scope,** and changes the **methods of its dissemination**, often requiring companies to ensure adequate preparation of employees at various levels. Consequently, AI systems may significantly influence the **ways of generating, organising, analysing, and distributing knowledge**, and indirectly also the **methods and locations of decision-making**. This is one of the reasons why issues related to knowledge management in the AI era are a particularly inspiring research topic.

Another significant function of HR processes relates to **managing personnel development**. In the case of intelligent systems, such development takes the form of

system updates, additional training of existing algorithms and, at times, **modifying their architecture and operating modes**. AI methods, algorithms and techniques are in the process of very dynamic development – it is therefore advisable to acquire the same from specialized suppliers, and when deciding to independently create and develop a solution, to engage the applicable processes within data science teams. This is particularly important if AI is to become the basis of competitive advantage: most of even highly advanced solutions are often published and described in publicly available resource bases, sometimes complete with the relevant software. Consequently, a determined competitor might be able to gain a significant edge over a company that may have implemented an intelligent solution but did little to continue developing it afterwards.

Another element in the development of intelligent solutions relates to the adaptation of business processes to match the potential of AI and identification of new areas in the organisation where subsequent pilot implementations would be worth attempting. In a sense, this roughly corresponds to the processes of talent management, in particular in terms of finding roles within the organisation that could be assigned to a given person in order to best utilise their potential. Ideally, the adaptation process should also include analyses of market trends and development strategies (so-called roadmaps) of key suppliers of intelligent solutions.

One of the most important responsibilities of HR departments is to **motivate** and **retain** employees. In the context of machines, this particular task is considerably easier: it is difficult, at least so far, to imagine a machine that is feeling moody or demotivated. A far greater challenge stems from **motivating employees to cooperate with and/or take full advantage of intelligent solutions**. Efforts oriented towards achieving that goal ought to be included in the scope of any AI implementation: the genuine conviction of employees that the purpose of the system is to enhance their own potential as well as the productivity of the entire organisation ought to be considered as one of key drivers of change.

Finally, as already mentioned, HR development strategies need to be aligned with the organisation's master strategy. This naturally also translates to strategies relevant to implementing intelligent systems. In this case, however, one should consider establishing some sort of a feedback system: i.e. not only manage AI with the view to contributing to the achievement of strategic goals, but also update strategies in order to take maximum advantage of in intelligent system's potential. Such a feedback loop may prove to be a viable driving force for a modern company.

The purpose of the presented metaphor comparing issues related to AI adoption to those observed in HR management is to support leaders in effective management of change generated by an AI implementation. When approaching new concepts - and ML algorithms or autonomous systems are certainly examples of such - drawing an analogy to a well known process tends to facilitate a better understanding of the nature of projects and consequently reduce potential anxiety.

A summary of the analogies between HRM and AIRM has been provided in the table below.

		HR	AIR
recruitment <> acquisition	**need analysis**		need analysis
	identification	external: advisory services, body-leasing, employment	identification of providers or technologies
		internal	independent development
	selection	competence assessment	evaluation
		candidate selection	selection
	onboarding <> implementation	obligatory training (H&S, etc.)	installation
		sector-specific training	configuration, additional training
		granting access to systems	integration with other systems
		on-the-job training	workstation training
evaluation		of employee competences	of system capacity
		of employee performance	of system performance
		of employee development potential	of system development potential
work environment management	**workstation organisation**	groupwork and communication tools	workstation adaptation (e.g. place on the production line, warehouse infrastructure, etc.)

			adaptation of business processes, work methodologies, and workstations to the capacity of autonomous systems
		ensuring workstation safety	ensuring safety (physical, digital, and legal (e.g. GDPR)
	knowledge management	knowledge generation	integration with analytical systems
		knowledge storage and organisation	adaptation of decision-making processes and methods
		knowledge dissemination	training on the interpretation of information and knowledge provided by intelligent systems
	information and change management	management of transformation and change	management of transformation and change
HR development <> system development	**development programmes**	training	updates
		mentoring	implementation of adaptation and self-improvement mechanisms
		coaching	additional training of algorithms, etc.
		identification of optimum tasks and roles	identification of optimum tasks and roles, process engineering
		professional development and career paths	strategies of system development
motivation	**motivation**	motivating employees to work	motivating employees to utilise the potential of machines, human augmentation
	retention	retention of key employees	system maintenance

Table 5. Summary of analogies between HR and AI resources management

Having established the **good practices** applicable to implementations of autonomous systems, **values** that can potentially be generated, **as well as the AIR Management metaphor,** we now have the basis to propose a method of effectively formulating business cases and adoption strategies for this type of solutions. Drawing upon the approach described in [@wodecki2018], one can recommend the following order of performing the respective necessary steps:

1. Needs analysis
    - **Identification of the area** where the greatest value is believed to be obtainable with the use of AI
    - **Identification of the value** that the given intelligent system can generate
    - Specification of **tasks** that the intelligent system ought to perform in order to secure said value
    - Development of a system for monitoring quality and system performance
2. Obtainment
    - Identification of suppliers and/or technologies who allow the performance of said tasks
    - Decision on whether to procure from a third-party vendor or develop internally
    - Selection (evaluation, choice)
3. Implementation
    - Adaptation of the working environment
    - Installation or production
    - Configuration, additional training
    - Integration with other systems, on-the-job training
    - Implementation of a system for monitoring the values generated
4. Consolidation of change
    - Organisational adaptation (new ways of conducting activities, business processes, knowledge management, etc.)
    - Continuous monitoring of the system-generated value

- Employee incentivisation to generate new initiatives oriented towards ensuring the maximum utilisation of the new technology

5. Development strategy
    - Alignment of AI management strategies with the company's business strategy
    - Implementation of mechanisms streamlining AI projects portfolios (in and long term) relative to the organisation's strategic goals
    - Implementation of development programs (dedicated to both systems and employees)

It is noteworthy that the method proposed above is not a recommendation for an AI management methodology, but rather a certain highly generalised procedural model. At the level of individual projects, application of agile methodologies still seems to be the most recommendable option.

Implementations of intelligent systems are still within the realm of *terra incognita:* even though a growing number of organisations elect to initiate such projects, due to their technical complexity, very rapid advances made in the field, as well as fairly common apprehension and anxiety of employees, the risks involved continue to be considerable. The method presented above is a product of in-depth analyses conducted on several hundred case studies, it employs good practices, identifies potential benefits, and relies on a sound analogy to the well known processes of HR management. At the end of the day, the recommended approach is also relatively simple: it is worth treating machines similarly to humans – hire them, pay for them, motivate and develop them so that they can generate tangible values.

Prospects for the Future

Potential impact on the future of organizations, markets and humans

Autonomous technologies influence not only individual companies but also entire markets. Organisations communicate and exchange ever larger datasets which come on more and more diverse forms. The pace of technological advancement is constantly growing, thus creating niches for new companies, new business models, and consequently new value structures.

In this chapter, we will take a closer look at network structures that may emerge as a consequence of the new (although rooted as far back as in the 1970s) trend in the development of AI – so-called distributed AI. Firstly, let us discuss the basic premise of such architecture and the opportunities, challenges, and concepts related to business network solutions that may feasibly stem form the same.

Distributed AI

Intelligent solutions are increasingly offered in the Software-as-a-Service model. In greatly simplified terms, the idea is that the user sends a query from their device, an AI installed and implemented on 'cloud' servers processes the query and returns the answer to the customer. The lead time depends on the complexity of the problem, system load, and connection speed. Such an architecture is viable in cases where the device itself has insufficient computing power to independently run an AI service and where ultrafast response times are not critical. Most users do not even know or care where and on what devices the intelligent services are installed: what matters is that they are rendered reliably and relatively fast.

The same looks slightly different from the perspective of the providers. They must not only ensure availability of adequate software and infrastructure but also guarantee the efficiency, security, and continued development thereof. In the IT world, efficiency and

security are most often ensured by way of controlled **distribution** of resources: this way, in the event of a failure, other devices can take over from the malfunctioning ones, and any lost data can be retrieved from backups kept on other machines. The local data processing and storage facilities in such distributed structures are known as nodes, and the entire system is described as **distributed computing**.

Companies offering intelligent services and employing the model of distributed computing often take advantage of infrastructures provided by other companies specialising in rendering cloud computing services, e.g. Amazon, Microsoft, or Google. However, in most cases, they own the software constituting the foundation of the intelligent service itself (sometimes referred to as cognitive), e.g. conversation processing or image recognition solutions. Consequently, the arrangement only entails decentralisation of the infrastructure: the AI service itself is offered by a single provider.

Notably, the suppliers have a vested interest in increasing the intelligence of their solutions by continuously enhancing their precision, expanding the spectrum of possible applications, etc. However, such development requires one not only to continuously upgrade algorithms (modify architectures, test and implement new AI methods) and increase computing power, but also to ensure access to large amounts of good quality data and feedback (e.g. on the correctness of medical imaging diagnoses). All those factors determine the quality of the service offered and contribute to the overall competitive advantage. Clearly, it is in AI providers' best interest to maximise the number of processable queries: this not only translates to higher financial profit but also, if used correctly, to improvement of overall product quality.

On the other hand, a growing number of customers purchasing intelligent services are beginning to notice the inherent dangers of 'intelligence outsourcing'. The benefits of using AI in the SaaS model are clear: very short time-to-market, no investment expenditures, pure variable costs facilitating quick return on investment (ROI), guaranteed stability, and continuous development of the system by the provider's expert teams. Nonetheless, this model is not without certain intrinsic risks, particularly once the service becomes the key source of competitive advantage: expert knowledge external to the organisation (it belongs to the provider), lack of understanding or control over

algorithms that are crucial to the company's operations, dependence on the providers (and their survival in the case of start-ups), concerns related to data security and the risk of failing to comply with legal regulations (e.g. the GDPR). It is therefore hardly surprising that a growing number of companies are beginning to consider the possibility of securing their own needs for intelligent services, which, however, requires considerable investment.

One possible solution to the above problems affecting both providers and their customers may be provided by the mentioned concept of decentralising AI services (distributed AI: DAI). Presented below is its basic premise, as well as opportunities and challenges it creates. Subsequently, the concept of so-called cognitive networks will be discussed – a new type of value structures that may emerge as such architectures become more prevalent.

Let us begin with a short presentation of the possible aspects of AI decentralisation. The deliberations will be organised to reflect the cycle of developing and marketing an intelligent service (cf. a very good analysis by J. Rodriguez [@rodriguez2018, @rodriguez2019]). The first dimension of decentralisation pertains to **data necessary to train a model**. In the 'centralised' approach, data are provided by a single company or organisation, e.g. a manufacturer interested in implementing a predictive maintenance system. In this context, decentralisation simply means that training data are sourced from a variety of companies or individual users. A good example could be a provider of solutions interpreting medical imaging who obtains described medical images from a number of hospitals.

Decentralisation can also affect the stage of **constructing the AI model architecture**. In the traditional, 'centralised' approach, such models are typically developed on the basis of e.g. specific application scenarios, by a single team of data scientists employed either by a company specialising in providing such models or at the customer's own research centre. In this case, decentralisation means that several independent experts (or teams) are commissioned to solve the problem. Once all of them return their contributions, the best option is selected, or a synthetic 'ensemble' model is created to combine the benefits

of all of the solutions. The approach is currently already used by financial companies employing the methods of so-called algorithmic trading, e.g. World Quant or Numer.ai.

The subsequent stages relate to **training** and **optimising the model**. Similarly to the cases discussed above, in the centralised approach these tasks are entrusted to dedicated teams, whereas decentralisation entails involving a number of experts (data scientists).

Distributed Ai is no longer just a theoretical concept. Even now, AI algorithm marketplaces are already operating (e.g. Algorithmia or Singularity.NET), a wide range of auxiliary technologies are being intensively developed (e.g. OpenMind, Ocean or Algorithmia DanKu), and a growing number of companies are actively using such solutions (e.g. the mentioned finance companies Numer.ai and World Quant, and even Google – which utilises its proprietary *federated learning* method to optimise the performance of the GBoard digital keyboard).

The mentioned **algorithm marketplaces** are a particularly interesting concept – the point is to allow independent AI developers to market their solutions, and interested clients to easily access the same. Practically the only thing that the creator of an algorithm (e.g. for text excerpt analysis) has to do is follow the relatively simple process of uploading it to the system which will then automatically install it on its servers, create so-called application programming interfaces (API), and make it available on the platform. Next, a potential user can browse the marketplace for an interesting solution, test it online, and implement it in his own application using any of the common API interfaces. The fee usually depends on the number of services performed (e.g. analyses or queries), and in the case of a sudden increase in the number thereof, the system is capable of scaling up automatically (e.g. by allocating more power, memory or disc space). Unlike in the traditional cloud computing approach (where the provider has to independently manage the ICT infrastructure), this option requires virtually no involvement on the part of the creator. Such implementation methods are usually referred to as *Serverless AI*. AI algorithm marketplaces can be the source of a new value structure – this problem will be discussed more closely in a further section of this book.

As already mentioned, decentralised AI addresses many problems encountered in the 'centralised' model of marketing services under the SaaS formula. The first of the same is

often referred to as *the rich get richer* issue: AI algorithms provided by large companies receive more data with every implementation completed, thus constantly improving, which in turn attracts new customers and inflates the entry barrier faced by potential new players. This situation can easily lead to the emergence of oligopolies, a problem that is increasingly noticed by regulators in many countries.

Another issue stems from the general public's misconceptions about how algorithms actually work, how messages displayed to social media users are personalised, on what basis loans are grated (or not), and what factors are taken into account by algorithms responsible for sometimes very large transactions taking place on financial markets. Such algorithms play a vital role in modern societies, often influencing decisions of fundamental importance – hence, their centralisation and the related lack of transparency (*who trains them and how? how exactly are they used? what is their acctual potential?*) raise growing concerns.

The final problem associated with centralisation is not so much a threat as it is a lost opportunity. Despite the obvious benefits of centralised systems, there are areas where they are simply ineffective. Decentralisation of intelligent system development could increase the related potential, not least by allowing contributions from less prominent creators or data science teams.

Business ecosystems utilising open architecture (e.g. Stack Overflow or GitHub programmer environments) generate numerous **benefits** for their users. The already mentioned algorithm marketplaces create opportunities for creators to sell their services, giving them the ability to continue their development work. In turn, application developers gain convenient access to inexpensive expert solutions. The organisers of the market naturally also benefit by collecting their agency fee. In short, we can observe most of the networking effects typically associated with B2B platforms.

At the same time, DAI is not without its inherent and rather numerous **challenges**. First and foremost, there are concerns related to **data security and confidentiality**. How does one guarantee the same for information (in compliance with regulations such as the GDPR) that is made available to other entities for the purposes of training algorithms? Is it even possible to train an algorithm using encrypted data? How does one prevent attacks

in decentralised networks? And is it possible to ensure transparency of AI models without the need for a central, trusted organisation? Similarly, the notion of **decentralised algorithm training** immediately runs into the question of whether it is possible to train, provide and evaluate a model in the absence of an overseeing authority providing some form of supervision. **Decentralising the processes of algorithm implementation and improvement** raises concerns about the viability of calculations performed and feedback collected from large numbers of end devices (e.g. mobile hones), as well as the possibility of adapting AI models to the needs of individual customers and continuously improving the same based on the response/opinion of large numbers of end users. Finally, there are questions about the exact **economics** of such markets, in particular about effective methods of motivating key users to actively participate in the system.

The answers to those questions fall outside the scope of this book. It is noteworthy, however, that practically all of those issues are currently being addressed by a very active community, both on the technological and economic level. Nowadays, the DAI movement is no longer exclusive to scientists and programmers as a growing number of companies and business organisations, e.g. members of the DAIA (Decentralized AI Alliance), are becoming involved as well.

Cognitive networks

The presented decentralised AI architecture prompts the possibility of extending the very same notion to value structures. In classic management concepts briefly presented in Chapter 1, one could observe a gradual evolution from chains, through constellations, to value networks. However, whatever the case, the value-generating nodes of such structures were always associated with classic organisations. Still, one cannot but wonder what would happen if the value was provided by algorithms. And how this would impact the role of organisations and rules of competition applicable to such markets? The analysis presented below is, for now, only a hypothesis which is why it has been included in the chapter discussing research trends. However, given the existence of phenomena such as DAI, it seems important to present its general premise and possible consequences.

The idea of cognitive networks is as follows. Let us imagine a situation where more and more companies are starting to employ artificial intelligence either to enhance their operating capacity or customer understanding, or as an element of their own offering. In time, AI would become a source of competitive advantage, not just for those companies but also for their partners and competitors. This would likely stimulate an 'AI race' between the market participants, as well as a rapid development of AI methods and technologies as such. In order to maintain their position on the market, companies would have to develop entirely new competences: without the ability to independently and rapidly improve their own AI algorithms, they would have to learn how to quickly and effectively identify the best cognitive services on the market, test and integrate the same with their own products, while at the same time maintaining the ability to flexibly switch to other, even better intelligent systems as they become available.

On the other hand, the growing demand for AI would create a market for providers of cognitive services which would also become increasingly competitive. Competitive advantage in this market segment would stem directly from the cost-to-quality of service ratio. In turn, quality would stem from such characteristics as precision, speed, efficiency, security, and customisability. In order to maintain their position on the market, AI providers would also have to develop new skills, e.g. related to the algorithms' ability to self-improve (the more analyses are conducted, the better the algorithm), or the company's ability to quickly incorporate AI solutions ('skills') offered by other market players in its own new products.

Interactions between customers and AI providers would take place via marketplaces similar to the already mentioned Algorithmia or SingularityNET. In time certain specific support roles would likely develop in such markets: 'competence' proxies (helping customers to find the best possible offers), providers of security services (be it digital, financial or legal), and possibly even market coordinators (which might contradict the idea of decentralisation). At the end of the day, we would end up with network structures wherein trade would be conducted in cognitive services: advanced analyses, image recognition, and even (thanks to the availability of 5G technologies) industrial robot control – hence the suggested name of cognitive networks.

Before we proceed to describe key competences and competition rules that would be applicable to such structures, let us first briefly summarise the primary elements of their architecture. The **actors** in this context would be the customers purchasing AI services, AI providers, and support organisations (market coordinators, competence proxies, data providers, security providers, etc.). Due to the changeable nature and high complexity of the processes involved, both customers and providers would be represented on the market by digital agents: algorithms equipped with knowledge of the goal function, trained to behave in ways maximising the same, and possessing specific competences allowing them to purchase/offer AI services. Consequently, market **interactions** would be automatized – they would basically entail negotiations between digital proxies (a process that is already observed on online advertising markets).

The network **topology** would most likely be modular: similarly to the internet, this would entail the establishment of multiple subnetworks capable of communicating with each other. Customers would release information (properly secured of course) generated by their domain systems: IoT sensors, production or warehouse control systems, e-commerce solutions, etc. In turn, providers would submit analyses or recommendations to the customers or directly control some of their subsystems. Consequently, one might expect to see deep integration of customer and provider systems that would take advantage of the full spectrum of available technologies: communications (Bluetooth, Wi-Fi, LTE, 5G), sensors, digital twins, or augmented reality. Furthermore, the system would be complemented by continuously active reinforcement learning algorithms used by customers and providers alike to train their agents and optimise the operation of the entire network infrastructure.

The complex character of interactions taking place in cognitive networks could stimulate the development of qualitatively new support **services** addressed to both customers and providers. This could be divided into services related to **recommendations, flexibility, personalisation, and confidence building**. Firstly, the network will support customers in **identifying the best cognitive services** by accounting for such aspects as cost, quality, performance, flexibility (speed of connection and disconnection), transparency, or security – with a scoring system developed based on information obtained from

providers, other customers, and system statistics. Providers would be able to take advantage of parallel services recommending the best customers relative to such characteristics as the expected number of queries or quality of data supplied (i.e. factors impacting the potential for algorithm improvement).

The offer addressed to providers might be supplemented by the addition of **personalisation services**. Customers (and their agents) would leave a particular digital trail in the cognitive network. Analysing the same would allow one to identify their behavioural patterns as well as, potentially, current goal functions. Access to such information would allow providers to identify optimum customers as well as to better personalise cognitive services with the view to increasing their market potential. The package could be further enhanced with the addition of dynamic pricing services capable of matching price levels to the individual ability and needs of the customer.

Another class of services offered via the network would focus on **flexibility** support. Similarly to the way that dynamic pricing helps to optimise the profitability of providers, the ability to quickly switch between the best possible AI services will be one of the key sources of competitive advantage for customers. As a result, we can expect the emergence of a new segment of *intelligent chain management (ICM)* services. They will support partners (customers and providers) in quickly preparing data formats and API interfaces as well as providing additional training to algorithms specific to the given customer's business – while ensuring maximum effectiveness of the process of connecting with the AI provider. This could be compared to the already discussed concept of *on-demand automation* where logistics robots can be implemented practically within a single day. The entire package would be complemented by **trust and reputation management** systems. As already observed, communication and decision-making taking place within cognitive networks will be largely automatized (performed by adequately prepared digital agents). Combined with effective flexibility mechanisms, this may create considerable dynamics in the relationship: the customer will be able to quickly and frequently change his suppliers, and vice versa. There may be many risks associated with changes progressing at such a high pace – which motivates the introduction of mechanisms facilitating management of trust and reputation. The already operating B2B marketplaces or auction

systems rely heavily on user reviews. In cognitive networks, this function could pass on to agents representing the customers and providers, respectively, as well as data from the auction management systems. For the scoring to be reliable, one would need to develop solid standards for reputation evaluation, which could be facilitated by the consensus mechanisms currently used in blockchain systems.

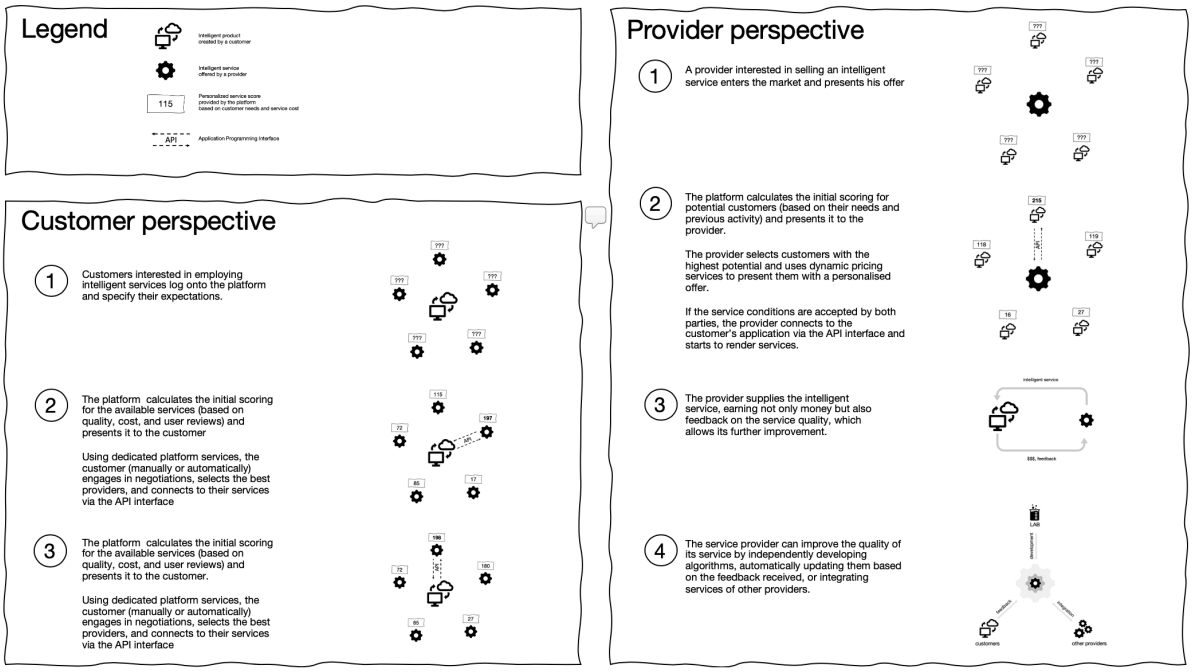

Figure 5. Working concept of the network services market from the perspective of customers and providers. Source: own elaboration.

The final component of the cognitive network architecture would be constituted by **control and optimisation mechanisms**. Communication withing a network requires solutions capable of standardising data exchange and transfer processes as well as effective coordination mechanisms (e.g. overload prevention, emergency communication, user log ins and log outs, etc.). Additionally, it would be advisable to implement mechanisms tasked with optimising the overall network operation, in particular its architecture, in a function of maximising the generated value, stability, and security.

A concept for the market implementation of such network services is illustrated in Fig. 5.

New competences

The vision described above encourages one to reflect on the new competences that would be key to effective operation in the context of the new value structures dominated by intelligent services. Presented below is a proposed identification of skills important to the providers, customers, and coordinators of the new network structures – many of which are so universal that one could safely recommend them even to companies not planning to actively participate in this type of markets.

Providers of intelligent systems, similarly to more traditional business models, will have to be **able to acquire new customers**. However, in their particular case such acquisition ought to account for factors related to strengthening the potential of their algorithms: it may turn out that a customer to whom we offer our service at a reduced price will ultimately generate a large number of queries, provide good quality data and feedback, and ultimately prove to be considerably more valuable than a customer that simply pays well. In short: when evaluating the value generated by the customer, one needs to consider the non-financial dimension related to improvement of the overall service quality.

In such a highly competitive and fast evolving environment, another key competence of the provider will also be the **ability to prepare and stimulate customers to independently generate value** with the use of the services offered. This could be illustrated on the example of providers of data science platforms: they offer solutions simplifying the creation and implementation of machine learning models, but it is up to the customers to formulate adequate business cases and manage such projects within their own organisations. For this reason, technology providers develop advisory and training ecosystems with the aim of increasing their customers' potential and likelihood of successful AI implementations in their respective organisations: the success of customers conditions the success of providers. Another factor encouraging the adoption of this approach is the significant modularity and reconfiguration potential of AI systems – customers can rearrange them like building blocks, in effect often generating solutions

with entirely new potential. It is the job of the providers to ensure that the building blocks used are theirs rather than the competitors' . . .

Another key competence of providers will be the ability to quickly adapt their products to the needs of a new customer segment. For instance, a company offering predictive maintenance services and addressing its solutions to the automotive industry must be ready to respond to a query form a heavy industry customer as well. It will be crucial to develop the capacity for fast and effective adaptation of the given service to the needs of a specific sector, most likely by relying on a combination of expert knowledge (heuristics) and autonomous mechanisms (e.g. taking advantage of machine learning algorithms).

The key competences applicable to **users of intelligent solutions** were already indirectly discussed in the preceding Chapter. The most important of the same include the ability to develop a portfolio of AI projects integrated with the overall strategy of the organisation, build strong business cases for such projects, effectively plan and execute implementations, and solidify the resulting changes. Auxiliary competences include a good grasp of this type of solutions and their characteristics, a solid data governance policy, and ability to identify the optimum providers or individual solutions. The ability to manage project engagement is also not without significance in this context, et the end of the day boiling down to adopting a suitable acquisition strategy: *which parts of the project should handle ourselves, and which are better left outsourced?* When aiming to use intelligent solutions based on network structures, it will also be important to ensure **flexibility**: the ability to quickly and effectively switch between various AI services.

Another interesting question in this context pertains to the future role and competences of **coordinators** of network structures. Apart from the already mentioned ability to stimulate the involvement of network participants (both in terms of financial incentives and by providing technical, advisory, ad economic support), much will depend on such persons' coordination skills. Notably, coordination will refer here not only to purely technical aspects (e.g. communication) but also to reconfiguring the roles and mutual interactions between partners with the view to creating new forms of value. It will also be important to effectively match customers with providers, which will be facilitated by mechanisms standardising the processed of data exchange and transfer, matchmaking

algorithms, and relevant legal regulations (e.g. contractual standards). Furthermore, coordinators will need to develop tools allowing in-depth analyses of vast quantities of data generated by market participants and have the ability to effectively use the same for the purposes of optimising and developing the network.

The presented concept of network structures wherein intelligent services constitute the primary object of exchange remains, for now, purely hypothetical. One hopes, however, that it may inspire readers to imagine interactions taking place on markets dominated by AI-based products, in particular in terms of the key roles and competences of the participants in such markets. At the end of the day, the most interesting reflection pertains to the place of the human in this new reality – this will be the focus of the subsequent section of this book.

Impact of AI on the role of humans in an organisation

The role of humans in organisations and societies intensively employing artificial intelligence has been the subject of numerous analyses and studies (cf. e.g. [@bostrom2014, @morini-bianzino]). How will autonomous systems affect the way we perform certain activities? How will they benefit our abilities? What professions are in danger of extinction? And consequently, what skills are worth developing? Seeing as the limitations of this book render it impossible to even briefly attempt an answer to these complex questions, let us at least consider some of the new roles that may emerge in organisations as a consequence of the continued proliferation of intelligent systems.

One of the first to almost certainly emerge in the near future will be that of **adopters**. Such persons, likely employed by both providers and customers, will serve the as 'interfaces' between the possibilities offered by intelligent technologies and actual business needs. The role of an adopter working for a company interested in utilising AI, e.g. for the purposes of facilitating better efficiency of business processes, will be to identify the best applicable solutions available on the market – i.e. he or she will be responsible for the stage of 'recruitment' as defined in the AIR Methodology. In particular, an adopter working on behalf of a customer will need to convert a business problem into a machine learning problem, be aware of the currently available solutions, be able to identify the

potential providers, and practically verify suitability of a solution (e.g. by conducting a pilot implementation). In turn, an adopter employed by a provider should be able to find new possible applications for the intelligent solutions included in the offer, where necessary adapt them to the specific needs of a new customer (in collaboration with his or her counterpart on the other side), and then provide implementation support and forward any experience gained in the process to R&D. In short, the chief responsibility of thus understood adopters will be to convert technological potential into viable business solutions.

Any AI product must first be developed and then adapted to specific real-world tasks and conditions. In both those processes, **trainers** (teachers) of machine algorithms will be indispensable. For the purposes of this analysis, the role will be understood somewhat less broadly than typically applicable when e.g. outlining the tasks of DataScience machine learning specialists: the purpose here is not to develop an algorithm from the ground up, but rather to prepare it for the performance of a specific task. Practically, the primary requirement here is good knowledge of training and model evaluation methods, with a somewhat lesser emphasis on the very complex and time consuming preparation of data for modelling purposes, selection of the algorithm and neural network architecture, and implementation of the same in the target IT environment. A machine trainer, having been provided with the model's quality parameters by the project manager or adopter, will be responsible for designing, performing, and evaluating the AI training process. If the AI system is supplied by a third-party provider, this will also entail collaboration with that company. Next, where needed, the trainer will ensure continued development of the solution performing tasks parallel to those related to **resource development** under the AIR Management model.

It is noteworthy that the 'training' does not necessarily have to relate solely to manual skills important e.g. in the work of robots. Machine trainers will, in particular, teach conversation systems to interpret the tone and meaning of customers' comments, but also to better comprehend the underlying context of a comment (e.g. by developing the ability to recognise sarcasm), develop empathy, gain a more in-depth insight into specific problems, or better evaluate the possible consequences of recommended actions. The

spectrum of viable didactic methods is also bound to develop, from the currently prevalent method of supervised learning, through reinforcement and imitation learning, to the eventual reliance on the machines' ability to independently supervise their own training (self-supervised learning).

Another important role may be played in organisations by so-called **explainers** (cf. [@morini-bianzino]). As the name suggests, they will be responsible for explaining the specific decisions and recommendations provided by intelligent systems: *What does the algorithm recommend? On what grounds? Why doesn't it recommend something else? What could be the consequences?* Furthermore, explainers will support decision-makers in arriving at concrete decisions based on said recommendations, in particular by assessing the usefulness of AI in a given situation and the advantages or limitations thereof relative to other available methods. In time, XAI systems (self-eXplainable AI) are likely to become a key technology supporting explainers in the performance of their work.

Finally, there will also be a niche for staff members tasked with the broadly understood maintenance of the AI solution (**sustainers**). They will be responsible for ensuring its correct operation, in particular in terms of ethical adequacy (e.g. ensuring objectivity or lack of discrimination) and correct collaboration with other staff members and systems. If needed, they will also be able to quickly identify malfunctions, suspend operation, and ensure process continuity.

The above list of possible roles is hardly exhaustive, many other tasks and competences are likely to be required from other employees of organisations. Leaders will need to learn to develop strategies with the view to fully utilising the potential of intelligent technologies, managers will need to adopt new ways of making decisions, and first-line personnel to learn how to increase their efficiency with the use of AI systems. Surely, there will also be other roles that we are not yet able to predict. At the same time, it will certainly be important to instil the capacity for critical thinking in the younger generation – one of the potential dangers that may stem from the replacement of humans in the performance of progressively more complex tasks relates to the risk of impairment to our cognitive ability (as the saying goes, *if you don't use it, you'll lose it*). As systems become more reliable, we find ourselves more inclined to increasingly rely on them, which is a danger

in itself – a healthy dose of common sense in this respect will have to be consciously maintained for many years to come. Even though one is fully aware of the fact that this particular notion of *healthy* is more than likely to evolve as well.

Research frontiers and inspirations for future research

Most of this book so far was devoted to presenting the practical applications of intelligent, autonomous, and often self-learning systems. The number of companies and implementations discussed may suggest that the industry has now reached full maturity. And in a sense, it has. Nonetheless, the observed very dynamic development of AI methods and technologies suggests that it will take a lot of time for companies operating in this sector to feel a considerable measure of stability.

Currently, there are numerous research teams, working at the most prestigious universities and technologically advanced companies and organisations, who are actively engaged in the intensive development of AI related methods and technologies, in particular in the field of reinforcement learning which is fast becoming the primary foundation of autonomous systems. Presented below are some of the arguably most interesting ongoing research projects that promise to significantly influence business technologies. The discussion will progress from primary studies, through applied research, and eventually to social sciences. The awareness of those projects will allow readers to better imagine trends that may emerge in the future. Hopefully it might also become a source of inspiration, as it did for the author of this book.

Primary research

Let us begin the presentation of primary research directions with training methods modelled after the learning mechanisms observed in the human brain. For years, this has been the central focus of research conducted by DeepMind – one of the leading research centres in the area of AI. Mott et al. [@mott2019] describe the use of attention mechanisms to control the agent's field of survey in the process of exploring and learning

a new environment. By employing said mechanisms, RL agents learn to control the focus of their attention, in particular in terms of differentiating between location and content ('where' vs 'what'), which was found to improve the efficiency of learning how to behave under new circumstances (human attention mechanisms are responsible in particular for 'rationalisation' of the use of limited cognitive resources: depending on the context, attention is shifted to the most important (rather than all) aspects of the situation). The paper reports on an interesting trend in research on various types of memory utilised by AI systems and the impact thereof on the speed of learning the ins and outs of operating in new contexts.

Another interesting direction in primary research pertains to the **possibilities of utilising human knowledge in machine learning**. Microsoft has been experimenting with a method described as **machine teaching** (cf. e.g. [@2019c]) whereby domain experts (e.g. energy management specialists) can simply and intuitively instil heir own knowledge and experience in intelligent systems. It is noteworthy that this approach differs significantly from supervised learning employed e.g. when training image recognition algorithms. In traditional supervised learning, the trained system (e.g. a neural network) is typically provided with 'training material' in the form of question and answer sets (e.g. a file with correctly described photographs): using the same to develop a model is the task of the network itself. Meanwhile, a human training a system will adopt a somewhat different approach: if explaining to an AI how to recognise e.g. a chair, the teacher will first provide a set of certain key features that need to be taken into account (e.g. having at least three legs) and only then proceed to illustrate this with examples. The added benefit of this approach stems from the fact that systems trained in this way are able to better justify their decisions or recommendations, which is crucial in the context of self-eXplainable AI solutions (XAI). Another interesting example of this trend is the SentiMate algorithm developed by the researchers at University College London (cf. [@kamlish2019, @knight]), which leant how to play chess by reading articles and online comments relating to various games, in each case additionally analysing the commenter's sentiment (and paying particular attention to very positive responses). Clearly, autonomous systems are now able to take advantage of human knowledge and experience in a variety of ways, both directly (by being taught) and indirectly (e.g. by reading) . . .

Another area of research focuses on **independent supervision of learning processes**. Yann LeCun, chief AI scientist at Facebook and a professor at NYU observed (cf. [@haoa]) that when we are children, we gain considerably more knowledge through exploration and experimentation that from instruction, and least of all from reinforcement learning. In Prof. LeCun's opinion, it is this direction (*self-supervised learning*) that holds the highest potential as a source of improvement in terms of the efficiency of AI learning processes. The team attempts to **integrate planning methodologies** (effective when applied to long-term tasks, provided an action plan is in place) **with reinforcement learning** (which is effective in individual teaching policies but struggles with long-term planning). The objective is to combine the strengths of both those methods with the view to effectively **solving problems that require long-term planning with limited feedback** and high-dimensional observations. To this end, the authors proposed an algorithm capable of decomposing a distant goal into partial targets and identifying sequences of simpler tasks collectively leading to the achievement of the ultimate goal. The decomposition of the principal goal is achieved with the use of planning algorithms, which in turn use abstractions of the environment facilitated by RL algorithms. This approach leads to a significant improvement of autonomous systems' efficiency in the performance of tasks consistent with the overlying goal that is only achievable in a long-term perspective.

In turn, Liangpeng et al. [@zhang2019] presented an algorithm supporting RL agents in calculating the **optimum exploration route** in the process of learning an environment (Value Iteration for Exploration: VIEC), which significantly reduces the duration of the learning process. Furthermore, Ramsted and Pal [@ramstedt2019] addressed the problem stemming from the often erroneous assumption that states of the environment within which an agent operates remain unchanged while a decision is being made. The proposed *Real-Time Actor Critic (RTAC)* algorithm is able to account for such changes significantly better than other RL algorithms and facilitates learning and **effective operation of systems in environments evolving in real time**.

Interesting findings in the area of autonomous systems' learning were also presented by Barreto et al. [@barreto2019]. Their approach focuses on training the **ability to combine**

known skills in order to develop new ones, necessary when solving complex, long-term problems. The solution posited in the publication entails defining and combining such skills in the so-called pseudo-reward space. The authors also point to the practical applications of the method in environments where certain actions require a combination of elementary skills such as resource management or robot navigation.

The research presented so far focused on developing so-called **higher cognitive functions of intelligent systems**: independent control over learning processes, exploration policies, hierarchisation of goals and tasks, and if necessary, combining existing abilities into new skills and learning to operate in fast-changing environments. Below, a few other inspiring studies will be discussed focusing specifically on the areas of hierarchisation and collective operation.

Task hierarchisation, one of the many attempted methods of providing autonomous systems with higher cognitive functions corresponding to those available to humans, refers to the ability to decompose a general, abstract goal and identify a number of specific contributing targets, actions necessary to accomplish those targets, and skills needed to perform those actions. A system equipped with this competence is capable of handling tasks that are defined only very generally and often in a fairly long-term perspective.

Multi-agency is employed in many of the applications already mentioned in this book: whenever the concept of an *intelligent swarm* appears in the description, it can be safely assumed that the providers of the given solution have very likely utilised this class of algorithms. Generally speaking, the idea is to prepare agents to collaborate in the performance of specific tasks, in particular by dividing duties in such a way that in total, they add up to the overall expected result. Particular applications include self-coordination of *intelligent swarms* operating both in the physical (e.g. groups of robots or drones) and virtual space (e.g. on-line team games).

RL methods allowing decomposition of complex tasks into many simpler ones are referred to as *hierarchical reinforcement learning*, and their applications in multi-agent environments is called *multi-agent hierarchical learning* (cf. e.g. [@zhang2019a, @yang2020]). It is certainly conceivable that research conducted in those areas will soon allow the establishment of virtual agent teams capable of creating management structures

composed of 'leaders', 'managers' and 'first-line workers' performing tasks at varying levels of abstraction. The 'leaders' will provide general goals, the 'managers' will be responsible for the performance of partial targets, and 'first-line workers' will perform the necessary tasks. This approach will be a natural next step in the evolution of already operating intelligent swarms of e.g. drones collaborating with each other to perform a common mission. One could easily imagine many possible applications for such solutions: from the mentioned drone swarms, through autonomous robots on a battlefield (cf. [@zotero-1663]), to swarms of nanorobots 'repairing' human bodies from within – presented below are several interesting directions in research that may soon make this possible.

Eccles et al. [@eccles2019] study new forms of communication (languages) developed by collaborating agents performing a common task. In order to achieve a collaborative goal, they need to effectively communicate in the course of their work. The authors aim to formulate a system of rewards that encourages agents to share valuable messages with their 'co-workers', while at the same time reviewing different forms of communication and their impact on the efficiency of task performance. Their results have been quite promising: it was discovered that agents allowed to independently use the developed language tend to show better efficiency than 'teams' which do not communicate with each other. In short: in order to improve the results of group work, virtual teams develop their own methods of communication.

Schroeder de Witt et al. [@schroederdewitt2019] generalise information exchange as knowledge sharing. Their algorithm - *Multi-Agent Common Knowledge Reinforcement Learning (MACKRL)* – helps teams to share so-called *common knowledge*, e.g. experiences or observations gathered, with the view to improving coordination. A key role in this approach is played by a hierarchy tree included in the policy: agents at higher levels of the hierarchy decompose tasks to obtain partial sub-tasks and delegate their performance to lower-level agents. The lowest tiers in this hierarchy are occupied by policies applicable to individual agents. They verify the effectiveness of their algorithm by applying the thus understood notion of micro-management to the performance of game missions in StarCraft II. This is certainly an interesting context for the proposed AIR Management

approach: one can clearly see an analogy with the hierarchy of management levels in an organisation.

Computer games, both off and on-line (where a number of players can play simultaneously), are among the most popular 'laboratories' used in the process of training and testing autonomous systems. The already mentioned DeepMind is one of the pioneers in this area: its research teams have been able to create e.g. an algorithm that achieved mastery in StarCraft II, a very popular (and difficult) on-line game [@zotero-1647]. To this end, they had to create a team of virtual agents capable not only of collaboration in the achievement of a common goal but also, in a sense, of creating a specific management hierarchy. Interestingly, the RL methods used in this project to allow training of entire teams of virtual agents, particularly so-called *population based training (PBT)* [@jaderberg2017], are currently employed by Waymo to improve the operation of autonomous cars (cf. [@knighta]).

A very inspiring study was also conducted by Du et al [@du2019]. They observed that what maters to teams of cooperating virtual agents solving complex problems is not only communication but also stimulation of various individual behaviours of the team members in a situation where, ultimately, the team receives a common, rather than individualised reward. In earlier studies, the problem was solved by either formulating separate goal functions for each of the agents or training a separate agent ('critic') tasked with individually rewarding each of the team members. Du at al. combined the two approaches by proposing a method where the critic is able to learn the individual incentive systems applicable to each of the collaborating agents and rewarding their behaviour based on that knowledge (*Learning Individual Intrinsic Reward (LIIR) in multi-agent reinforcement learning*). The method is tested in StarCraft II to evaluate the collaboration improvement. The approach further supports the concept of AIR Management: in order to effectively manage a team of intelligent machines, it is worth learning the intrinsic value system of each of them and then apply it in the motivation system . . .

Another fascinating example of using RL algorithms for the purposes of training collective behaviour was described by Baker at el. [@baker2020]. The authors used the concept of

hide-and-seek to train competing teams of virtual agents. After many iterations it was discovered that the teams developed their own, rather intricate strategies for hiding themselves and locating the opponents (e.g. building bunkers out of boxes or other obstacles, or using the same boxes when traversing barriers built by the opposing team). Observation of those collective efforts revealed many interesting similarities to human behaviour.

Applied research

Let us begin the presentation of applied research with a very interesting project conducted under the name of *TransLearn* by Kuka, a robot manufacturer [@zotero-1687]. The project aims to address two problems: on the one hand, the very high cost and long duration of additional robot training that needs to be conducted at a given workstation, and on the other, the limitations of training a robot exclusively in a simulated environment (so-called reality gap). The aim of the project is to use reinforcement learning methods for the purposes of integrating learning processes conducted on simulators and real-world production lines. This is to yield the benefits of reducing the cost of robot programming and, ultimately, facilitating the performance of workstation adaptation by a domain expert rather than programmer (the goal of the project is to develop a method that will not require any programming skills from the 'trainer').

Another interesting problem blending the distinction between primary and applied research is so-called imitation learning. In Chapter 2, the method was presented as an approach significantly accelerating the process of adapting a robot to the requirements of a given workstation. It has been observed, however, that simple imitation e.g. of a hand or arm movement proves insufficient in situations where the action has a very specific and often rather complex purpose. Studies conducted by Ding et al. [@ding2019] fit in well with the general trend of searching for new ways of improving existing imitation learning methods in cases where both the reward and goal of the action are not clearly defined (i.e. there are no reinforcement signals). Their algorithm called *goalGAIL (Goal-Conditioned Imitation Learning),* which is an improvement on the earlier *Hindsight Experience Reply*

(HER) approach proves to be one of the most effective solutions in its class, with numerous potential applications, particularly in robotics.

Reinforcement learning also finds many applications in the programming and management of ICT infrastructures. Mao et al. [@mao] described PARK: an environment simulating many different IT systems to allow agent training with reinforcement learning. With the use of this simulator, researchers can train their algorithms in solving problems such as adaptive video streaming, spark cluster job scheduling, SQL database query optimisation, network congestion control, circuit design, multi-dimensional database indexing, server load balancing, or switch scheduling. Clearly, the spectrum of possible uses is rather broad: from optimisation of network infrastructures, through data streaming and optimisation of the computing power of databases, to designing integrated circuitry.

As for some more down-to-earth (literally) solutions, one could mention the MIT study on teaching robots delivering small parcels how to correctly identify front doors [@zotero-1696]. It turns out that the problem is anything but prosaic: access to the doors is often blocked by obstacles not documented on maps, e.g. fences or shrubs. The problem is solved with the use of advanced techniques such as *Simultaneous Localization and Mapping (SLAM).* The very fact that research has to be conducted on that issue is a clear indication that often 'problems' that may seem trivial to humans prove to pose considerable challenges to machines and require the involvement of highly qualified research teams.

There have been hundreds if not thousands of studies pertaining to the practical applications of AI in general, and reinforcement learning in particular, in various sectors of industry (cf. [@li2018b]). In the context of sales and marketing, researchers are working on perfecting recommendation systems, personalisation algorithms, methods of boosting search ranking positions, improving fraud identification in e-commerce, and even streamlining crowdsourcing incentives. In transport, RL methods are studied in the context of vehicle fleet management, distribution of notifications in ride-sharing systems, traffic light management, or improvement of airplane fuel economy by taking advantage of thermal currents. Studies on the applications of reinforcement learning in energy

industry help to manage the cooling of data centres, forecast and respond to power grid overloads, control the systems of intelligent power grids (so-called SMART GRIDs), or manage energy consumption in buildings. In the sector of finance, research is being conducted into the applicability of RL algorithms in option pricing, execution of stock market orders, or development of asset liquidation strategies (liquidation strategy analysis). In turn, in the healthcare sector, the potential of RL is studied in the context of medical diagnostics, generation of diagnostic reports, or implementation of so-called dynamic treatment strategies. AI is also employed in systems improving the effectiveness of didactic processes, from personalisation of the learning process (adapting the content, pace, and methods of learning to the given student's individual potential) to systems matching compatible students and teachers by analysing their behavioural and psychological profiles.

As evident from the above, the spectrum of research on various AI methods and their potential business applications is very broad.

Finally, the subsequent section will focus on some potentially interesting research directions observed in the context of social sciences.

Management and social studies

Autonomous systems and more generally AI have also inspired some rather interesting research in the context of management studies.

Firstly, due to the growing needs of businesses, one can expect many new research projects to emerge with the aim of studying the relationship between the quality of technical parameters in advanced solutions (e.g. algorithm precision) and their actual business value. The methodologies resulting from studies such may later aid managers in building business cases or formulating performance frameworks for AI projects. Research projects in this area ought to be implemented by interdisciplinary teams: they require the involvement of experts in the fields of financial controlling, business process optimisation, as well as AI methodologies and techniques, in each case capable of bringing both academic and business experience to the table. Therefore, the execution of such grants is a challenge in itself, but one that yields very tangible results.

Equally promising studies are those pertaining to the effectiveness of various methods of developing portfolios of AI projects, formulating business cases, and planning implementations. This topic has already been discussed with an indication of some qualitatively new approaches such as *skill-by-skill* or *by-confidence-level.* Due to the rapid development observed in this area, one can expect new methods to continue being developed – analyses of their effectiveness in various contexts may yield some interesting and valuable results.

It is also interesting to consider studies focusing on the impact of AI on the places and methods of decision-making, particularly in the strategic dimension. Increasingly effective prediction algorithms already allow decision-makers to not only rely on past events but also account for those that are the most likely to transpire in the future (however odd this may seem . . .). This has the potential of radically changing decision-making rulebooks, naturally provided that the decision-makers themselves (with their own experience and skills in tow) are willing to adopt such methods as a viable element of the process.

Another interesting area relates to research pertaining to simulating decentralised AI markets. A valuable contribution in this context is offered by studies focusing on the collaboration between virtual RL agents in decentralised networks (so-called d*ecentralized networked deep multi-agent reinforcement* learning) conducted by e.g. Qu et al. [@qu2019]. Indeed, the first simulators of decentralised markets, primarily utilising BlockChain technologies, are already starting to emerge and may serve as environments facilitating studies on the dynamics of markets such as the cognitive networks described above.

Finally, research on the impact of technology on social value systems, particularly with respect to young people entering the job market, is also a very important field of study. In considerably simplified terms, advanced recommendation systems are already able not only to identify user needs but sometimes also generate them. The answer to the question of how the same influences the value systems of young people is vital to employers, particularly in terms of motivation, formulation of tasks and evaluation of their

performance. The assumption that a young employee subscribes to the same values as their superior can be a rather costly misconception . . .

Conclusions

Artificial intelligence continues to be perceived as rather abstract by many people. They are aware of its existence but mostly associate it with vague media hype, fashion, or esoteric scientific obsession. Many inspiring conversation tend to come to a dead end with the question: *OK, but what does that mean for me practically? What can I do with it here and now?*

One of the defining features of top-quality products (one is tempted to write luxury but the word has far too many connotations other that quality) is the fact that they are invisible. We only take note of them when they malfunction. A high quality shoe is comfortable (we can almost forget we are wearing one). A good car is quiet (we cannot hear it) and simply works (we do not need to fix it). A good lecture is simple: we have no problems understanding it. And our favourite computer programmes tend to be the ones we never needed to learn: we just knew how to use them straight away.

A similar principle applies to advanced technologies: we only notice them once they disappear or stop working. They make other things more user-friendly (e.g. instead of typing in a password we just need to look at a phone to unlock it), lighter, faster, cheaper, or more easily attainable.

Few end users ever stop to ask how exactly something like that is possible. What allows the phone to recognise its owner? How does it understand voice commands and simply executes them? It just works, which is great. That is all . . .

But the things look much different from the perspective of a creator. We live in a time when it has become relatively easy do objectively difficult things (e.g. for end users), but creating something simple requires a considerable effort, for instance by employing very advanced technologies which increasingly take advantage of artificial intelligence. And all that is effectively imperceptible to the user. And with good reason: this is exactly what the best technology should be . . .

In the near future, we can expect to be surrounded by more and more autonomous 'things' capable of not only operating effectively but also continuously self-improving. A key

aspect of our further development may turn out to be the gentle art of wisely balancing the consumption of benefits offered by technologies with their active development and utilisation in generating new value. Just like learning to play the guitar allows one to hear entirely new sounds in a well known piece of music, development of AI systems helps to better understand and utilise intelligent solutions. Indeed, that is the primary purpose of this book: to raise the awareness of the ideas that underlie such technologies and the value they can generate, in order to encourage informed experimentation.